EFFECTS OF HIP ADDUCTOR EXERTIONS AND TWO EXERCISE SEQUENCES ON VASTUS MEDIALIS OBLIQUE (VMO) AND VASTUS LATERALIS (VL) EMG ACTIVITIES DURING SQUAT EXERCISE



SIRIRAT KAEWWICHAI

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Sirirat Kaewwichai
.....
Miss Sirirat Kaewwichai
Candidate

Wattana Jalayondeja
.....
Asst. Prof. Wattana Jalayondeja, Ph.D.
Major Advisor

Roongtiwa Vachalathiti
.....
Assoc. Prof. Roongtiwa Vachalathiti, Ph.D.
Co-advisor

B. Mahaisavariya
.....
Prof. Banchong Mahaisavariya, M.D.
Dean
Faculty of Graduate Studies

Raweewan Lekskulchai
.....
Asst. Prof. Raweewan Lekskulchai, Ph.D.
Chair
Master of Physical Therapy Program
Faculty of Physical Therapy and Applied Movement Science

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was submitted to the Faculty of Graduate Studies, Mahidol University
for the degree of Master of Science (Physical Therapy)
on
February 20, 2009

Sirirat Kaewwichai
.....
Miss Sirirat Kaewwichai
Candidate

.....
Assoc. Prof. Prawit Janwantanakul, Ph.D.
Chair

Roongtiwa Vachalathiti
.....
Assoc. Prof. Roongtiwa Vachalathiti, Ph.D.
Member

Wattana Jalayondeja
.....
Asst. Prof. Wattana Jalayondeja, Ph.D.
Member

B. Mahaisavariya
.....
Prof. Banchong Mahaisavariya, M.D.
Dean
Faculty of Graduate Studies
Mahidol University

Roongtiwa Vachalathiti
.....
Assoc. Prof. Roongtiwa Vachalathiti, Ph.D.
Dean
Faculty of Physical Therapy and Applied Movement Science
Mahidol University

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Sirirat Keawwichai

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EFFECTS OF HIP ADDUCTOR EXERTIONS AND TWO EXERCISE SEQUENCES ON VASTUS MEDIALIS OBLIQUE (VMO) AND VASTUS LATERALIS (VL) EMG ACTIVITIES DURING SQUAT EXERCISE

SIRIRAT KAEWWICHAI 4936645 PTPT/M

M.Sc. (PHYSICAL THERAPY)

THESIS ADVISORY COMMITEE: WATTANA JALAYONDEJA, Ph.D., ROONGTIWA VACHALATHITI, Ph.D.

ABSTRACT

This cross sectional study was proposed in order to determine the effects of hip adductor exertions and two exercise sequences on EMG activity of the vastus medialis oblique (VMO) and vastus lateralis (VL) muscles during squat exercises. Thirty healthy young adults without a history of anterior knee pain voluntarily participated in this study. Surface electromyography (EMG) were recorded from the VMO and VL muscles of the subject's dominant leg during squatting without hip adduction and with 35%, 70% and 100% isometric hip adductor exertion. The two exercise sequences were: 1) squeeze (hip adduction) entire time-squat exercise (SE-SE); and 2) squeeze at 60 degrees knee flexion-squat exercise (SF-SE). EMG activities were normalized by the maximum voluntary isometric contraction (MVC) of each muscle.

The results showed significant effects of hip adductor exertions ($p<0.001$) but no significant effects from the exercise sequences and no interaction ($p>0.05$) on the VMO, VL activities and VMO:VL ratio. The %MVC of the VMO during squatting with 70% and 100% of isometric hip adductor exertion was greater than squatting without hip adduction (38.35-62.13 vs. 35.77 %MVC), %MVC of VL and significantly increased from 0% to 100% hip adductor exertion (36.2-73.60 %MVC). The VMO:VL ratio during squatting without hip adduction was greater than squatting with 35% and 70% of hip adduction exertion (1.06 vs. 0.76-0.87), but not significantly different from squatting with maximum hip adduction (0.94-1.00). No significant difference between the VMO and VL activities while squatting without hip adduction and squatting with 100% hip adduction was seen.

The VMO and VL activities during squatting combined hip adduction with SE-SE and SF-SE and were not significantly different. Both sequences could be trained with no different effects on the muscle activities. Squatting with 70% and 100% isometric hip adductor exertion increased both the VMO and VL activities. Squatting without hip adduction and squatting with maximum isometric hip adduction seemed to promote balanced activities of the VMO and VL muscles, which could benefit VMO muscle retraining for abnormal patellar tracking.

KEY WORDS: VASTUS MEDIALIS OBLIQUE/ VASTUS LATERALIS/ SQUAT EXERCISE/ ELECTROMYOGRAPHY

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ผลของแรงหุบสะโพกและลำดับการหุบสะโพกสองแบบต่อคลื่นไฟฟ้ากล้ามเนื้อ VASTUS MEDIALIS OBLIQUE (VMO) และ VASTUS LATERALIS (VL) ขณะออกกำลังกายท่าย่อเข่า
(EFFECTS OF HIP ADDUCTOR EXERTIONS AND TWO EXERCISE SEQUENCES ON VASTUS MEDIALIS OBLIQUE (VMO) AND VASTUS LATERALIS (VL) EMG ACTIVITIES DURING SQUAT EXERCISE)

ศิริรัตน์ แก้ววิชัย 4936645 PTPT/M

วท.ม. (กายภาพบำบัด)

คณะกรรมการที่ปรึกษาวิทยานิพนธ์: วรรธนะ ชลายนเดชะ, Ph.D., รุ่งทิวา วัจฉละฐิติ, Ph.D.

บทคัดย่อ

การศึกษานี้เป็นการศึกษาภาคตัดขวาง เพื่อศึกษาผลของแรงหุบสะโพกและลำดับการหุบสะโพกสองแบบต่อคลื่นไฟฟ้ากล้ามเนื้อ (EMG) ของ vastus medialis oblique (VMO) และ vastus lateralis (VL) ขณะออกกำลังกายท่าย่อเข่า อาสาสมัคร 30 คนเป็นผู้ไม่เคยมีอาการปวดเข่าทางด้านหน้า ได้รับการวัดคลื่นไฟฟ้ากล้ามเนื้อ VMO และ VL ในขาข้างถนัด ขณะย่อเข่าธรรมดาและย่อเข่าร่วมกับหุบสะโพกด้วยแรงร้อยละ 35, 70 และ 100 ตามลำดับการหุบสะโพก 2 แบบ คือ หุบสะโพกตลอดการย่อเข่าและหุบสะโพกขณะย่อเข่า 60 องศา วิเคราะห์คลื่นไฟฟ้ากล้ามเนื้อด้วยเทคนิค root mean square (RMS) และ %MVC

ผลการศึกษาพบว่ามีเพียงแรงหุบสะโพกที่มีผลต่อค่า %MVC ของ VMO, VL และอัตราส่วน VMO:VL (p<0.001) ค่า %MVC ของ VMO ขณะย่อเข่าร่วมกับหุบสะโพกด้วยแรงร้อยละ 70 และ 100 มากกว่าขณะย่อเข่าธรรมดา (38.35-62.13 vs. 35.77 %MVC) ค่า %MVC ของ VL เพิ่มขึ้นเมื่อแรงหุบสะโพกมากขึ้น (36.2-73.60 %MVC) ค่าอัตราส่วน VMO:VL ขณะย่อเข่าธรรมดามีค่ามากกว่าขณะย่อเข่าร่วมกับหุบสะโพกด้วยแรงร้อยละ 35 และ 70 (1.06 vs. 0.76-0.87) แต่ไม่แตกต่างจากขณะย่อเข่าร่วมกับหุบสะโพกด้วยแรงมากที่สุด (0.94-1.00) ขณะย่อเข่าธรรมดาและย่อเข่าร่วมกับหุบสะโพกด้วยแรงมากที่สุดค่าเฉลี่ย %MVC ของ VMO และ VL ไม่แตกต่างกัน

คลื่นไฟฟ้ากล้ามเนื้อ VMO และ VL ขณะหุบสะโพกตลอดการย่อเข่าหรือหุบสะโพกขณะย่อเข่า 60 องศา มีค่าไม่แตกต่างกัน การย่อเข่าร่วมกับหุบสะโพกด้วยแรงร้อยละ 70 และ 100 สามารถเพิ่มการทำงานของกล้ามเนื้อ VMO และ VL ได้ การย่อเข่าธรรมดาและย่อเข่าร่วมกับหุบสะโพกด้วยแรงมากที่สุดทำให้เกิดสมดุลระหว่างคลื่นไฟฟ้ากล้ามเนื้อ VMO และ VL ซึ่งสามารถนำไปฝึกกล้ามเนื้อ VMO ให้ทำงานเพื่อให้สะบ้าเคลื่อนไหวอย่างสมดุลได้

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LIST OF ABBREVIATIONS

VM	=	Vastus medialis
VMO	=	Vastus medialis oblique
VML	=	Vastus medialis longus
VL	=	Vastus lateralis
VLL	=	Vastus lateralis longus
ITB	=	Iliotibial band
LE	=	Lower extremity
ASIS	=	Anterior superior iliac spine
PFPS	=	Patellofemoral pain syndrome
PFJ	=	Patellofemoral joint
TFJ	=	Tibiofemoral joint
Q angle	=	Quadriceps angle
US	=	Ultrasound
TENS	=	Transcutaneous electrical nerve stimulation
EMG	=	Electromyography
RMS	=	Root mean square
%MVC	=	Percentage of maximum voluntary isometric contraction
CKC	=	Closed kinetic chain
OKC	=	Opened kinetic chain
BMI	=	Body mass index
SE-SE	=	Squeeze entire time-squat exercise
SF-SE	=	Squeeze at 60 degrees knee flexion-squat exercise
SE	=	Squat without hip adduction exercise

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CHAPTER I
INTRODUCTION

The vastus medialis oblique (VMO) muscle is only dynamic medial stabilizer which counteracts the lateral force produced by the vastus lateralis (VL) muscle and the iliotibial band in order to maintain normal tracking of the patella (1). Insufficiency of the VMO muscle contraction plays an important role in abnormal patellar tracking which leads to abnormal loading patterns on the patellofemoral joint and surrounding soft tissues. Consequently, the dysfunction, degenerative change within the patellofemoral joint and impaired the soft tissues may be produced (1-4). In principle, training of the VMO muscle is focused on preventing and recovering of the lateral tracking of the patella to eliminate the patellofemoral dysfunction in patellofemoral pain syndrome (PFPS), knee extensor mechanisms injury, ligament of the knee injury and quadriceps muscle weakness (5-11).

Insufficiency of the VMO muscle contraction can be investigated by electromyography (EMG). It can evaluate imbalance activities of the VMO and VL muscles. Imbalance between those muscles commonly is presented in term of imbalance in magnitude and delayed activation of the VMO muscle relative to the VL muscle resulted from an altered motor control pattern of these muscles (10, 12, 13). Ideally, magnitude EMG activity ratio of the VMO and VL muscles should be about 1.0 for the patella to be balanced within the intercondylar groove. However, numerous studies showed that the VMO:VL ratio less than 1.0 in PFPS caused by decreasing activity of the VMO muscle or increasing activity of the VL muscle. This is related to abnormal patellar alignment (2, 14, 15).

Due to fibers of the VMO muscle attach to the adductor muscles, activating these muscles may increase the VMO recruitment (5, 7, 8, 16). To enhance activation

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of the VMO muscle, combination of active knee extension and isometric hip adduction (proposed as a stable origin of the VMO muscle) was suggested (7, 17). Besides, there is a stretch to the VMO muscle which would alter the length tension properties, thus enhancing contraction force (16). The VMO muscle should be trained with closed kinetics chain (CKC) exercises such as squatting to promote motor control in activities which are functional and specific. Due to CKC exercises have been widely indicated for knee rehabilitation. They promote functional stability by producing co-contraction of the lower extremity muscles, normal proprioceptive input and functional movement (14, 15, 18, 19). The VMO muscle contains high proportion of slow fibers, indicating the VMO muscle performs an antigravity role (20). Hodges and Richardson (1993) suggested that weight bearing position such squatting would promote interaction between the adductor muscles and the VMO muscle (8). Increasing VMO activity and balancing activities of the VMO and VL muscles have been existed during squatting (8, 9, 21). Several authors recommended that short arc CKC knee extension exercise such as squat in healthy individual and patellofemoral pain patient should be increased to 60 degrees knee flexion (8, 15, 19, 22). This promotes greater recruitment of the VMO muscle in relation to VL muscle with no pain (19, 22). Hodges and Richardson (1993) suggested the squat additional with hip adduction exercise should be executed with 60 degrees knee flexion (8). Therefore, the squat with isometric hip adduction exercise should be set at 60 degrees knee flexion which provided greater muscle activities.

Some studies suggested that differences in the amount of hip adduction effort may affect the variability in the VMO activity when performing knee extension together with hip adduction exercise (11). Previous studies showed an increase in the VMO activity when performing squat exercise combined with isometric hip adduction, nevertheless, most studies focused on only maximal hip adduction (6, 9, 21, 23). Some studies showed that the squat combined with maximum isometric hip adduction provides high VMO activity (6, 8, 21) and balancing between the VMO and VL activities (8, 9, 21). This strenuous exercise may exacerbate pain and lead to fatigue of hip adductor and vasti muscles. In clinical settings, this exercise was complained as a non comfortable and difficult exercise. Only previous study

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investigated the VMO and VL activities during squat combined with sub-maximum hip adduction but those data were not normalized. Due to anthropomorphic differences between different recording sites; thus, comparisons of EMG activities between muscles should be normalized (24).

The squat with hip adduction exercise is applied by two different sequences. The first one is to maintain hip adduction at entire range of squat. It is proposed to 1) stabilize the origin and 2) stretch the VMO muscle. This can provide greater VMO activity at entire range of exercise (8). Another type of exercise is to perform hip adduction after squatting to the specific knee angle. It is believed that origin of the VMO muscle can be stabilized and stretched at the end of squatting (6). It is easy and comfortable to do thus it has been widely applied. However, no previous study concerned differences between these two exercise sequences which might affect recruitment of the VMO and VL muscles.

Therefore, this study purposed to examine effects of hip adductor exertions and two types of exercise sequence on EMG activity of the VMO and VL muscles during squat exercise.

Purposes of the Study

General objectives

The purposes of this study were to examine the effects of hip adductor exertions and two exercise sequences on EMG activity of the VMO and VL muscles during squatting exercise in healthy individuals.

Specific objectives

1. To examine the effects of hip adductor exertions, 0%, 35%, 70% and 100% hip adductor exertion, on the normalized root mean square (RMS) of the VMO and VL EMG activities.

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2. To examine the effects of two types of exercise sequence, squeeze entire time-squat exercise (SE-SE) and squeeze at 60 degrees knee flexion-squat exercise (SF-SE), on the normalized RMS of the VMO and VL EMG activities.

Parameters of the Study

Independent variables

1. Hip adductor exertions
   - 1.1 0% hip adductor exertion (squat without hip adduction exercise; SE)
   - 1.2 35% hip adductor exertion
   - 1.3 70% hip adductor exertion
   - 1.4 100% hip adductor exertion
2. Two types of exercise sequence
   - 2.1 Maintain hip adduction entire range of squat (squeeze entire time-squat exercise (SE-SE))
   - 2.2 Perform hip adduction at 60 degrees of knee flexion (squeeze at 60 degrees knee flexion-squat exercise (SF-SE))

Dependent variables

1. Normalized RMS (%MVC) of the VMO muscle
2. Normalized RMS (%MVC) of the VL muscle
3. Normalized RMS (%MVC) of the VMO:VL ratio

Scope of the Study

This cross sectional study focused on healthy young male and female subjects. EMG of VMO, VL and VMO:VL ratio in the dominant leg were analyzed during squatting with 0%, 35%, 70% and 100% of hip adductor exertion. Differences

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in EMG of VMO, VL and VMO:VL ratio between two exercise sequences were also studied.

Hypotheses of the Study

1. There was significant effects of hip adductor exertions on the normalized RMS (%MVC) of the VMO, VL and VMO:VL ratio.
2. There was significant differences between two types of exercise sequence (SE-SE and SF-SE) on the normalized RMS (%MVC) of the VMO, VL and VMO:VL ratio.

Advantages of the Study

1. To recommend percentage of hip adductor exertions during squat that enhance activity of the VMO muscle and promote balance activity of the VMO and VL muscles. It may benefit to retrain the VMO muscle for recover the abnormal patellar tracking.
2. To recommend the exercise sequence of squat associated with hip adduction exercise that enhance activity of the VMO muscle and promote balance activity of the VMO and VL muscles. It may benefit to retrain the VMO muscle for recover the abnormal patellar tracking.
3. To recommend a new use for elastic ball with pressure biofeedback for squat with hip adduction exercise or isolated hip adduction exercise.

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CHAPTER II
LITERATURE REVIEW

2.1 Anatomy of the Structure in the Patellofemoral Area

2.1.1 Patellofemoral joint (PFJ)

The knee joint complex is formed by the tibiofemoral joint (TFJ) and the PFJ. Both joints are localized in the same joint capsule, known as the knee joint capsule (figure 2.1). The PFJ is arisen from the articulation between the posterior surface of the patella and the trochlea (intercondylar) groove of the femur, anterior aspect of the distal femur (figure 2.1). The major movement of the PFJ during motions of the knee flexion and extension is non-axial movement, superior-inferior gilding movement of the patella along the trochlea groove (25-27).

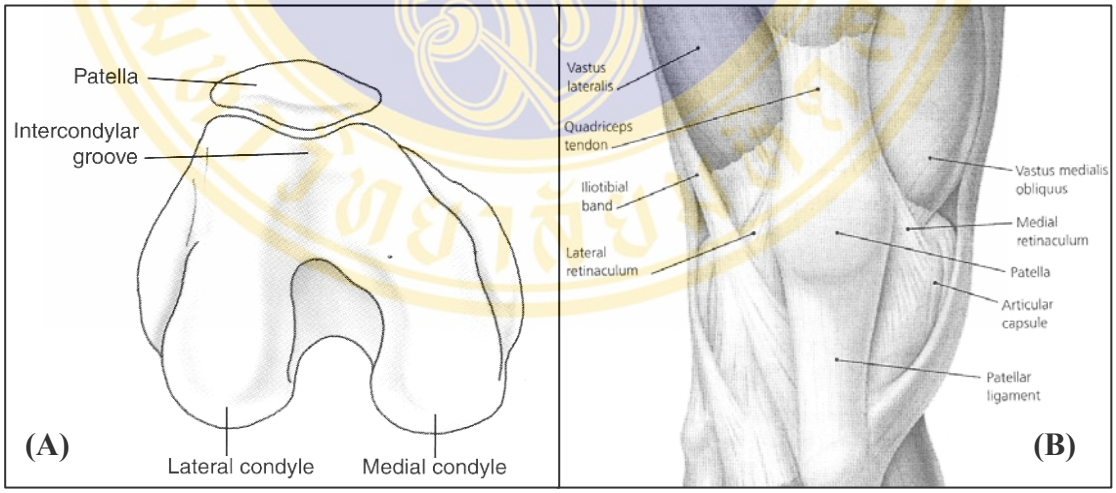


Figure 2.1 (A) The patellofemoral joint of right knee. (B) The anterior view of the right knee.

The patella is the largest sesamoid bone in the human body which completely embeds in the patellar tendon (figure 2.1). Its shape is triangular that apex

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point distally. Its articular surface is divided into seven facets, which exists as proximal, middle, and distal horizontal pairs on the medial and lateral sides of posterior surface, and medial segment of medial facet known as the odd facet (figure 2.2). The articular surface of the patella is covered by the thickest hyaline cartilage of any joint in the body, which to serve the highly compressive force of the patella during knee joint movement. The functions of the patella include two major functions (see 2.2) (27).

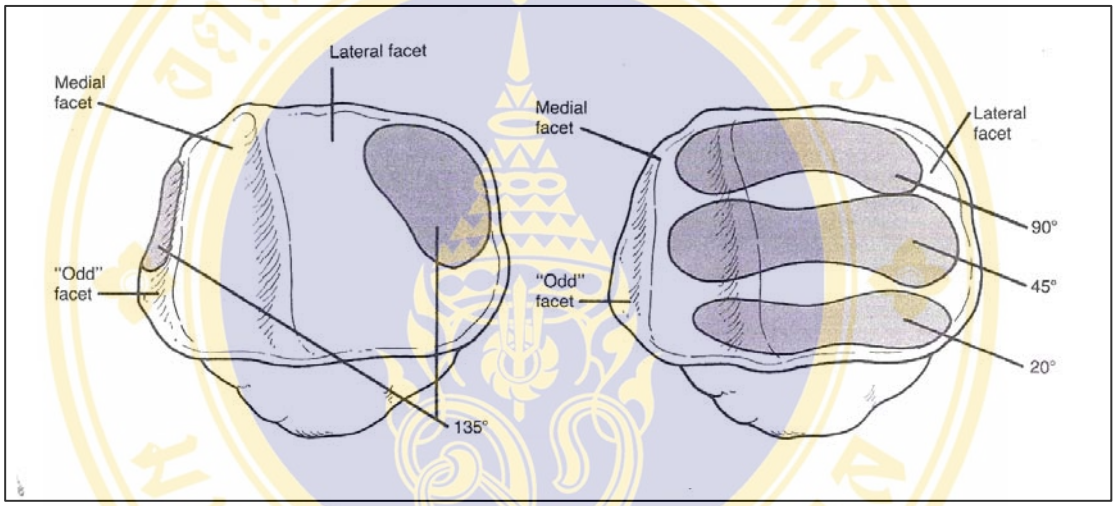


Figure 2.2 Articular surface of the patella.

2.1.2 The muscular structure

The quadriceps muscle influences the function of PFJ. It is consisted from four major parts; rectus femoris, vastus medialis, vastus lateralis and vastus intermedius. It locates on anterior aspect of the shaft of the femur, and inserts to the patella via the quadriceps tendon. Its innervation is the branches of the femoral nerve, which is formed by the second, third and fourth lumbar spinal nerves. It acts as a prime mover for knee extension. The rectus femoris also act as prime mover for hip flexion and secondary mover for hip abduction. On the other hand, the quadriceps muscle act as active stabilizer of the patella, especially the vastus medialis and vastus lateralis muscles (26, 27).

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2.1.2.1 Vastus medialis muscle

The vastus medialis originates from the entire length of posteromedial aspect of the shaft of the femur to the lower half of intertrochanteric line, the medial lip of the linea aspera and the tendon of the adductor magnus, which inserts into medial border of the patella through the quadriceps tendon (26). It is innervated by the deep branches of the femoral nerve that supply at middle portion of the muscle after it passes through the adductor tunnel (24). The primary function is a knee extension and also acts the secondary function as the stabilization of the patella (26). The vastus medialis is divided into two components, the vastus medialis oblique (VMO) and vastus medialis longus (VML) (figure 2.3). Several authors reported that this muscle have fascia plane dividing it into two portions, separately nerve innervation and different fiber orientation (28, 29). However other studies suggested that distinguishing of these two muscles is difficult (30). Also they act as separately specific functions. The VML extend knee from resting, whereas, the VMO active throughout knee extension to maintain the normal patellar tracking (31).

2.1.2.2 Vastus medialis oblique (VMO) muscle

The VMO arises from the tendon of adductor magnus and directly inserts into the medial border of the patella. Its fibers orientate medially in 55 degrees relative to the femoral line. Some authors indicated that it is innervated separately by branches of the sapheneous nerve (26, 27). Its primary function is acting as medial stabilization of the patella by pulling the patella medially. The medial force from the VMO counteracts with lateral force from the vastus lateralis (VL) to ensure the normal patellar tracking during knee extension (31-33).

2.1.2.3 Vastus lateralis (VL) muscle

The VL originates from the lateral surface of the femur just below the greater trochanter and from the upper half of the linea aspera. It inserts via quadriceps tendon into lateral and superior border of the patella. Its fibers orientate laterally by 15 degrees against the femoral line. Its primary action is knee extension. Also it creates lateral force to counteract with medial force from VMO, consequently serve normal patellar tracking during knee extension. As the others head of the

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quadriceps, it is innervated by branches of the femoral nerve, which come from the second and third lumbar nerve roots (26, 27).

2.1.3 The quadriceps and patellar tendon

Commonly, the "quadriceps tendon" is a tendon located between the quadriceps muscle and the superior pole of the patella (figure 2.1). Whereas, the tendon is positioned between inferior pole of the patella and the tibial tuberosity refer to the "patellar tendon or ligament" (figure 2.1). These tendons expand to form the medial and lateral retinaculum, and merge with knee joint capsule (31). Although the patella is embedded within these tendons but does not separate them into two structure. Furthermore, the connective tissues of the quadriceps and patellar tendons are continuous and that consist of the same structure (27).

2.1.4 The patellar retinaculum

The medial and lateral retinaculums are the expansion of the quadriceps muscle. The patellar retinaculum, shape of the patella and the femoral trochlea provide the passive stabilization of PFJ. The lateral retinaculum merges with the fibers of the lateral collateral ligament and the iliotibial band to form the lateral part of the knee joint capsule (figure 2.1). Also the medial retinaculum joins with the medial collateral ligament and pes anserine tendon to form the medial part of the knee joint capsule (figure 2.1) (25, 27, 31).

The lateral retinaculum is divided into superficial and deep layer. The superficial layer arises from the fibrous expansion of the anterior portion of the VL and the fibers from the oblique portion of the iliotibial band, which merges with patellar tendon. Due to the lateral retinaculum mostly consists from the iliotibial band, the excessive lateral tracking of the patella are occurred by iliotibial band tightness (31).

The medial retinaculum has less effects to the patellar position, due to its fibers is thinner than the lateral retinaculum. The primary restrainer of the lateral patellar translation is provided by the medial patellofemoral and medial

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patellomeniscal ligaments. Theses lie beneath the medial retinaculum. However, several authors suggested that the medial-lateral retinaculum is not functionally useful (25, 27, 31).

2.1.5 The patellar fat pad and bursa

The several fat pad and bursas related to the PFJ provide joint lubrication and shock absorption. These include the infrapatellar fat pad, prepatellar bursa, supra patellar bursa, superficial/subcutaneous infrapatellar bursa and deep infrapatellar bursa (figure 2.3). The infrapatellar fat pad is encapsulated collection of adipose tissue; locate over the anterior aspect of the intercondylar fossa of the tibia and deep to the patellar tendon. It separates the deep infrapatellar bursa from the synovial membrane of the knee joint capsule and cavity. The prepatellar bursa positions between the skin and anterior surface of the patella that serves free movement of the skin over the patella during knee joint motion. The suprapatellar bursa locates between the knee joint capsule and the patella that lies beneath the quadriceps tendon. The superficial infrapatellar bursa exists between skin and the patellar tendon. The deep infrapatellar bursa lies between the patellar tendon and the anterior aspect of the tibial tuberosity (27).

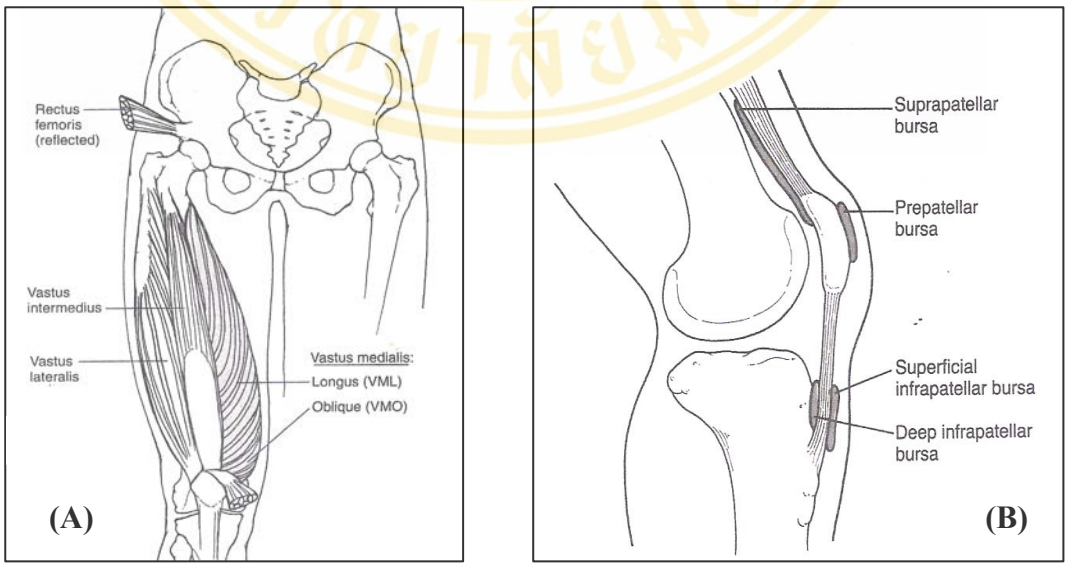


Figure 2.3 (A) The VMO and VML of the vastus medialis muscle. (B) Location of the bursas (anterior aspect of the knee).

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2.2 Biomechanics of the Patellofemoral Joint

2.2.1 Patellofemoral motion

Motion of the patella during knee flexion-extension consists of both translation and rotation. Translation of the patella includes two directions; proximal-distal and medial-lateral. When flexed knee from full extension, the patella slightly medial translate at the beginning. The patella starts laterally translation at 30 degrees knee flexion, and continues to increase as knee flexed. If the knee is flexed greater than 45 degrees, it will be plateau. However from full extension to 140 degrees of flexion, both medial and lateral facets of the femur articulate with the patella (31). Rotation of the patella contains three directions which are medial-lateral tilt (around supero-inferior axis), flexion-extension (around medio-lateral axis) and medial-lateral rotation (around antero-posterior axis) (Figure 2.4). The patella appears to be in slight lateral tilt that increases a little as knee flexes (26). Hehne (1990) indicated that the patella rotate externally and only the medial femoral facet articulate with the patella when knee is flexed greater than 90 degrees (34).

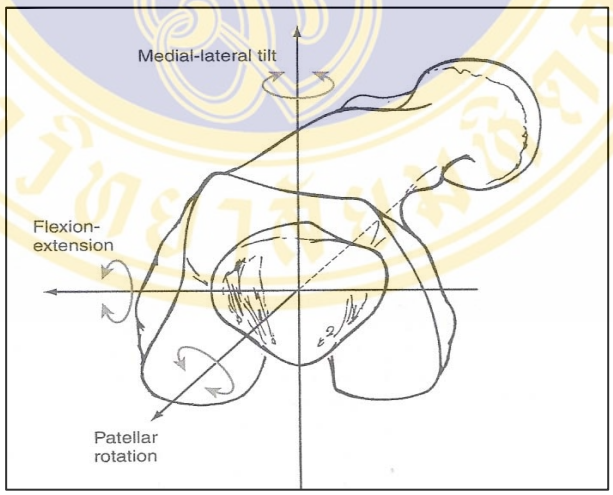


Figure 2.4 Three-dimensional motions of the patella and the cardinal axis.

2.2.2 Function of the patella

The patella increases contact area between the patella tendon and femur that providing wider distribution of compressive stress on the femur. Also it increases lever arm of the quadriceps, which can create high forces. At full flexion, the patella is in the intercondylar groove thus it provides the least length of quadriceps muscle

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force lever arm, approximately 10% of the total length. As knee is extended into 45 degrees, the lever arm of quadriceps is increased rapidly due to the patella rises from the intercondylar groove. The lever arm is lengthened as approximately 30% at knee extension 45 degrees. When knee extension is beyond 45 degrees, the lever arm of quadriceps decreases slightly; therefore, the quadriceps muscle force must be risen for remaining the torque of the knee. The quadriceps muscle required higher forces to extend the knee at last 15 degrees approximately 60% of total capacity. If the patella is removed (patellectomy), the lever arm of the quadriceps muscle can be reduced as 33% to almost 70%. The quadriceps muscle must produce force as much as 30% more than normally for extend knee that may be further than the capacity of the muscle in some patients (25, 35).

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2.2.3 Contact area of the patella

Movement of the patella leads to change in location and total area of contact between the patella and the femur that alter stress applied to the joint surface. When the knee is in extension or slight flexion, the inferior pole of the patella will contact with the femur. The area of contact on the patella steadily increases and moves proximally on the patella as the knee is flexed to approximately 90 degrees (figure 2.5) (26, 35). Goodfellow et al (1976) indicated that the increasing contact area compensates for the larger PFJ reaction force when knee flexion increase (3). During later flexion, only the medial femoral condyle contacts with the odds facet of the patella. The abnormal patellar movement during knee flexion-extension leads to abnormal stress and loading pattern on the patella which may contribute to degenerative change within the PFJ and other patellofemoral disorders. Also it provides abnormal load to the surrounding soft tissue that causes impairments and damages (35).

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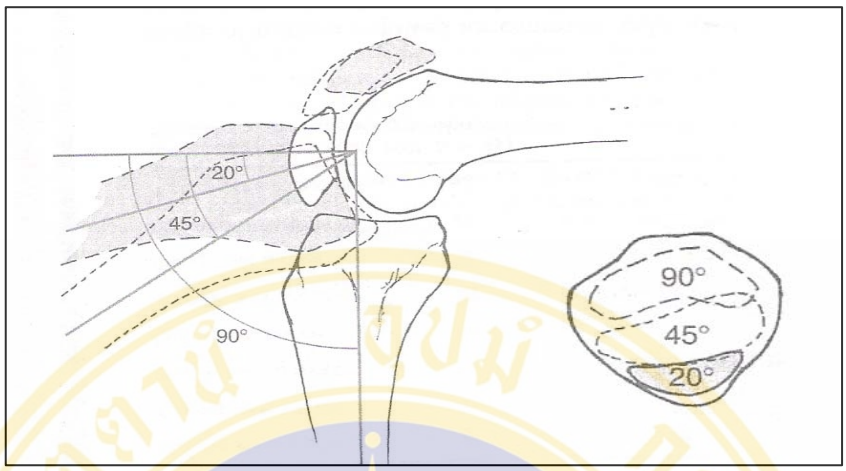


Figure 2.5 The area of contact on the patella during knee flexion from 20 to 90 degrees.

2.2.4 Statics and dynamics of the patellofemoral joint

During dynamic activities the PFJ reaction force is directly influenced by magnitude of muscle force. In general, greater muscle force provides higher joint reaction force. In the PFJ, the quadriceps muscle force increases with knee flexion. Due to during upright standing with knee flexion increases, the center of gravity will shift further away from the center of the rotation of the PFJ. Therefore, the quadriceps muscle force must be required for counterbalance the increasing flexion moments. When quadriceps muscle forces rise, PFJ reaction force will be greater (35).

Furthermore, knee flexion affects the PFJ reaction force by influencing the angle between the quadriceps tendon force and the patellar tendon force. As a result, the angle between these two tendons increases with greater the tibiofemoral flexion. Therefore the resultant of these two forces, magnitude of the PFJ reaction force, is larger when knee flexion increases (figure 2.6) (35).

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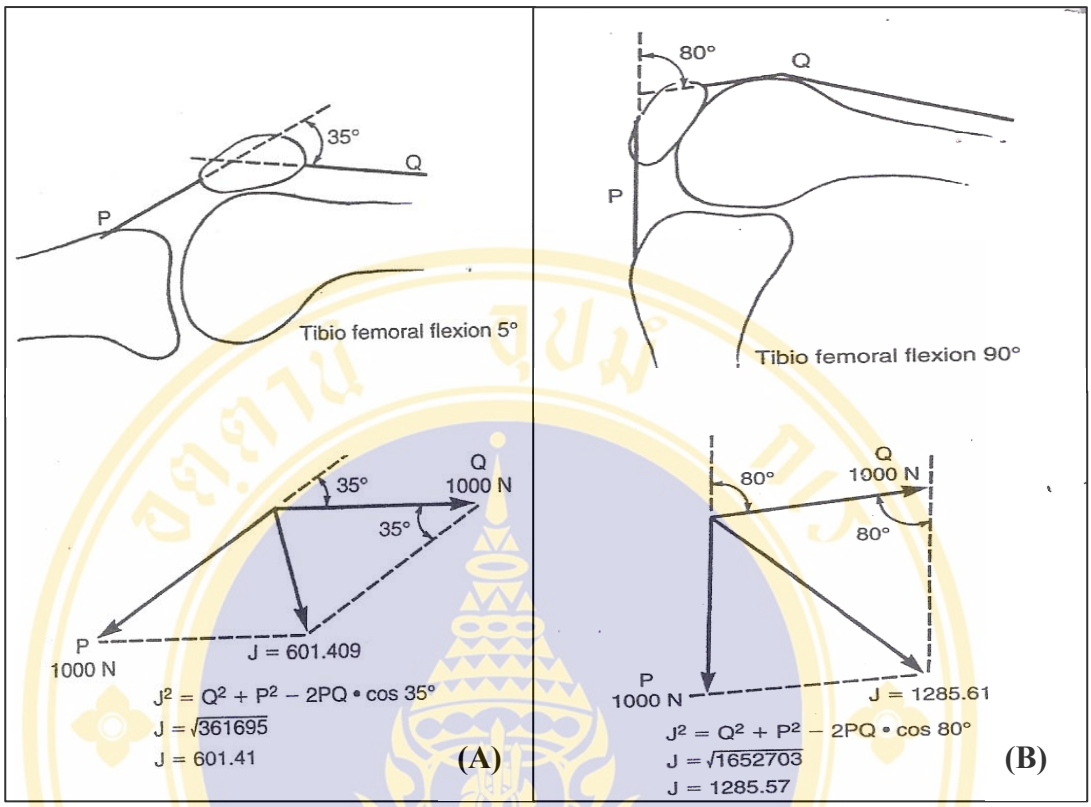


Figure 2.6 Calculation of the patellofemoral joint reaction force during tibiofemoral flexion at 5 degrees (A) and 90 degrees (B): *(P)* force of the patella tendon, *(Q)* force of the quadriceps tendon, *(J)* the patellofemoral joint reaction force.

In other word, PFJ joint reaction force is greater during activities that require larger knee flexion. This force is reached 2.5 to 3 times of body weight with the knee flexed 90 degrees (figure 2.7). Several dynamics activities involve varying amount of knee flexion (table 2.1) that provides large changing of the reaction force acting on the PFJ. During walking, the peak value of knee flexion occurs in middle of stance phase, which the PFJ reaction force is created as one-half body weight. Stair climbing requires knee flexion up to 60 degrees that may provide this reaction force as 3.3 times body weight (35). During squatting this force may up to 7 times of body weight (32).

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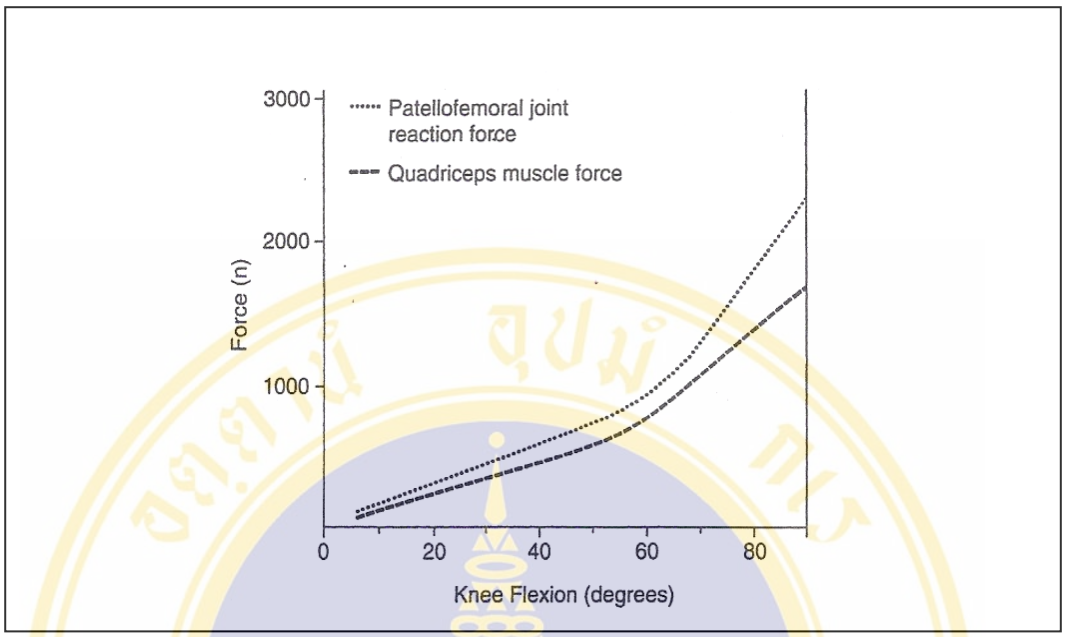


Figure 2.7 Patellofemoral joint reaction force and quadriceps muscle force during knee flexes to 90 degrees.

Due to greater magnitude of quadriceps muscle force and PFJ reaction force during activities requiring a large amount of knee flexion, therefore the patients with patellofemoral disorders have experience of pain when perform these activities (Table 2.1) (35).

Table 2.1 Range of knee joint motion during common activities

Activities	**Range of motion from knee extension to knee flexion (degrees)**
Walking	0-60
Ascending stairs	0-83
Descending stairs	0-90
Sitting down	0-93
Tying a shoe	0-106
Lifting an object	0-117

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2.3 Patellar Tracking

Patellar tracking refers to the patella articulation with the femur and its motion during knee flexion-extension. Ideally, the patella must perfectly track in the middle of the intercondylar groove thus medial and lateral forces on the patella should be equal (25, 26). However, normal patellar tracking has tendency to lateral deviate slightly due to its anatomical configuration (31). The force acting on the patella is directed too laterally as illustrated in Figure 2.8 (26).

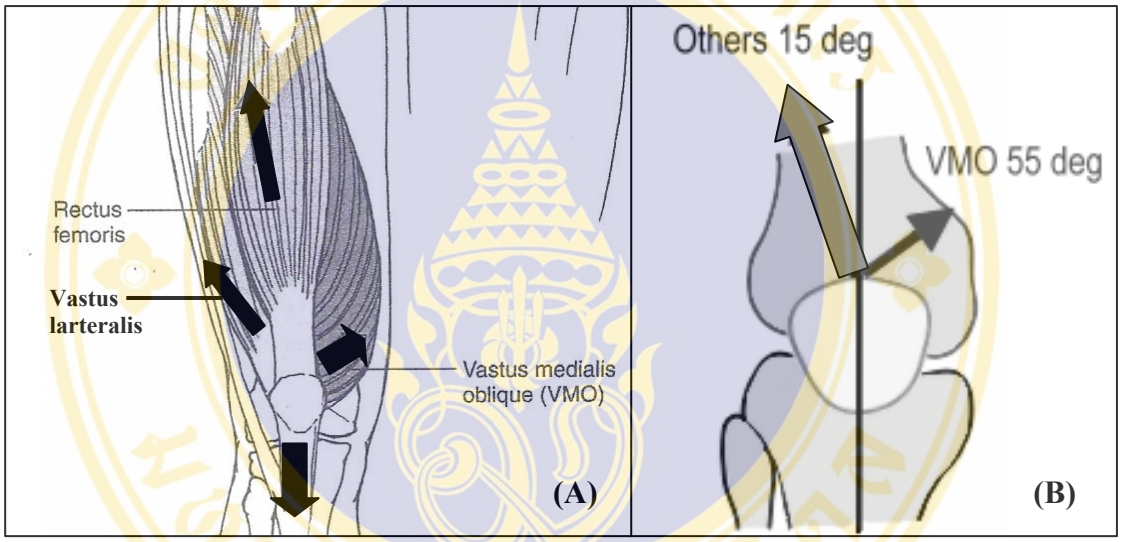


Figure 2.8 (A) Dynamics stabilizer of the patellofemoral joint. (B) The medial and lateral pulling acting on the patella.

Patellar tracking is controlled by dynamic and static stabilizer of the PFJ. Dynamic stability of the PFJ is provided by the quadriceps tendon, patellar tendon, VMO, VL, and iliotibial band (figure 2.8). The VMO is the only muscle that provide medial pulling on the patella, thus it is an important dynamic stabilizer of the PFJ. Static stability of the PFJ is provided by the patellar shape, femoral trochlea, knee joint capsule, medial and lateral patellar retinaculum, and patellofemoral ligament (25, 26, 32).

The rectus femoris and vastus intermedius are centrally positioned on the femur and drag the patella along the long axis of the femur. This moves the patella deviate laterally from the tibia. The VL draws the patella in direction of slightly

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lateral with respect to the femur. Also the patella is pulled downward by the patellar tendon. Therefore, the addition of these forces should direct laterally on the patella, which is known as quadriceps pulling (figure 2.8). Lateral pulling of the quadriceps muscle force can be estimated by Q angle, which indicates as angle between a line drawn from the anterior superior iliac spine (ASIS) to center of the patella and other line is drawn from center of the patella to center of the tibial tuberosity (26).

However, lateral movement of the patella must be protected by the following major systems. First; expansion surface on the lateral condyle of the femur that serves as a buttress against lateral displacement of the patella. The second is the medial retinaculum. Finally, the vastus medialis muscle, particularly the fibers of the VMO, which is the only dynamic medial stabilizer of the patella (26).

Balance activity between the VMO and VL muscles is considered desirable to maintain normal patellofemoral track, since forces of these muscles have an effect on the patella. Estimation based on physiological cross-sectional area, the VL may contribute 40% of quadriceps muscle and the vastus medialis may assist as approximately 20-35%. The VMO is an only one part of the vastus medialis. Although cross-sectional area of the VMO is smaller than the VL, nevertheless, the fibers of VMO oriented more obliquely than VL (figure 2.8). Therefore, some authors suggested that the VMO and VL will counteract in order to provide normal patellar tracking (26).

Due to patellar tracking is controlled by the static and dynamic stabilizer; imbalances in forces of these stabilizers can alter patellar tracking. Abnormal patellar tracking may present as lateral patellar tracking (excessive deviation laterally) more frequently, or medial tracking (excessive deviation medially) (26). However, both lead to abnormal distribution of forces along the patellofemoral articular surface, quadriceps tendon, patellar tendon and adjacent soft tissues (26, 31). In other word, the patellar tracking is organized by three major systems. There are the osseoligamentous structures, muscles and neuromuscular control (36). Grabiner et al (1994) suggested that these three systems are interdependent components of the

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patellar tracking, thus the compensation of one structure must arise because of deficit in the others. Consequently, patellar tracking may be altered, which affects the magnitude and distribution of the forces acting on the PFJ (4). Patellar tracking problems frequently influence by tightness or hyperactivity of lateral structures (VL, iliotibial band, lateral retinaculum) or weakness of the medial muscles (vastus medialis and VMO) or the lengthening of medial soft tissue structure (medial retinaculum) (24, 37).

Assessment of patellar tracking can be evaluated by observation in motion path of the patella during knee flexion-extension. Abnormal in term of lateral patellar tracking indicate as “J” sign; sudden medial deviation of the patella in early flexion when subject perform a single leg squat. Otherwise, if the knee is extended from flexion to full extension in supine or seated position, the patella will present laterally deviation (“J” sign) at the terminal phase of extension (figure 2.9) (32).

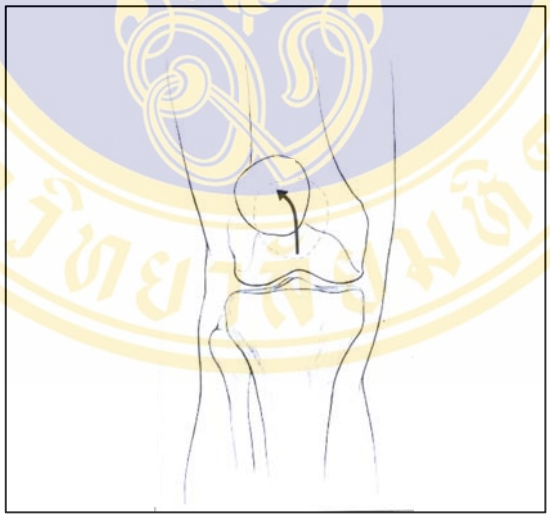


Figure 2.9 Lateral patellar tracking (“J sign”) during knee extension from 90 degrees flexion (broken line) to full extension (unbroken line).

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2.4 Patellofemoral Pain Syndrome (PFPS)

2.4.1 Definition of patellofemoral pain syndrome (PFPS)

The term “patellofemoral pain syndrome (PFPS)” defines as clinical presentation of anterior knee pain. This refers to pain within the anterior aspect of the knee or retro-patellar area, excepts anterior knee pain due to intra-articular pathologies, patellar tendinopathy, meniscal injury, peripatellar tendinitis or bursitis, patellar fat pad inflammation, plica syndromes, Sinding Lasen Johanson’s disease, Osgood Schlatter disease, Hoffas disease, neuromas, osteochondritis dissecans of the femoral condyles, inflammatory or degenerative arthritis, tumor and other rare pathologies (28, 38, 39).

PFPS have been known as chondromalacia patella, runner’s knee, dashboard knee, patellofemoral arthralgia, patellar pain, patellar pain syndrome, patellofemoral pain and patellofemoral joint pain (38, 40). However, the term “chondromalacia patella” is appropriate to diagnose pain which originates from pathological changes in the retropatellar cartilage. This is only the subset of PFPS (41). The word “patellofemoral” and “syndrome” seems appropriate to describe the structure affecting and a combination of signs and symptoms (42).

2.4.2 Incidence and prevalence of PFPS

PFPS is the most common source of knee complaints. The incidence in general population is reported to be as high as 25% (13, 43, 44). It increased in athletes and more common in female. It often affects people between 10 and 35 years of age (45). Eleven percent of musculoskeletal complaints in the office setting are caused by anterior knee pain which most commonly results from PFPS (46). Fairbank et al (1984) reported that 10% of young boys and girls (by age twenty) complain about pain in the patellofemoral joint, and almost half of them have stopped being physically active (47). Patients with clinical presentation of PFPS might report the reduction in their activity level (48). Some affected their working life (49). The study in Asian population by Guo et al (1995) which surveyed for chondromalacia patellae in 2743 Chineses reported that the prevalence rate of PFPS was 36.2% and even higher in women group (50).

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2.4.3 Characteristics of PFPS

The most common symptom in patients with PFPS is pain that is usually described as dull, aching or sharp pain. The area of pain is frequently localized at area of retropatellar, peripatellar or lateral side of patellar; however, it can be medial or inferior to patellar (32, 38, 39, 51). According to sources of pain in patients with PFPS can not be direct from patellofemoral articular cartilage, which is non-innervated tissue. The source of pain may describe as lateral patellar retinaculum, synovium and subchondral bone. The lateral patellar retinaculum is highly potent source of pain, its rich of innervation. Some researches reported the evidence of injury to small nerve fiber in the lateral retinaculum including neuroma formation, nerve fribosis and increasing of vascularity. This makes tenderness at the lateral retinaculum in patients with PFPS. However, patellar articular cartilage may be indirectly source of pain. It has a lot of nerve innervation. Its injuries can lead to the chemical and mechanical irritation of synovial, which. Also its damaging may cause the increase of intraosseous pressure of subchondral bone and venous obstruction. This will lead to pain (31, 39, 40).

Furthermore, common symptoms of PFPS include crepitus, giving way, stiffness, catching or popping sensation and swelling. Symptom of crepitus is not always present. It can be present without any symptoms. The symptom of knee giving way may result from sudden quadriceps inhibition due to pain or deconditioning, or ligamentous instability related to patellar dislocation or subluxation. However, it should be distinguished from of menical lesion and ligamentous injuries that should be evaluated first. Stiffness sensation usually complains when knee is flexed for long period of time in PFPS. Catching or popping sensation may be described but it is not resulting from intra-articular pathology (32, 38-40, 51).

The symptoms of patients with PFPS are usually insidious onset and intermittent period, especially the symptom of pain. These symptoms are easily aggravated by overuse such as increasing physical activity or sport activity. Also those symptoms may be exacerbated during prolong sitting with knee flexed (sometimes called “movie goers knee” or “theater sign”) that describe as an increasing

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PFJ compression force when knee flexion. The symptoms are often triggered by conditions that provide high patellofemoral load such as ascending or descending stair, up or down hill, climbing, running or squatting (32, 38-40, 51).

2.4.4 Etiology of PFPS

In fact, cause of PFPS has been indicated as unknown. It is known that the PFPS has multifactor causing, which result of various pathophysiological processes. Three major contributing factors could be concluded; (1) malalignment of the lower extremities and/or the patella, (2) muscular dysfunctions, and (3) overuse/overload (figure 2.10) (32, 33, 40, 42, 51).

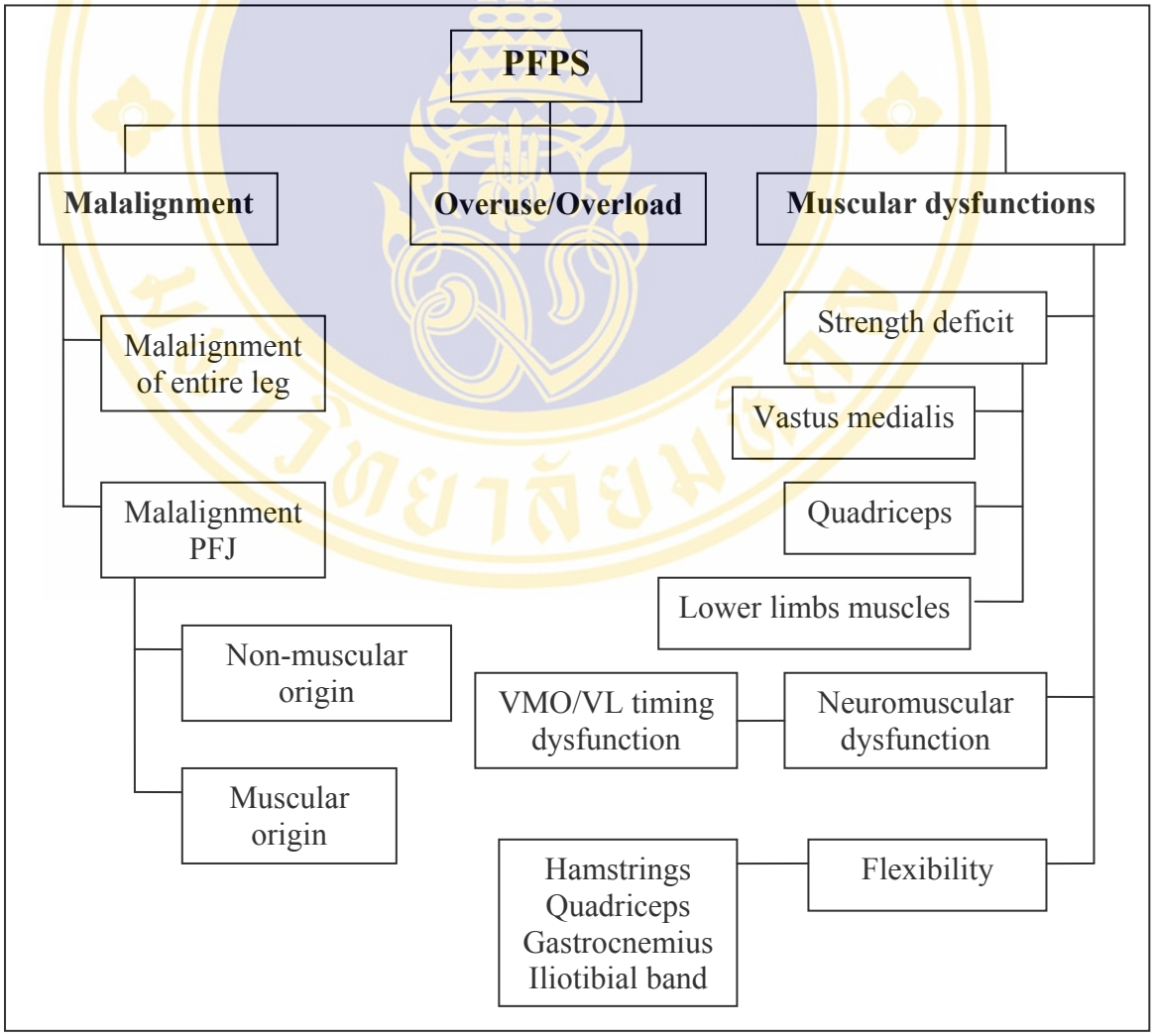


Figure 2.10 Etiology of the PFPS. *Adapted from Witvrouw et al (2005).*

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2.4.5 Treatment of PFPS

2.4.5.1 Relative rest

The first step for reducing pain associated with PFPS should be decrease knee activities that diminish load on the PFJ and the surrounding soft tissues. Ice and others type of cold application should be added for relieving the symptoms.

2.4.5.2 Non-steroidal anti-inflammatory drugs (NSIADs)

NSIADs have been prescribed at the initiation of the treatment of pain during daily activities in case of the symptoms can not be controlled with cold applications. Although NSIADs are commonly suggested for patients with PFPS, but the evidence support their effectiveness is scare (52). Side effects of NSAIDs may be greater than adverse effects of ice application or rehabilitative exercises (33).

2.4.5.3 Physical therapy

Physical therapy management of PFPS involves analysis of the problems in order to identify the contributory factors and further correcting these problems. Primary physical therapy interventions should focus on reducing pain by anti-inflammatory therapy such as relative rest suggestion, cold applications, ultrasound (US), transcutaneous electrical nerve stimulation (TENS), and pain reduction exercises. Second, abnormal biomechanics of the patellar tracking system should be restored by active interventions i.e. quadriceps strengthening, VMO retraining and additional exercises. Also passive interventions should be applied such as stretching, mobilization, taping, foot orthoses, knee sleeves or braces (31-33, 40, 42).

According to the treatment plan for PFPS, McConnell based program was discussed generally. The program addresses each component that contributes to patellofemoral pain. Lower limb mechanics are assessed and improved. Dynamic control of the PFJ is optimized by specific muscles training, particularly the VMO. Taping is used to correct the patellar alignment contributes to decrease pain,

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increase EMG activity of the VMO and adjust quadriceps torque development (18, 51).

Although several authors evaluated efficacy of physical therapy interventions, such as VMO retraining (53, 54), standard open kinetic chain exercise (14, 53), and other trials including the McConnell based program (55, 56). The evidence indicated that physical therapy can reduce pain in PFPS patients. There remains inconclusive about the intervention effectiveness (14, 53-56). However, Cowan et al (2001) and Crossly et al (2001) studied a randomized, double blind, placebo controlled trial of a McConnell-based physical therapy that included six treatments for one week in 71 PFPS patients (37, 57). The results showed that a McConnell-based physical therapy program significantly improves pain, functions and EMG onset of VMO relative to VL when compared with placebo treatment.

2.4.5.4 Surgery

Surgery has been described for the treatment of patients with PFPS when rehabilitative management has failed. Aims of surgery are correcting malalignment, abnormalities of the knee extensor mechanism or injured cartilage (33, 40).

2.5 Patellofemoral Joint Malalignment

2.5.1 Clinical assessments

Malalignment the PFJ is an important manifestation of its abnormal mechanics which leads to PFJ disorder. The PFJ alignments can be assessed by observation and palpation. This is easy and convenient to use in clinical setting. McConnell in 1986 and 1996 suggested that the assessment of the PFJ should perform in four components; glide, medio-lateral tilt, antero-posterior tilt and rotation (18, 51). These components should assess both static (when quadriceps muscle is relaxed) and dynamic alignment (when quadriceps muscle is contracted).

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2.5.1.1 Glide component

Optimal static alignment refers to midpole of the patella is middle (+/- 5mm.) to the medial and lateral femoral epicondyles. Abnormal static alignment is lateral patellar glide. This is indicated as midpole of the patella positions closer to the lateral femoral epicondyles (31). Static alignment can be assessed by the patellar mobility test. This described displacement of the patella in relation to its width on 4 quadrants (figure 2.11). Tightness of the lateral structure will be indicated in case of less than one quadrant medial displacement. Patella hypermobility means more than three quadrants displacement (32, 58). Abnormal dynamic alignment refers to the patella moves laterally when the quadriceps is contracted. This may result in the lateral structures tightness. This provide decreasing the VMO tension (31).

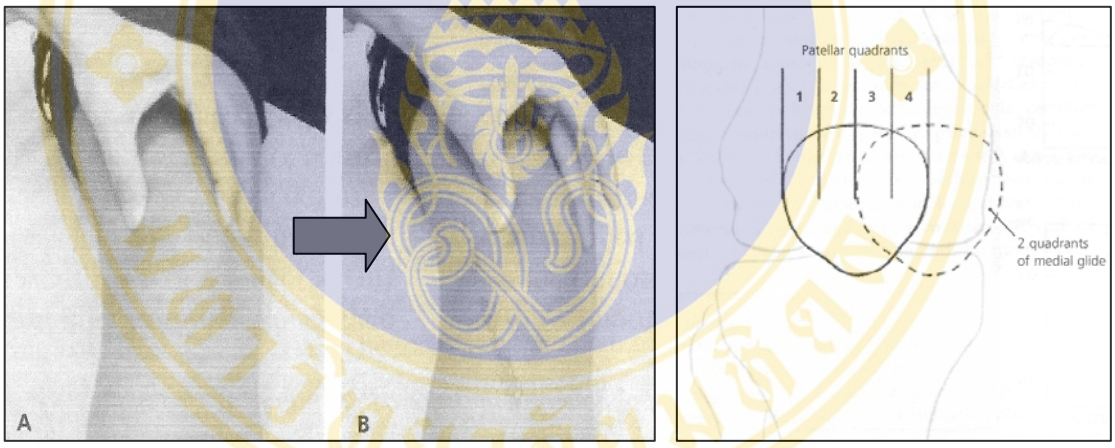


Figure 2.11 Patellar mobility test; medial glide testing of the right knee.

2.5.1.2 Medio-lateral tilt

Optimal static alignment refers to equal height of medial and lateral borders of the patella and posterior edge of both borders can be palpated. Abnormal static alignment frequently shows lateral tilt. This refers to medial border is higher than lateral border and posterior edge of both borders is difficult to be palpated (31). In addition, static alignment of this component can be assessed by the patellar tilt test; the examiner grasps the patella with the thumb and the index fingers (figure 2.12). If the medial aspect of the patella cannot be raised to at least the horizontal position (0 degree), the test is positive. This indicates tightness of lateral structures (32). Abnormal dynamic alignment may show high lateral tilt, this results from medial

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patella displacement. Finding lateral patellar tilt indicates tightness of lateral retinacular structures (31).

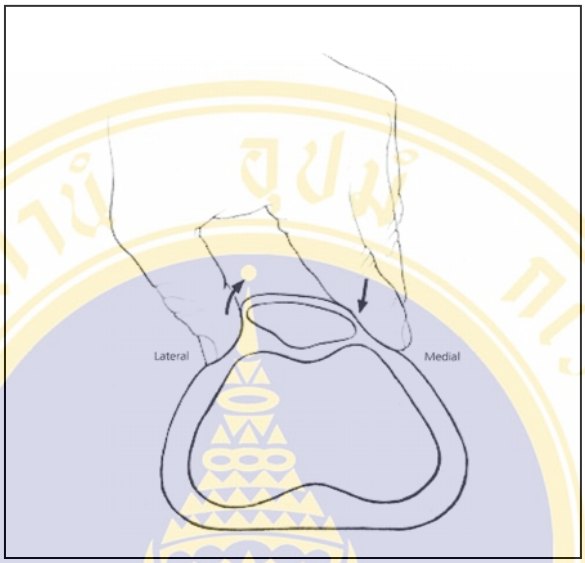


Figure 2.12 Patellar tilt test.

2.5.1.3 Antero-posterior tilt

Optimal static alignment refers to equal height of superior and inferior borders of the patella. Abnormal static alignment usually shows posterior tilt; inferior pole of the patella embeds in the infrapatellar fat pad. Abnormal dynamic alignment usually presents an increasing posterior tilt, especially with knee hyperextension. Posterior tilt of the patella may causes fat pad irritation and lead to inferior patella pain. This is usually triggered by knee extension (31).

2.5.1.4 Rotation component

Optimal static alignment refers to the long axis of the femur and that of the patella are parallel. Abnormal static alignment frequently shows external rotation; inferior pole of the patella laterally aways from the long axis of the femur. This abnormal indicates tightness of retinacular structures (31).

2.5.2 Radiographic assessments

Assessment of the PFJ alignment may be measured by magnetic resonance imaging (MRI) and computer tomography (CT) techniques. These yield

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clinically useful information. The investigation of PFJ alignment should include proximal-distal position of the patella, sulcus angle, congruence angle (CA) and patellar tilt angle (PTA) (26).

2.5.2.1 Proximal-distal position

Proximal-distal position of the patella is described by a ratio of distance between the patella and the tibia to height of the patella (figure 2.13). Increasing in the ratio (increased distance between the tibia and the patella) indicates patellar alta. Decreasing in ratio refers to patellar baja (infera) (26). Both of them lead to abnormal loading on the PFJ, which associate with anterior knee pain (59). The patellar alta may associate with recurrent subluxation of the patella (59).

2.5.2.2 Sulcus angle

The angle is formed by lines drown from the deepest point of the femoral sulcus to the highest point on each condyle (figure 2.13). Thus depth of the sulcus varies over the femoral surface. Increasing in the sulcus angle indicates that the sulcus is relatively shallow. This associates with a history of recurrent PFJ subluxation (26).

2.5.2.3 Congruence angle (CA)

The CA is a measurement of how well the patella fits into the intercondylar groove of the femur. This angle is formed by a line bisecting the sulcus angle and another line that projects from the apex of the sulcus angle through the lowest point on the patellar ridge (figure 2.13) (26). This angle may evaluate during relaxation and/or contraction of the quadriceps. Therefore, the congruence angle difference (CAD) may be more suitable for indicating translation of the patella than CA. The CAD refers to CA during contracted compares with uncontracted quadriceps (CA during contraction – CA during relaxation). Positive value of the CAD indicates the increase in lateral translation (lateralization) of the patella. Conversely, negative value indicates medialization. The association between an increasing in the CA and anterior knee pain and patellofemoral disorder was found (17, 42).

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2.5.2.4 Patellar tilt angle (PTA)

The PTA is formed by a line touching the anterior surfaces of the medial and lateral femoral condyles and a line drawn through the largest width of the patella (figure 2.13) (26). The patellar tilt angle difference (PTAD) should be evaluated. This compares PTA during quadriceps contraction and relaxation. The PTAD represents patellar rotation over supero-inferior axis when quadriceps contraction. Positive value or increasing in PTAD indicates rising of patellar lateral tilt (17). However, the patella has tendency in slight lateral tilt position in normal condition (26). This is suggested that lateral patellar tilt and lateral translation associated with patellofemoral disorder and anterior knee pain (17, 42).

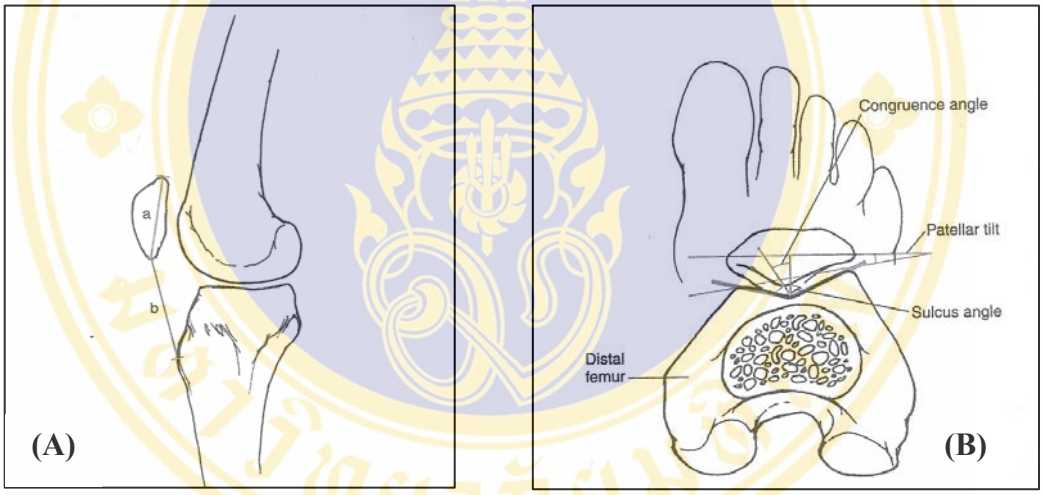


Figure 2.13 (A) The proximal-distal alignment of the patella is explained by a ratio of length of the patella (a) and the distance between the tibial tubercle and the distal of the patella (b). (B) The congruence angle (CA), the patella tilt angle (PTA), and the sulcus angle.

2.6 The Entire Leg Malalignment Affect PFJ Mechanics

2.6.1 Excessive foot pronation

Foot pronation means to combination of eversion, dorsiflexion and abduction of the foot. The excessive foot pronation may be known as flat foot or pes planus. This often occurs in individuals who lack of a supportive medial arch. It may lead to a compensatory internal rotation of the tibia or femur (femoral neck

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antiversion). These upset the patellofemoral mechanism (33). McConnell (1986) reported that subtalar pronation was the single most significant predictor of patellofemoral pain in adolescent boys (51). Therefore, arch supports or custom orthotics are frequent management in patients with PFPS (32, 33, 42).

2.6.2 High Q angle

The quadriceps angle (Q angle) is formed by the interaction of a line drawn from the ASIS to center of the patella and other line drawn from center of the patella to center of the tibial tuberosity (figure 2.14). Estimate values of Q angle in healthy population vary normal range from approximately 10-20 degrees. This was greater in women than men (15-20 vs. 10-15degrees). This angle is specified that active contraction of the quadriceps muscle applies a lateral pull on the patella as it pulls the patella proximally and extends the knee. Increasing in this angle indicates as greater lateral pull on the patella that provides larger stress between lateral facet of the patella and lateral trochlear of the femur. Therefore, patellofemoral pain and anterior knee pain have an association with high Q angle. Many investigators believed that the Q angle is the most important single factor contributing to patellofemoral pain. The Q angle could be altered by torsion of the tibia and femur and rotatoty malalignment of the foot. Some authors reported that larger Q angle associated with greater femoral antiversion, external tibial rotation and lateral displacement of the tibial tubercle (26, 32, 33, 40, 42).

2.6.3 Genu valgum

Frontal plane alignment of the knee can be described as the angle that formed by lines through shaft of the femur and shaft of the tibia. Knee valgus or genu valgum is defined as an abduction of the tibia relative to the femur. The normal frontal plane alignment of the knee is slightly valgus; range between 5 to 15 degrees. An excessive valgus (knock-knees) angle directly associates with increasing lateral pulling force of the quadriceps muscle. Many factors contribute to an increasing genu valgum angle including excessive pronation of the foot, a laxing of medial collateral ligament of the knee and excessive femoral neck antiversion. The valgus angle can be

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reduced by increasing the base of support while standing (feet are placed apart) (figure 2.14) (25, 26, 42, 51).

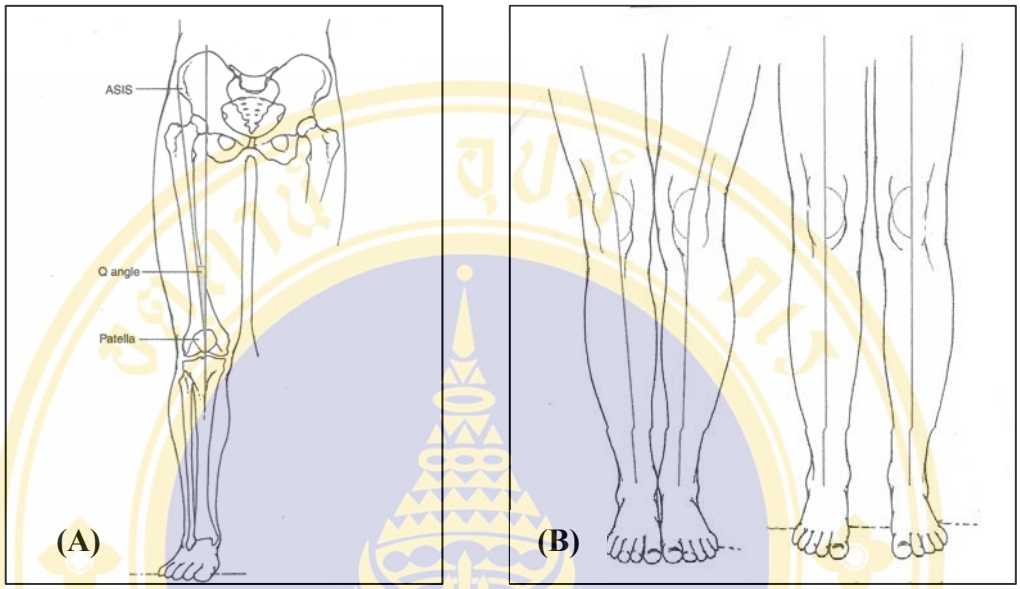


Figure 2.14 (A) The Q angle. (B) Valgus of the knee when feet are placed closely and apart.

2.6.4 Genu recurvatum

The sagittal plane alignment of the knee during standing can be indicated by an angle between the vertical alignment of the femur and tibia. In general, the angle is 180 degrees or both alignments are parallel. Genu-recurvatum is the postural alignment which is known as a hyperextension of the knee in standing. This frequently results from muscle imbalances at the ankle or knee. However, this affects to the PFJ mechanics and leads to the patellofemoral disorder (25, 26, 42).

2.6.5 Femoral neck antiversion

Transverse plane alignment of the proximal femur can be described as antiversion-retroversion (figure 2.15). In adult, the femoral neck positions anteriorly with respect to the plane of the femoral condyles in approximately 15 degrees of antiversion. Excessive femoral neck antiversion places the head of the femur farther anteriorly in acetabulum than normal. Its primary compensation is medial rotation of the hip, which proposed to move the femoral head in a normal location within the

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acetabulum. This compensation results in an in-toed posture during standing. If excessive antiversion presents for long period of time, secondary compensation in the tibia may occur. The tibia may rotate laterally, which turns the foot laterally with respect to the knee. Thus the in-toed standing disappears. Both compensation for femoral antiversion may affect patellofemoral mechanics that leads to patellofemoral disorder (26, 42).

2.6.6 External tibial torsion

The transverse plane alignment of the knee during knee extension can be described as tibial torsion. Normal tibial torsion is indicated as a transverse axes of the proximal tibia and distal of the femur are parallel (figure 2.15). External tibial torsion means that the tibial condyles are rotated laterally with respect to the femoral condyles. This may be the cause of other malalignments. They affect PFJ mechanics and lead to patellofemoral disorder (26).

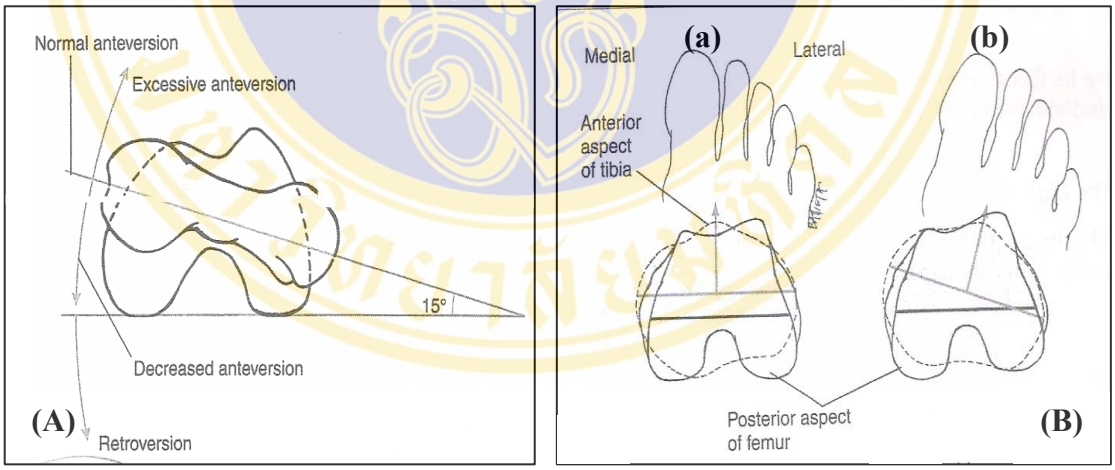


Figure 2.15 (A) Femoral neck antiversion-retroversion. (B) Normal alignment (a) and lateral torsion in transverse plane of the knee (b).

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2.7 The Muscular Dysfunctions Affect PFJ Mechanics

2.7.1 Strength deficit

2.7.1.1 Overall quadriceps muscle weakness

Weakness of the quadriceps muscle could present in terms of combine or isolate weakness. These have adverse effect to functions of any weak part and normal part. There alter the PFJ mechanics, which lead to patellofemoral pain. Numerous studies reported that PFPS patients present significant loss of quadriceps strength, particularly during eccentric quadriceps contraction (15, 55). Many PFPS patients present atrophy of the quadriceps muscle (59). Witvrouw et al (2005) suggested that the contralateral asymptomatic leg of PFPS patients may have abnormal sign of PFJ. This should be treated when the muscle strength of the quadriceps muscle of the symptomatic leg (measured by the isokinetic method) loss more than 15% (42).

2.7.1.2 Selective strength deficit of the VMO muscle (see 2.8)

2.7.1.3 Weakness of the hip muscles

Crossley et al (2000) suggested that a stable pelvis minimizes unnecessary stress on the knee. Therefore, training of the gluteus medius (posterior fibers) may decrease hip internal rotation and the valgus vector force that occur at the knee. This may improve the VMO activation (31). Dysfunction of hip external rotator muscles related to compensatory foot pronation. This leads to alter the PJF mechanics. Thus simple stretch on this muscle may improve the muscular efficiency (31). Based on the principle the VMO originates on hip adductor muscles. Some authors suggested that hip adductor strengthening may necessary for improvement of the PFJ mechanics (7).

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2.7.2 Muscular tightness

Several authors suggested that loss of flexibility of the iliotibial band, hamstrings, gastrocnemius, and quadriceps muscle strongly affect PFJ mechanics (60).

2.7.2.1 Iliotibial band

Tightness of the iliotibial band provides excessive lateral force on the patella. This could externally rotate the tibia. This upsets the balance of the PFJ mechanics which leads to excessive lateral tracking of the patella during knee motion (33).

2.7.2.2 Hamstring

Tightness of the hamstring muscle requires greater force being generated by the quadriceps. These provides an increasing of the patellofemoral joint reaction force (59, 60). Due to the hamstring muscle acts as the knee flexor. Tightness of the hamstring muscle increases posterior force on the knee. This increases pressure between the patella and the femur (12). In addition, raising knee flexion causes an increasing ankle dorsiflexion. This provides inadequate the talocrural joint motion. Thus, the subtalar joint assists with compensatory foot pronation. This affects to the PFJ mechanics (51).

2.7.2.3 Calf muscles

Due to a dorsiflexion at the talocrural joint can not occur in case of tight calf muscles, thus the subtalar joint adopt compensatory foot pronation (51). In addition, the posterior force on the knee is increased by tight calf. These affect to the PFJ mechanics (61).

2.7.2.4 Quadriceps muscle

Pattern of the quadriceps muscle tightness may include isolated or combined tightness. The isolate tightness of the rectus femoris muscle is common. The quadriceps muscle tightness directly limits knee flexion (26). Consequently, this contributes to increase in the force of contact between the patella

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and the femur. In addition, this affects efficacy of the knee extension, which require greater force for knee extension. More quadriceps force occurred provides greater patellofemoral joint reaction force (35).

2.8 VMO Insufficiency / VMO-VL Imbalance

The relationship between the VMO and VL has a major role to maintain the patellar tracking. The VMO functions are only medial dynamic stabilizer of the patella which counteracts the lateral pulling produced by the VL and iliotibial band (24, 28, 32). Dysfunction of the VMO causes lateral patella shifts and increases lateral patella pressure, resulting in predisposing to lateral subluxation or dislocation of the patella and to degenerative disorder of the patellofemoral joint (62-64). However, assessment of insufficiency of the VMO muscle could not be directly assessed by the strength of the VMO or VL in isolation. Therefore, the EMG recording has been used to measure the relative activation of the VMO and VL muscles (31).

One proposed mechanism for abnormal patellar tracking is an imbalance in the activation of the VMO and VL. The imbalance may include either reduced activation of the VMO or enhanced activation of the VL (10, 65, 66). This could be caused by a reduction in the force producing capability of the VMO that shows in form of an imbalance in the magnitude of VMO and VL activation (10, 66, 67). However, it may be caused by alteration of temporal control of VMO and VL activities this presents in form of delayed activation of the VMO relative to VL (12, 68).

Strong evidences showed that patients with PFPS have been disturbed in balance activation between the VMO and VL in both magnitude (10, 65, 66, 69, 70) and onset activation (12, 70-72). According to magnitude of EMG activities of the VMO and VL, there is general agreement that in individuals with no-painful knees, the EMG ratio of the VMO and VL (VMO:VL ratio) should be around 1.0 for the patella to be balanced within the intercondylar groove (7, 17, 64, 66, 69). Several authors found that VMO:VL ratio was less than 1.0 in people with patellofemoral pain

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(10, 17, 66). This was discussed that the individuals with PFPS have been decreased magnitude of EMG activity of the VMO relative to VL when compared with healthy subjects. However, some authors argued whether the appropriate number of the VMO:VL ratio for the healthy subjects should be 1.1 (10, 72) or 0.85 (70).

According to the onset activation of the VMO and VL, it was found that the EMG onset of the VL occurred earlier than the VMO in individuals with PFPS. The EMG onset of these muscles occurred simultaneously in healthy subjects (12, 68, 71). It is hypothesized that the VMO must be activated earlier than VL to optimal patellofemoral tracking in asymptomatic population. This hypothesis based on the VMO has smaller cross-sectional area and the VL has predominantly lateral direction force (4).

Muscle fiber type of the VMO is composed of the large portion of white fibers in the natural (67, 73). This fiber percentage is decreased in case of PFPS, who presented the VMO insufficiency. The EMG studies also showed that the VMO activity is tonic in healthy people and that is phasic in PFPS (67).

One purposed mechanism of the VMO-VL imbalance is described by a motor unit synchronization. It was found that the coordination of motor units between the VMO and VL in people with anterior knee pain is reduced when compared with people without knee pain (74).

It is hypothesized that the VMO and VL has major role to maintain the patellar tracking particularly in the first 30 degrees of knee flexion. This is due to stability of the patella does not support by the femoral trochlea (31). However, many researches proved that the VMO is active throughout the full range of the knee extension (28, 75). In addition, the study of motor unit synchronization by Mellor and Hodges (2006) concluded that the basic coordinative mechanism between the vasti muscles is controlled consistently throughout knee range of motion. It is not augmented at specific angles where the requirement for dynamic control of stability increases (76).

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Imbalance in flexibility of any part of quadriceps should be considered. Overall quadriceps muscle decreased flexibility that directly provides patellofemoral stress during daily life activities (26). Isolated flexibility of any parts of quadriceps should be evaluated. If the flexibility problem presents, abnormal distribution of patellofemoral stress during knee motion may occur. In addition, this may affect efficacy of muscles contraction which leads to abnormal patellar tracking and abnormal PFJ stress. For example, decreasing in the VL flexibility may also lengthen the VMO. This leads to the increase in lateral patellofemoral stress, stress between lateral facet of the patella and medial aspect of lateral femoral trochlea (24).

Treatment program for individuals with imbalance in activation of VMO-VL (such as PFPS, general quadriceps muscle weakness, knee extensor mechanism injury, ligament or meniscus of the knee injuries) should focus on retraining of the VMO. The retraining aims to restore dynamics patellar stability by correcting patellar tracking through the appropriate timing and intensity of the VMO and VL activation. In addition, flexibility of overall and isolated parts of the quadriceps should be evaluated and improved.

2.9 VMO Retraining

Recognition of imbalance between the VMO and VL muscles leads to focus on retraining of the VMO muscle in treatment of PFPS. Individuals with general quadriceps muscle weakness, knee extensor mechanism injury, ligament or meniscus of the knee injuries may have VMO insufficiency, thus they should be trained the VMO muscle.

The VMO retraining aims to correct abnormal patellar tracking through the suitable timing and intensity of the VMO that relative to the VL activation. Several studies suggested that EMG activity of the VMO and VL muscles in normal population is relatively balanced in a wide variety of static, dynamic, weight bearing and non-weight bearing activities (31, 51). The VMO plays an important role to stabilize the PFJ, thus endurance training by progress the number of training

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repetitions should be considered (31). The treatment should be conversed from the exercises to functional activities which refer to the patient's daily activities or sport activities (31, 51).

There are some characteristics that should be considered before deciding how the patient should perform VMO retraining such as CKC vs OKC exercises, position of the hips and knees. Closed kinetic chain (CKC) exercises such as step-up, step-down, squats and lunges have been widely indicated for knee rehabilitation. It promotes functional stability by producing co-contraction of the lower extremity muscles and normal proprioceptive input (5, 10, 14, 77). In CKC exercises, the patellofemoral joint compressive force is enhanced with greater knee flexion. However, this force is distributed by higher contact area between the patella and the femoral trochlear groove. Among the CKC exercises, the squat exercise is prescribed as safe and effective due to the stabilizing effect of the quadriceps and hamstring muscle co-contraction (6, 14). The VMO contains larger proportion of slow fibers, indicating VMO performs an antigravity role (67, 75). Therefore, VMO should be retrained with CKC thus enabling motor retraining in activities that more functional and specific (31, 51).

More evidences suggested that the VMO is not preferentially activated compared with VL during the three common OKC quadriceps exercises: 1) inner range knee extension or quadriceps sets (5, 69, 78) 2) straight leg raises (64, 78) 3) isometric knee extension (5, 64, 66, 78). Some investigators reported that the VMO is likely activated greater than VL during CKC knee extension exercise such squat (66, 78, 79), step-up (10, 65, 66, 78), step-down (5, 10, 65, 66) and lunge (5, 10). Stensdotter et al (2003) found that the onset of EMG activity of the VMO and VL muscles in closed chain knee extension was more simultaneous than in the open chain (80). Hodges and Richardson (1993) recommended weight bearing position for enhancing interaction between the VMO and the adductor muscles. This provided greater the VMO recruitment. Therefore, it could be concluded that the VMO muscle should be trained in CKC exercises such as squatting.

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The earlier literature suggested that patellar position is primarily controlled by action of the VMO, especially in the last 30 degrees knee flexion (61, 81, 82). The recent studies reported the VMO is active throughout the full range of the knee extension. Coordinative mechanism between the VMO and VL muscles must be controlled consistently throughout knee range of motion (75, 76). Some authors suggested the appropriate knee flexion during exercise in symptomatic knee pain should not be more than 45 degrees knee flexion. This avoid high patellofemoral compressive force (60, 77). However, it is recommended that short arc CKC knee extension exercise such squat in healthy individual and patellofemoral pain patient should be increased to 60 degrees knee flexion (8, 15, 19, 22). This promotes greater recruitment of the VMO muscle in relation to VL muscle more than 45 degrees knee flexion with no pain (19, 22). Knee pain was usually induced when squatting with 75-90 degrees knee flexion in individual with patellofemoral pain (19). Hodges and Richardson (1993) suggested the squat additional with hip adduction exercise should be executed with 60 degrees knee flexion (8). Therefore, the squat with isometric hip adduction exercise should be set at 60 degrees knee flexion which was safe and provided greater muscle activities.

The position of the hips should be neutral or slight external rotate (31). There was aimed to decrease the Q angle and lateral force acting on the patella (18, 26, 51). However, the recent study reported that there were no significant difference in the VMO:VL ratio during isometric quadriceps contraction in long sitting with variation of hip position, which were neutral, internal and external rotation (83).

Some authors suggested that pain has an inhibitory effect on the quadriceps muscle contraction thus patellar taping can be applied to ensure pain-free exercise in symptomatic patients (18, 40, 84). Barak et al (2006) reported that the VMO should be retrained the isometric, concentric, eccentric contraction and may progress to isokinetic exercise due to role of the VMO involves in all types of contraction (85).

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In order to many factors could affect the patellofemoral mechanics and consequently influence to recruitment of the VMO. Therefore some previous studies suggested that the VMO should be trained with complicated knee extension exercise such combined knee extension exercise with hip abduction/gluteal contraction (23, 86) and internal or external rotation of the hip (5, 64). Due to inability of the gluteus medius to control internal rotation of femur during CKC and VMO inhibition may lead to excessive lateral tracking of the patella. Thus there were hypothesized that addition of hip abduction or gluteal contraction (action of gluteus medius) with knee extension exercise may promote VMO activation (23, 86). However, Hertel et al (2003) reported that the VMO activation when combined knee extension with hip abduction was not preferentially enhanced compared with VL (23).

In order to stress at the knee caused by hip rotation during knee extension may affect to patellofemoral mechanics and activation of the VMO. Thus there were hypothesized that combination hip internal or external rotation with knee extension exercise may effect to recruitment of the VMO muscle (5, 64). However, Cerny (1995) and Kast and Jewett (1993) reported that the addition of hip internal or external rotation with knee extension exercise did not provide a beneficial effect on the VMO activation (5, 64).

Bose et al (1980) and McConnell (1986) suggested that the addition of hip adduction while performing knee extension might facilitate the VMO activity because of the large amounts of the VMO fibers originate from the tendon of the adductor magnus (see 2.10) (29, 51).

2.10 Knee Extension Combined with Hip Adduction Exercise

More evidences interested that activation of the VMO might be enhanced by combination knee extensions with hip adduction (4-6, 8-11, 64, 87-89). Rationales served the connection between hip adduction and recruitment of the VMO muscle can be focused in three hypotheses. First; the fibers of the VMO attach to the adductor magnus muscle (28, 29) hence activating this muscle may increase VMO fiber

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recruitment. Second; base on anatomical configuration of these two muscles performing hip adduction may provide a stable origin and an alteration in length tension properties of the VMO that enhance its recruitment (7). Third; some authors believed that the innervations of VMO and adductor magnus muscle present connections (7, 73).

The relationship between hip adduction and knee extension on the VMO activity is clearly different between OKC and CKC. Additional isometric hip adduction with OKC knee extension exercises has not been shown to selectively increase the VMO activity (5, 11, 64, 87-89). Conversely, incorporate simultaneous hip adduction and squat exercise such as semi-squat or wall squat have been associated with increases the VMO activity (6, 8, 10, 21).

Several studies investigated effects of hip adduction on the VMO and VL EMG activities during squatting (CKC) with hip adduction exercise (5, 6, 8, 9, 21, 23). However, most studies focused on only maximal hip adduction (6, 8, 9, 21, 23) or did not control hip adductor exertion (5). In case of focusing in maximum hip adduction, some investigators instructed subjects to adduct their hip with maximum effort but they did not measure or control it (6, 21, 23). These might be cause of the non-significant greater the VMO:VL ratio when perform squat with hip adduction exercise in some studies (6, 21, 23). Some studies found an increasing in VMO activity (6, 21) when addition hip adduction during squat exercise. Although, some authors found that this exercise no preferentially enhanced VMO:VL ratio during this exercise (21). However, their data showed no significant difference in VMO and VL activity during perform squat with hip adduction exercise whereas, it presented significant difference during perform squat without hip adduction exercise (9, 21). Thus, they believed that squat with hip adduction exercises could promote a greater muscle balance between the activity of the medial and lateral quadriceps muscle and could benefit the treatment of PFPS patients (8, 21).

Although some studies suggested that combining maximal hip adduction (subjective and objective evaluation) with squat exercise enhanced activity of the

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VMO (6, 8, 9, 21). However, the strenuous exercise may exacerbate pain and lead to fatigue of hip adductor and vasti muscles. Application of this exercise in clinical is complained as a non comfortable and difficult exercise. Rice et al (1995) reported that differences in the amount of hip adduction effort may effects to variability in the VMO activity when perform knee extension with hip adduction exercise (11). However, no previous study investigated effects of hip adduction effort on VMO and VL recruitment when performing this exercise.

In clinical application, this exercise is applied by two difference sequences. One maintained hip adduction at entire range of squat, which is proposed to stabilize the origin of the VMO at the beginning and continuously for entire range of exercise. Another one performed hip adduction when squatting to the specific angle. The latter exercise is widely accepted because it is easy and comfortable to do. Furthermore, it is believed that can also stabilize the origin of the VMO and stretch its length at specific angle of knee flexion (6). However, no previous study compares the vasti recruitment between these exercise sequences.

2.11 EMG Biofeedback for VMO Retraining

EMG biofeedback is devices that detect the neuromuscular contractions and provides real time feedback information to the subjects during training sessions. Therefore, it could facilitate the integration of sensory cues and motor recruitment of the muscles (90). Some authors reported that EMG biofeedback can assess the conditions and it is the most direct way to treat the case (18, 91).

The surface EMG biofeedback in clinical practice is extremely useful in facilitating VMO activation because it can provide immediate information on the muscle activation pattern to the subject and reinforcement when the correct pattern is achieved (90, 92). Previous studies reported that EMG biofeedback effective for retraining the VMO such as improvement of balanced activity between the VMO and the VL or increasing VMO:VL ratio (90-93), correcting patellar malalignment which

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position the patella medially (91, 94), altering the PFJ force (93), decreasing pain (90, 93, 95), improvement of knee functions (90) and recovery quadriceps functions (96). However, they usually applied EMG biofeedback during progressive exercise program (91, 93) or combined with other conventional treatments (90, 95). Thus those positive results were argued that might be effect by confounding factors or multi treatment effect.

The instruction for subjects during retraining the VMO by using EMG biofeedback has been varied in previous study such as to increase the VMO activity while maintain stable VL activity (92, 93), to increase the VMO activity while minimize the VL activity (91, 92) and to balance activity between the VMO and VL activities (90). LeVeau and Rogers (1980) found that their subjects reported difficult reducing VL activity without decreasing VMO activity. However, subjects succeed in increasing VMO activity while maintaining VL activity below the threshold level. Therefore, they suggested that up-training the VMO was more successful strategy than down-training the VL (92).

EMG biofeedback has weak points as providing non-normalized data. The EMG activity might vary when it is measured repeatedly over a long time frame. Therefore, the evaluation of VMO retraining by EMG biofeedback should be concerned. In addition, the EMG biofeedback tools are expensive in present.

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CHAPTER III
MATERIALS AND METHODS

3.1 Subjects

Healthy male and female age between 18 and 29 years voluntarily participated in this study. They presented right dominant leg.

Exclusion criteria

3.1.1 History of anterior knee pain reported on at least two common aggravating activities: ascending/descending stair, squatting, running, hopping/jumping or prolong sitting.

3.1.2 History of patellar dislocation or subluxation.

3.1.3 History of meniscus, ligament of the knee, quadriceps or patellar tendon injuries.

3.1.4 History of knee or lower limb surgery.

3.1.5 Having hip adductor muscles or iliotibial band tightness determined by the adduction contracture test and the Ober's test.

3.1.6 Having abnormal varus, valgus or hyperextension knee.

3.1.7 Having BMI more than 23 kg/m^2.

3.1.8 Having menstruation period on the testing day.

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3.2 Instrumentations

3.2.1 Electromyography (EMG)

Surface EMG with 8 channels (Naroxon INC., USA © 2006) (Figure 3.1) was used for collecting EMG activity of the VMO and VL muscles. The sampling frequency of 1500 Hz and the cut-off frequency of 500 Hz were set.



Figure 3.1 Electromyography (EMG) unit; (A) computer, (B) receiver card, (C) EMG with transmitter card, (D) channel 1, (E) channel 2 and (F) EMG electrodes.

3.2.2 Surface EMG electrodes

Pregelled, round, AG-AgCl surface electrodes (Blue Sensor, Olystyke, Denmark) with a 10 mm contact area (Figure 3.1) were used.

3.2.3 The elastic ball with pressure biofeedback

This in-house instrument was used for providing the resistant against hip adduction. It consists of an elastic ball, a valve pump with a three ways hallow rubber tube, and a sphygmomanometer (Figure 3.2). The elastic ball was covered by globous cloth to limit the changes of shape and volume of the elastic ball during exercises. The valve pump and the three ways hallow rubber tube transferred pressure from the elastic ball to the sphygmomanometer when subjects put the hip adduction on the ball.

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Subjects got feedback of their hip adduction effort from display of the sphygmomanometer.

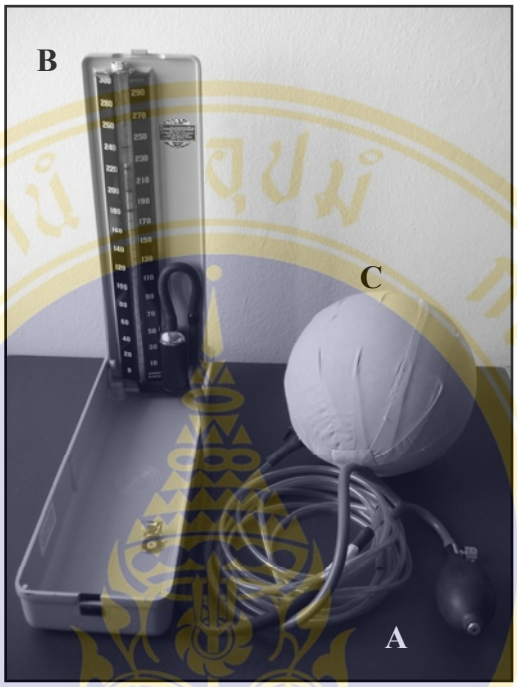


Figure 3.2 Elastic ball with pressure biofeedback; (A) three ways hallow rubber tube and valve pump, (B) sphygmomanometer and (C) elastic ball.

3.2.4 Stabilization strap (Figure 3.3)

3.2.5 Resistive band (Figure 3.3)

3.2.6 Single use blade (Figure 3.3)

3.2.7 Alcohol solution (Figure 3.3)

3.2.8 Hypoallergenic tape

3.2.9 Universal goniometer (Figure 3.3)

3.2.10 Elastic bandage (Figure 3.3)

3.2.11 Measurement tape (Figure 3.3)

3.2.12 Knee extension table (Figure 3.4)

3.2.13 Triangular stick (Figure 3.4)

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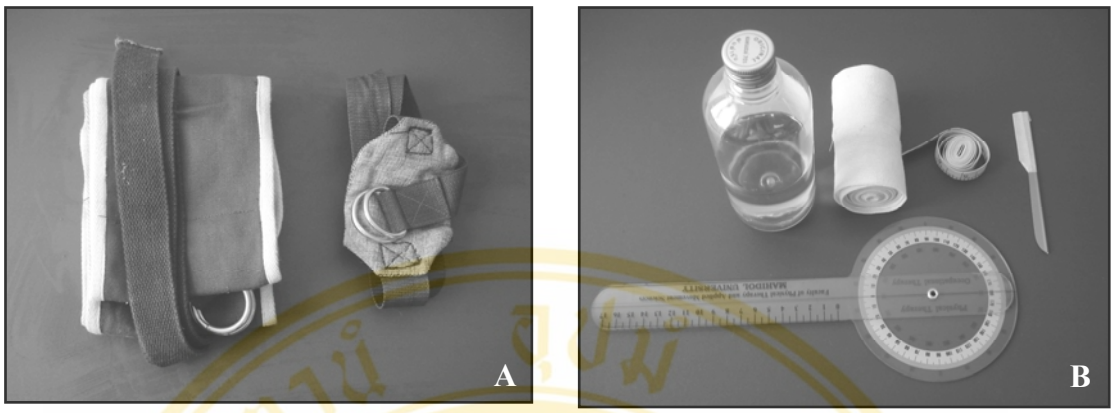


Figure 3.3 (A) Stabilization strap and resistive band. (B) Universal goniometer, alcohol solution, elastic bandage, measurement tape and single use blade.



Figure 3.4 (A) Knee extension table. (B) Triangular stick.

3.3 Subject Preparation

The demographic data (gender, age, weight, height, BMI, and dominant leg) and medical history were recorded by interviewing (appendix C.1). The following tests were executed to screen out subjects who had tightness of the hip adductor muscle or the iliotibial band and any abnormal alignment of the knee e.g. valgus-varus or hyperextension knee.

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3.3.1 Adduction contracture test

Tightness of the hip adductor muscles was indicated by the angle less than 90 degrees forming between the femoral line of the tested leg and the line joining between both anterior superior iliac spines (ASIS) in supine. When the investigator attempted to abduct the hip at 90 degrees, the pelvis would be shifted up on the tested side or down on the other one (97).

3.3.2 Ober's test

The subject lay in side lying with the lower leg flexed at the hip and knee for stability. The investigator stabilized the subject's pelvis and passively extended knee. Then, hip abduction and extension of the upper leg were done. The iliotibial band tightness referred to the subject's upper leg remained abduct and was not fall to the bed when the investigator slowly lowers the upper leg (97).

3.3.3 Knee alignment test

The varus knee was determined as medial side of the knees did not touch together when standing with the feet closed. If subjects could not put their feet touch together while standing with the knees closed, they were described as valgus knee (97).

The hyperextension knee was reported as the angle formed by the lateral femoral line (line joined the greather trochanter and the lateral femoral condyle) and the lateral line of the leg (line joined head of the fibula and the lateral malleolus) during standing more than 10 degrees (25, 98).

3.4 Testing Procedures

This study was approved by the Mahidol University Institutional Review Board. Study procedures were explained to each subject. They signed a consent form prior to participation.

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3.4.1 Warm up and training session

A warm up session included a stretching of the quadriceps, hamstrings, iliotibial band, hip adductor and calf muscles (Figure 3.5). After that, the sequences of each exercise testing were explained. Subjects were allowed to practice each exercise testing until they could do each of them correctly.

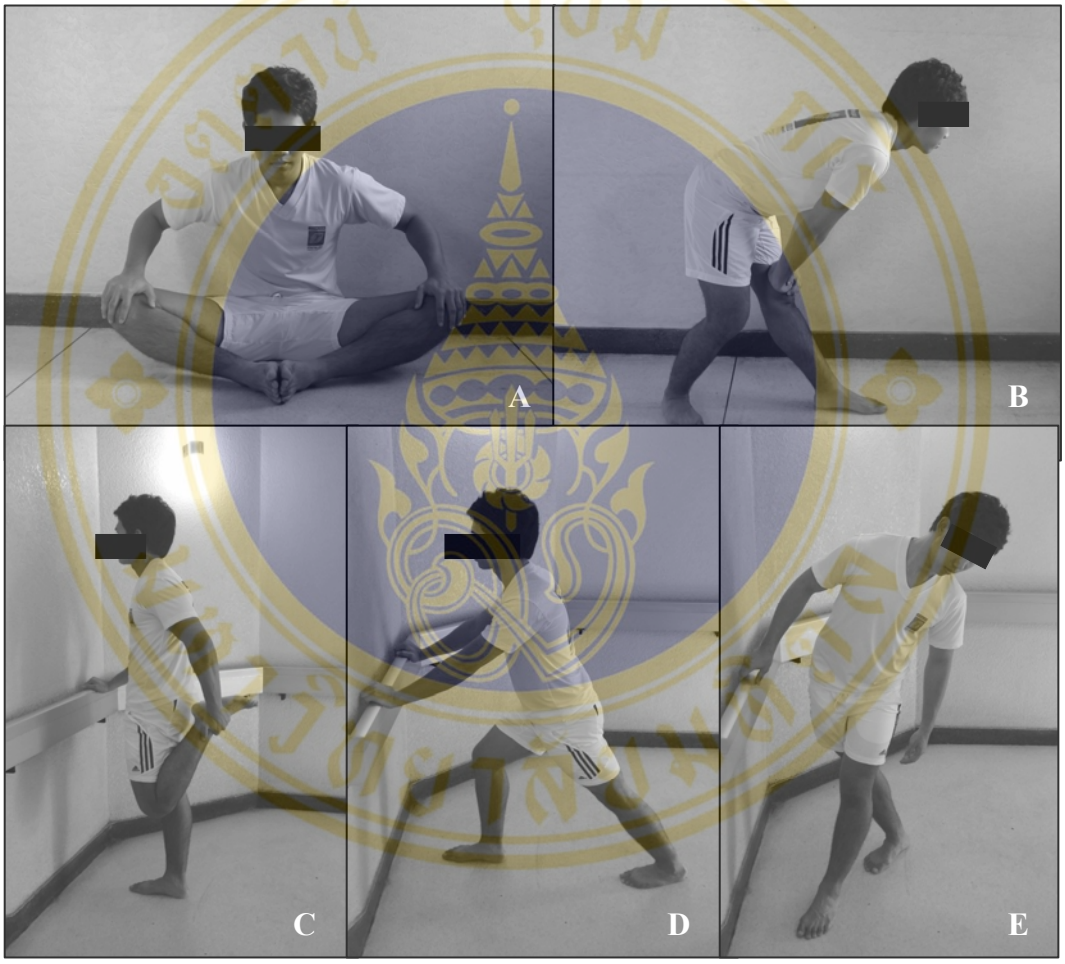


Figure 3.5 Stretching position; (A) hip adductor, (B) hamstring, (C) quadriceps, (D) calf and (E) iliotibial band.

3.4.2 Starting position of the exercise testing

Subjects stand with their back against a wall, placing their feet approximately 10 cm far from the wall. They placed their feet apart at their shoulder width while their hands were across the chests. Both hips were slight externally

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rotated (Figure 3.6). This was aimed to decrease the Q angle and lateral force acting on the patella (18, 51).



Figure 3.6 Starting position of the exercise testing.

3.4.3 Testing for maximum hip adductor exertion

The maximum hip adductor exertion was measured by the pressure biofeedback during squeezing on the elastic ball. Subjects in the starting position (Figure 3.6) which the elastic ball was positioned between subject's knees. They flexed their knees to 60 degree flexion (their buttocks touched a triangular stick that was fixed on the wall) and consequently performed isometrically squeeze their thighs together as hard as possible (Figure 3.7). Verbal commands for 2 seconds were consistently encouraged. Prior testing, subjects could rehearse by performing 2 sub-maximum and 1 maximum isometric hip adduction and followed by 2 min rest. During testing, subjects were instructed to maintain their back light touch to the wall. This test was performed 3 times with 2 min resting interval. A maximum value from this procedure was used to calculate the percentage of hip adductor exertion.

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Figure 3.7 Position of maximum hip adductor exertion testing.

3.4.4 EMG electrodes placement

EMG electrodes were positioned on the dominant leg of subjects. Skin was shaved and swabbed with alcohol to minimize the contact impedance. An inter-electrode distance of 22 mm was set. Electrodes placement were executed during sitting with slightly flex position of the testing knee. Electrodes of the VMO were fixed at approximately 3 cm medial to and 4 cm superior to the superomedial patella border and oriented at 55 degrees to the vertical line (VMO muscle fibers line). The electrodes of the VL were placed 10 cm superior to the superior border of the patella and 6-8 cm lateral to the perpendicular line from superior border of the patella regarding to the muscle bellies during contraction. The electrodes were also oriented 15 degrees to the vertical line (Figure 3.8) (24, 99). A reference electrode was positioned over the anterior aspect of the thigh. The electrodes were secured with hypoallergenic tape and ran over with an elastic bandage. Subjects were asked to squatting with 60 degrees knee flexion for rechecking the positions of all electrodes whether they remained over the muscle bellies.

Before starting the test, the EMG activities were recorded while the investigator was pressing the ball over the electrode during relaxation. If there was

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any change in EMG activity from this pressure, the setting was redone until no activity occurs.

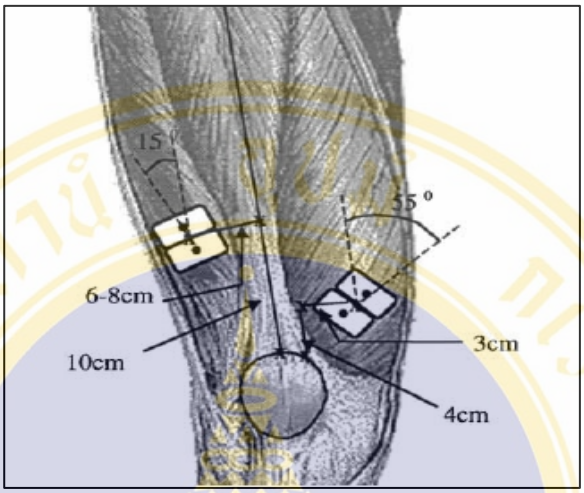


Figure 3.8 Surface EMG electrodes placement for the VMO and VL muscles.

3.4.5 Maximum voluntary isometric contraction (MVC) recording

The EMG activity of maximum voluntary isometric contraction (MVC) was collected to normalize the EMG recording obtained during the exercises testing. Subjects sat on the knee extension table in approximately 90 degrees of hip flexion and 60 degrees of knee flexion as the knee angle exercises testing. They were secured with the stabilization straps around the pelvis. The resistant pad was positioned at 2.5 cm above the medial malleolus which provided the resistant against knee extension (Figure 3.9).

Subjects repeated 2 sub-maximal and 1 maximal isometric knee extension contractions as a warm-up at this position. After 2 min resting, subjects were instructed to extend their knee as hard as possible and the encouragement continues by verbal commands for 2 seconds, the EMG data were recorded at the same time. The EMG of MVC was collected for 3 times with 2 min apart. Two of three records that showed a one second window of a stable period of EMG activity were selected to analysis.

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Figure 3.9 Position of maximum voluntary isometric contraction (MVC) testing.

3.4.6 Exercises testing

Exercises testing in this study included 7 conditions as follows.

(1) Squat without hip adduction exercise (0%squat exercise (0%SE)).

(2) Squeeze entire time-squat exercise (SE-SE) with 35% hip adductor exertion (35%SE-SE).

(3) Squeeze entire time-squat exercise with 70% hip adductor exertion (70%SE-SE).

(4) Squeeze entire time-squat exercise with 100% hip adductor exertion (100%SE-SE).

(5) Squeeze at 60 degrees knee flexion-squat exercise with 35% hip adductor exertion (35%SF-SE).

(6) Squeeze at 60 degrees knee flexion-squat exercise with 70% hip adductor exertion (70%SF-SE).

(7) Squeeze at 60 degrees knee flexion-squat exercise with 100% hip adductor exertion (100%SF-SE).

Subjects performed squatting with 60 degrees of knee flexion in each exercise. The SE-SE was executed by the following sequence (Figure 3.10);

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(1) Assumed the starting position (see 3.4.2)
(2) Squeezed thighs against the elastic ball with 35%, 70% or 100% hip adductor exertion
(3) Bent knees to 60 degrees of knee flexion and held for 2 seconds

The SF-SE was executed by the following sequence (Figure 3.10);

(1) Assumed the starting position (see 3.4.2)
(2) Bent knees to 60 degrees of knee flexion
(3) Squeezed thighs against the elastic ball with 35%, 70% or 100% hip adductor exertion and held for 2 seconds

The 0%SE was executed by the following sequence;

(1) Assumed the starting position (see 3.4.2)
(2) Bent knees to 60 degrees of knee flexion and held for 2 seconds

The EMG was recorded at the whole exercise time. During all testings, subjects were instructed to maintain their back light touch the wall and they got the feedback for hip adductor exertion from the display of sphygmomanometer (Figure 3.10). All subjects performed these exercises testing in random order. Each exercise was performed for 3 trials with at least 2 min rest between the two tests. Before starting all exercises testing, the initial pressure of the elastic ball was set at the same level as 180 mmHg.

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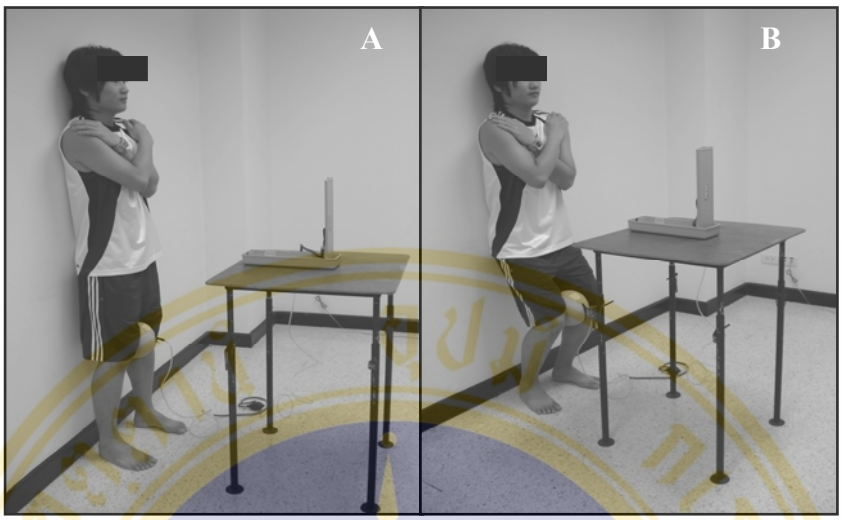


Figure 3.10 (A) Starting position. (B) Position of squat with hip adduction exercise.

3.4.7 EMG data acquisition

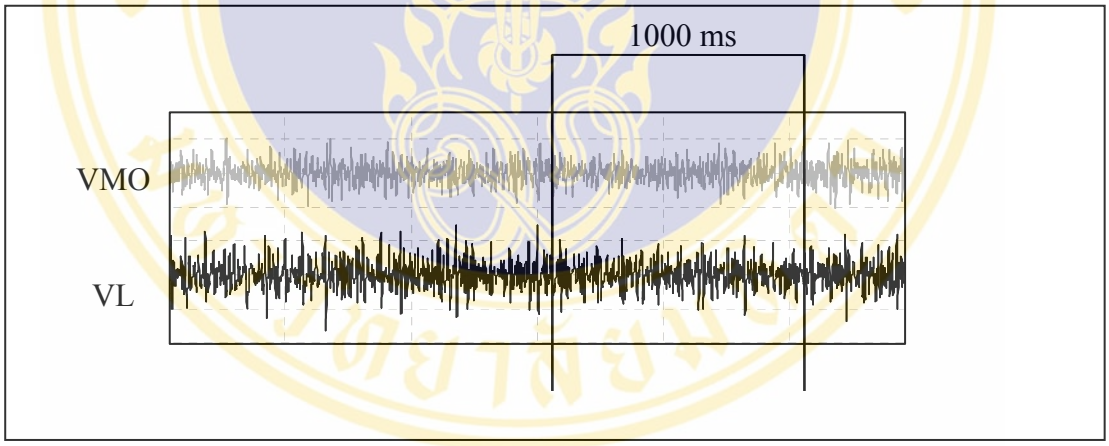


Figure 3.11 Calculation period of the VMO and VL EMG activities.

The EMG data from the VMO and VL muscles were collected at the sampling frequncy of 1500 Hz. The raw EMG was viewed by the Myoresearch XP Master edition 1.06.64. Two of three records that showed a 1 second window of a stable period of EMG activity at 60 degrees of knee flexion (hold phase) were selected (Figure 3.11). The data were exported to Microsoft Excel 2003. A macro was designed to calculate the RMS value of the selected phase and performed normalization of the data as in the following formula. The RMS values of each trial were calculated as 20 milliseconds bin interval and averaged continuously for 1

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second. The EMG data of each muscle were analyzed in terms of their RMS values. They were normalized by the averaged RMS value from the two 100%MVC of knee extension. The normalized RMS of the VMO, VL and VMO:VL ratio were calculated from individual trial and averaged from two repetitions.

$$\text{Normalized RMS} = \frac{\text{RMS during exercise} - \text{RMS during rest}}{\text{RMS during MVC} - \text{RMS during rest}} \times 100$$

3.5 Statistical Analysis

Statistical analysis was performed using Statistical Package for Social Science (SPSS) version 13.0. Statistical normality was tested by Kolmogorov-Smirnov Goodness of Fit test. Mauchly's test of sphericity showed statistical significance in hip adductor exertion factor, the multivariate test was performed to determine effects of two types of exercise sequence (SE-SE and SF-SE) and hip adductor exertion (0%, 35%, 70% and 100%) on the mean normalized RMS (%MVC) of the VMO, VL and VMO:VL ratio. The within-subjects contrasts was used to identify specific significant difference. Paired t-test was used to determine difference between mean normalized RMS of the VMO and VL muscles under each exercise condition. The level of significance was preset at *p*-value less than 0.05.

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3.6 Diagram of the Study

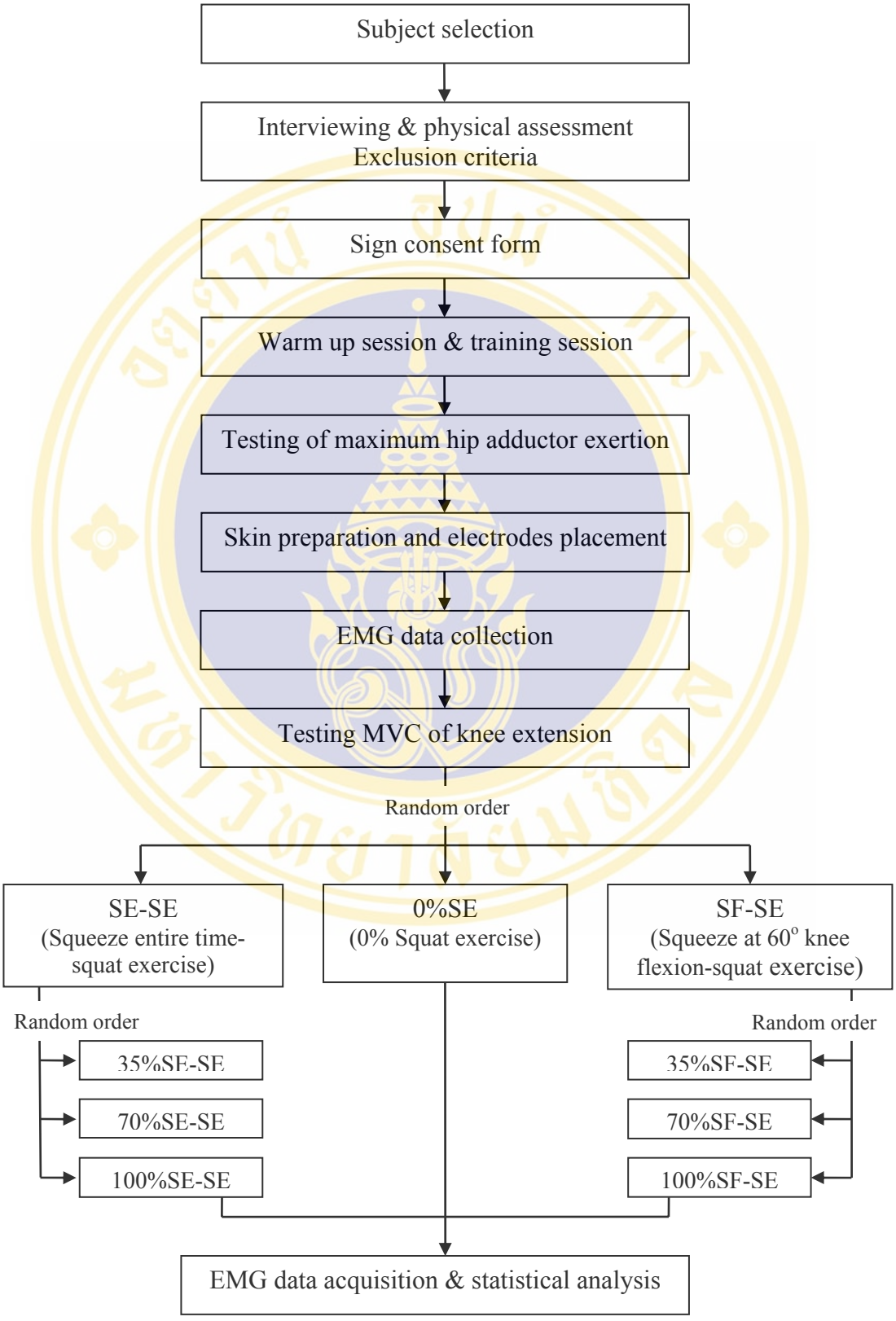


Figure 3.12 Diagram of the study.

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CHAPTER IV
RESULTS

Effects of hip adductor exertions (0%, 35%, 70% and 100%) and two exercise sequences (SE-SE and SF-SE) on EMG activities of the VMO and VL muscles were examined in healthy persons aged 18-29 years.

4.1 Characteristics of Subjects

The sample size of 30 subjects was calculated as demonstrated in Appendix F. The characteristics of subjects including age, weight, height, BMI, gender, maximum hip adductor exertion (presented in mmHg of a pressure cuff) and RMS of 100%MVC of the VMO and VL muscles are presented in Table 4.1.

Table 4.1 Characteristics of subjects, maximum hip adductor exertion and root mean square (RMS) of 100% maximum voluntary isometric contraction (MVC) of the vastus medialis oblique (VMO) and vastus lateralis (VL) muscles (n = 30)

	Mean ± SD
Subject's characteristics	
Age (yrs)	21.10 ± 2.55
Weight (kg)	56.77 ± 8.44
Height (cm)	166.17 ± 8.55
BMI (kg/m^2)	20.47 ± 1.83
Gender	Male = 15, Female = 15
Maximum hip adductor exertion (mmHg)	258 ± 20.49
RMS of 100%MVC	
VMO (mV)	267.33 ± 137.21
VL (mV)	165.79 ± 84.54

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4.2 Effects of Exercise Sequences and Hip Adductor Exertions on Normalized RMS (%MVC) of the VMO, VL and VMO:VL ratio

4.2.1 Data distribution

The results of Kolmogorov-Sminov Goodness of Fit test showed normal distribution ($p > 0.05$) in all parameters (Appendix E).

4.2.2 Descriptive analysis

Means and standard deviations (SD) of normalized RMS (%MVC) of the VMO, VL and VMO:VL ratio are shown in Table 4.2.

Table 4.2 Mean and standard deviation (SD) of normalized root mean square (RMS) of the vastus medialis oblique (VMO), vastus lateralis (VL) and VMO:VL ratio

Parameter	Hip Adductor Exertion	Exercise Sequence	
		SE-SE (Mean ± SD)	SF-SE (Mean ± SD)
VMO	0%	35.77 ± 14.23	35.77 ± 14.23
	35%	30.43 ± 11.25	28.15 ± 12.86
	70%	44.02 ± 17.01	38.35 ± 17.21
	100%	62.14 ± 27.25	60.68 ± 27.32
VL	0%	36.27 ± 14.25	36.27 ± 14.25
	35%	41.71 ± 18.37	40.99 ± 17.60
	70%	55.40 ± 23.70	50.55 ± 21.63
	100%	73.60 ± 35.00	68.36 ± 29.01
VMO:VL ratio	0%	1.057 ± 0.404	1.057 ± 0.404
	35%	0.800 ± 0.326	0.760 ± 0.353
	70%	0.865 ± 0.347	0.845 ± 0.366
	100%	0.935 ± 0.410	0.994 ± 0.469

SE-SE = squeeze (hip adduction) entire time–squat exercise

SF-SE = squeeze (hip adduction) at 60 degrees knee flexion–squat exercise

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4.2.3 Sphericity test

Due to this study was within subjects design, sphericity test was determined first. Mauchly's test was used to investigate sphericity of the normalized RMS (%MVC) of the VMO, VL and VMO:VL ratio. Statistical significant were indicated for all parameters in both hip adductor exertion factor and interaction between exercise sequence and hip adductor exertion ($p < 0.05$) as showed in Table 4.3.

Table 4.3 Mauchly's test of sphericity of normalized root mean square (RMS) of the vastus medialis oblique (VMO), vastus lateralis (VL) and VMO:VL ratio (the values in this table are shown in form of *p*-value)

Factor	*p*-value		
	VMO	VL	VMO:VL ratio
Exercise sequence	.	.	.
Hip adductor exertion	<0.001**	<0.001**	<0.001**
Exercise x Hip adductor exertion	<0.001**	<0.001**	<0.036*

* = significant difference at $p < 0.05$

** = significant difference at $p < 0.001$

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4.2.4 Effects of exercise sequences (SF-SE and SE-SE) and hip adductor exertions (0%, 35%, 70% and 100%) on normalized RMS (%MVC) of the VMO, VL and VMO:VL ratio

Due to the sphericity test of all parameters showed statistical significance in both hip adductor exertions factor and interaction between exercise sequences and hip adductor exertions as showed in Table 4.3. Therefore, multivariate test was used to determine effects of exercise sequences (SE-SE and SF-SE), hip adductor exertions (0%, 35%, 70% and 100%) and interaction effect between exercise sequences and hip adductor exertions on mean normalized RMS of the VMO, VL and VMO:VL ratio, respectively. The results of all parameters presented that there were statistical significant effects of hip adductor exertions ($p < 0.001$) but no statistical significant effects of exercise sequences ($p > 0.05$) and interaction between two factors ($p > 0.05$) as showed in Table 4.4.

Table 4.4 Effects of exercise sequences and hip adductor exertions on normalized root mean square (RMS) of the vastus medialis oblique (VMO), vastus lateralis (VL) and VMO:VL ratio determined by multivariate test (the values in this table are shown in form of *p*-value)

Effect	*p*-value		
	VMO	VL	VMO:VL ratio
Exercise sequence	0.221	0.072	0.999
Hip adductor exertion	<0.001**	<0.001**	<0.001**
Exercise x Hip adductor exertion	0.244	0.331	0.524

** = significant difference at $p < 0.001$

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4.2.5 Comparison of normalized RMS (%MVC) of the VMO, VL and VMO:VL ratio between 0%, 35%, 70% and 100% hip adductor exertion

There were statistical significant effects of hip adductor exertions on all parameters (VMO, VL and VMO:VL ratio) calculated by multivariate test. Then within-subjects contrasts were regarded to investigate the difference of VMO, VL and VMO:VL ratio among 0%, 35%, 70% and 100% hip adductor exertion. The results are shown in Table 4.5 and Figures 4.1 – 4.3.

Table 4.5 Comparison of normalized root mean square (RMS) of the vastus medialis oblique (VMO), vastus lateralis (VL) and VMO:VL ratio between 0%, 35%, 70% and 100% hip adductor exertion tested by within-subjects contrasts (the values in this table are shown in form of *p*-value)

Parameter	Hip adductor exertion	*p*-value		
		0%	35%	70%
VMO	35%	0.007*		
	70%	0.048*	<0.001**	
	100%	<0.001**	<0.001**	<0.001**
VL	35%	0.014*		
	70%	<0.001**	<0.001**	
	100%	<0.001**	<0.001**	<0.001**
VMO:VL ratio	35%	<0.001**		
	70%	0.002*	0.014*	
	100%	0.141	<0.001**	0.002*

* = significant difference at $p < 0.05$

** = significant difference at $p < 0.001$

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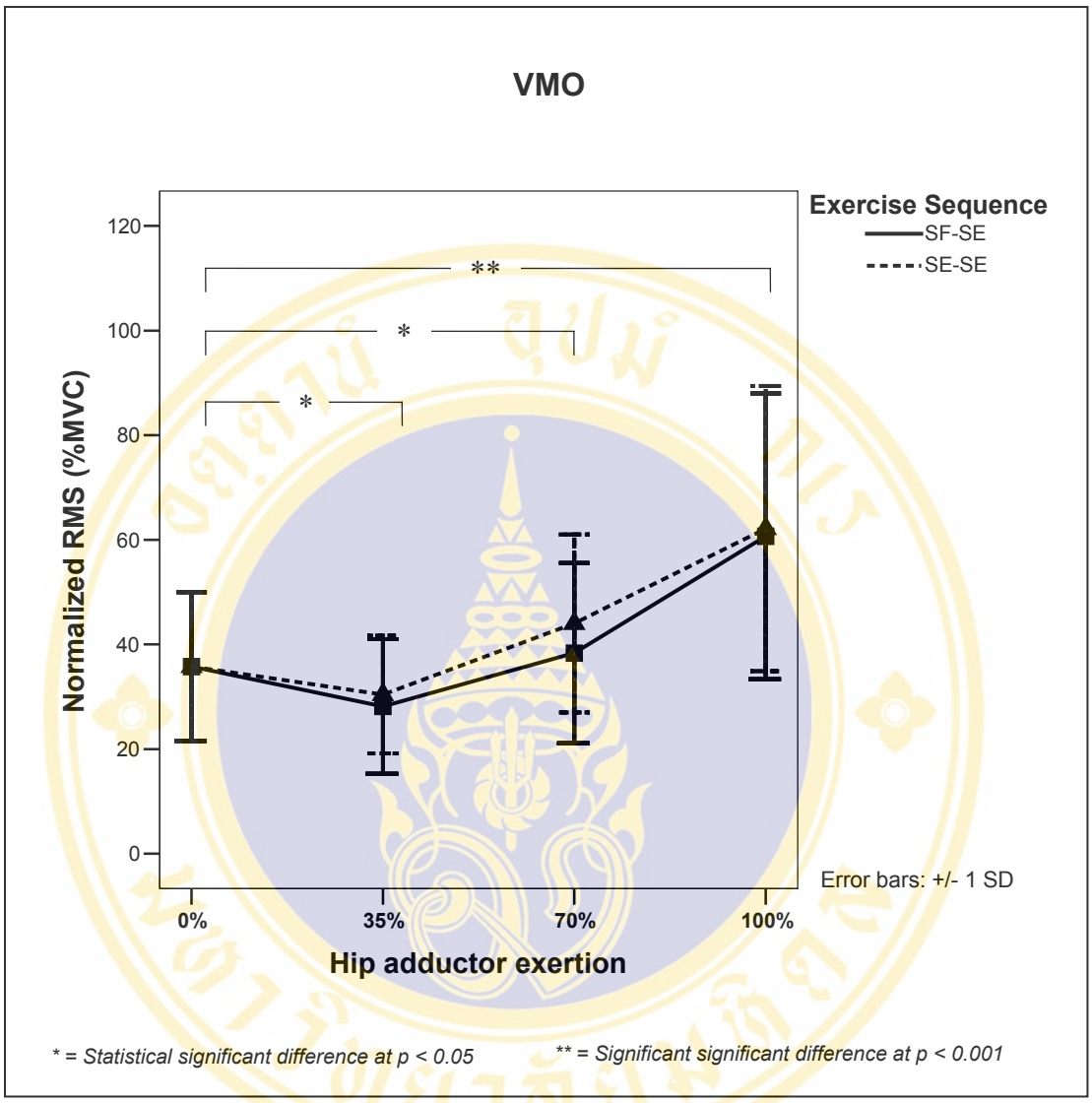


Figure 4.1 Comparison of normalized root mean square (RMS) of the vastus medialis oblique (VMO) muscle between 0% and 35%, 70%, 100% hip adductor exertion during squeeze (hip adduction) at 60 degrees knee flexion–squat exercise (SF-SE) and squeeze (hip adduction) entire time–squat exercise (SE-SE).

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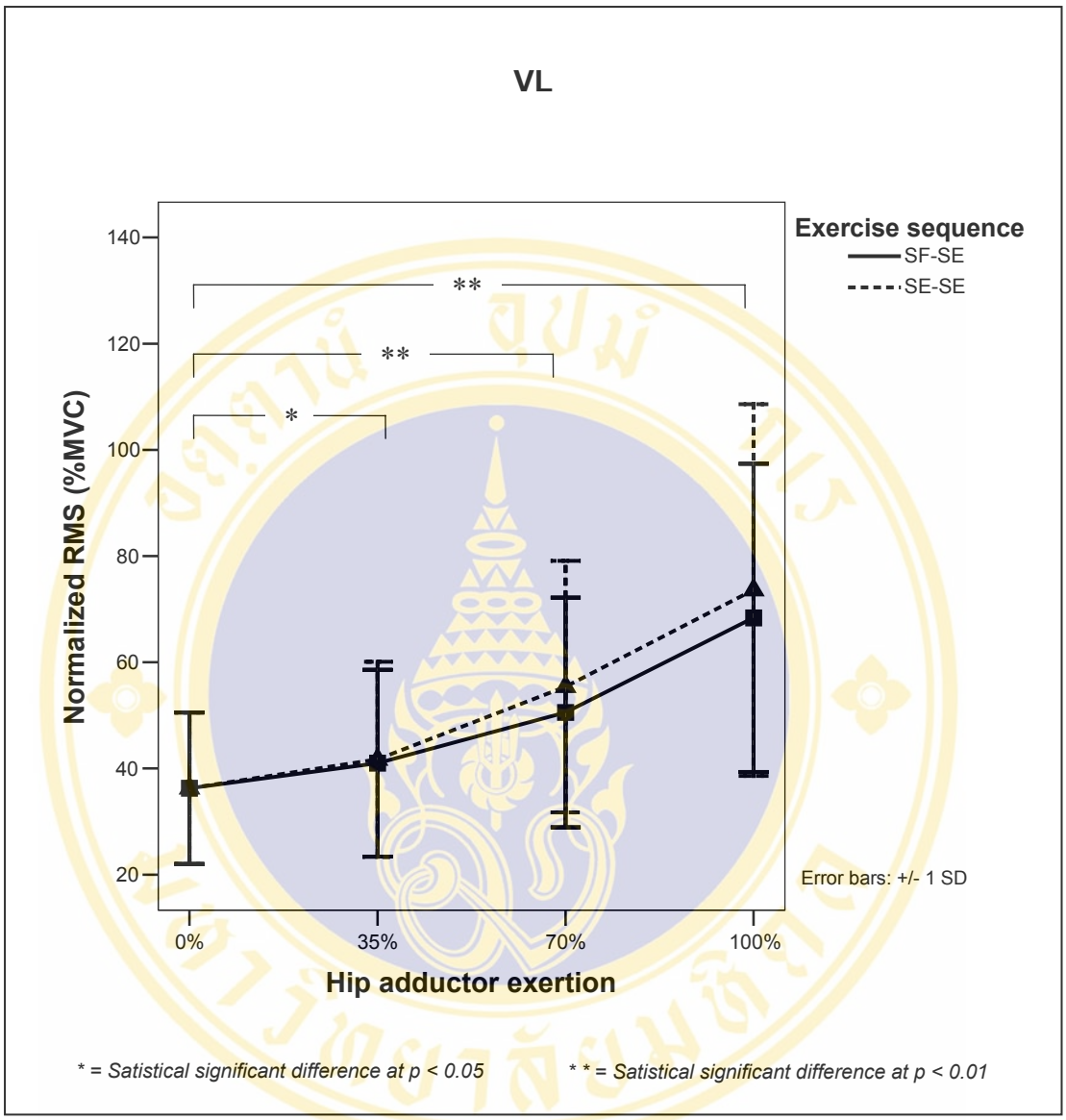


Figure 4.2 Comparison of normalized root mean square (RMS) of the vastus lateralis (VL) muscle between 0% and 35%, 70%, 100% hip adductor exertion during squeeze (hip adduction) at 60 degrees knee flexion– squat exercise (SF-SE) and squeeze (hip adduction) entire time–squat exercise (SE-SE).

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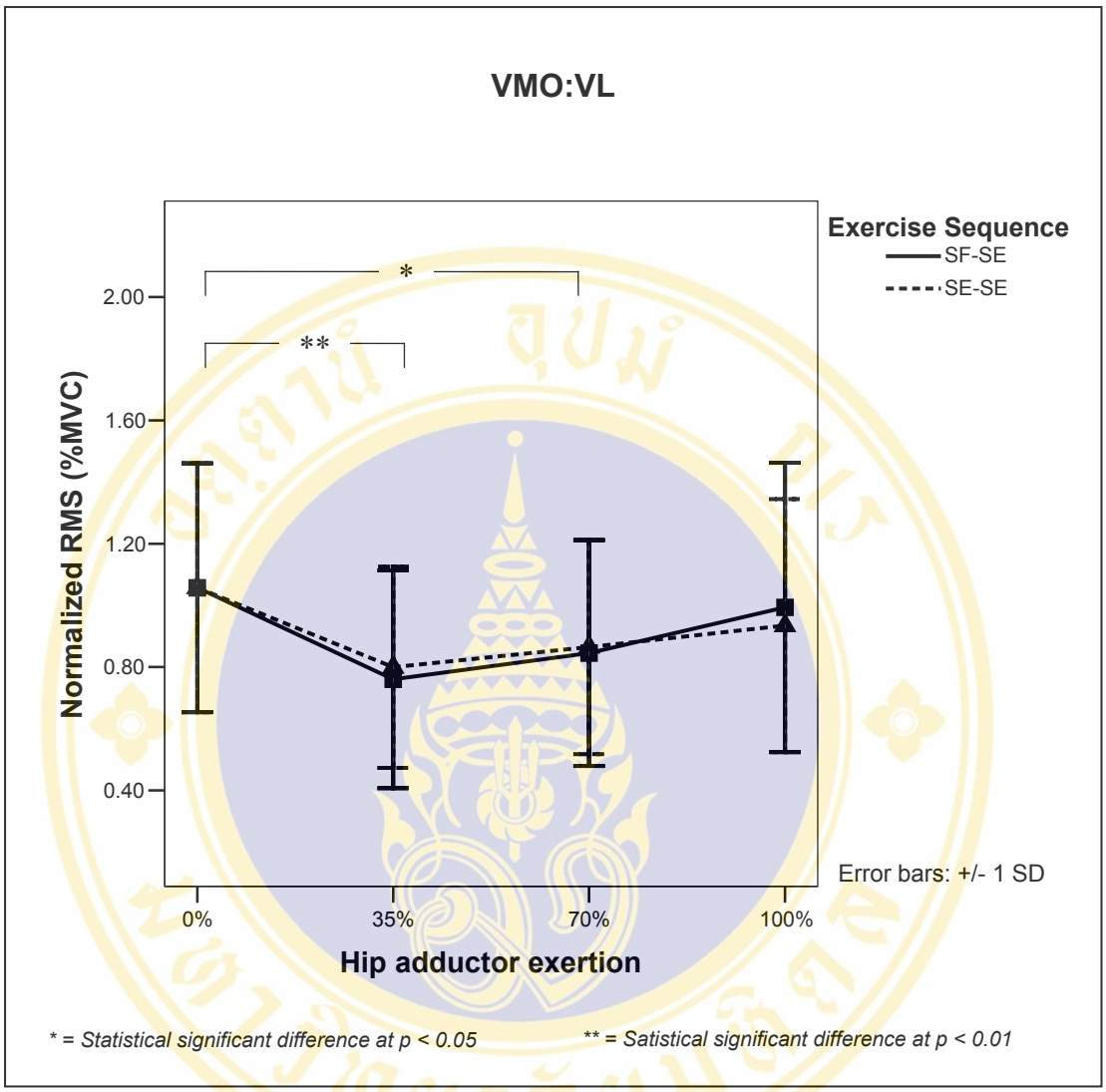


Figure 4.3 Comparison of normalized root mean square (RMS) of the vastus medialis oblique (VMO): vastus lateralis (VL) ratio between 0% and 35%, 70%, 100% hip adductor exertion during squeeze (hip adduction) at 60 degrees knee flexion–squat exercise (SF-SE) and squeeze (hip adduction) entire time–squat exercise (SE-SE).

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4.3 Comparison of Normalized RMS (%MVC) between the VMO and VL Muscles

Paired t-test was used to evaluate the difference between mean normalized RMS (%MVC) of the VMO and VL muscles in each condition. In Table 4.6 and Figures 4.4-4.5, there were statistically significant differences between normalized RMS (%MVC) of the VMO and VL muscles at 35%SE-SE (p = 0.001), 70%SE-SE (p = 0.015), 35%SF-SE (p < 0.001) and 70%SF-SE (p = 0.001).

Table 4.6 Comparison of normalized root mean square (RMS) between the vastus medialis oblique (VMO) and vastus lateralis (VL) muscles

Exercise	Hip adductor exertion	Mean ± SD		t	p-value [a]
		VMO	VL		
SE	0%	35.77 ± 14.23	36.27 ± 14.25	-0.190	0.851
SE-SE	35%	30.43 ± 11.25	41.71 ± 18.37	-3.684	0.001*
	70%	44.02 ± 17.01	55.40 ± 23.70	-2.599	0.015*
	100%	62.14 ± 27.25	73.60 ± 35.00	-1.716	0.097
SF-SE	35%	28.15 ± 12.86	40.99 ± 17.60	-4.006	<0.001**
	70%	38.35 ± 17.21	50.55 ± 21.63	-3.526	0.001*
	100%	60.68 ± 27.32	68.36 ± 29.01	-1.365	0.183

a = paired t-test

* = significant difference at p < 0.05

** = significant difference at p < 0.001

SE = squat without hip adduction exercise

SE-SE = squeeze (hip adduction) entire time–squat exercise

SF-SE = squeeze (hip adduction) at 60 degrees knee flexion–squat exercise

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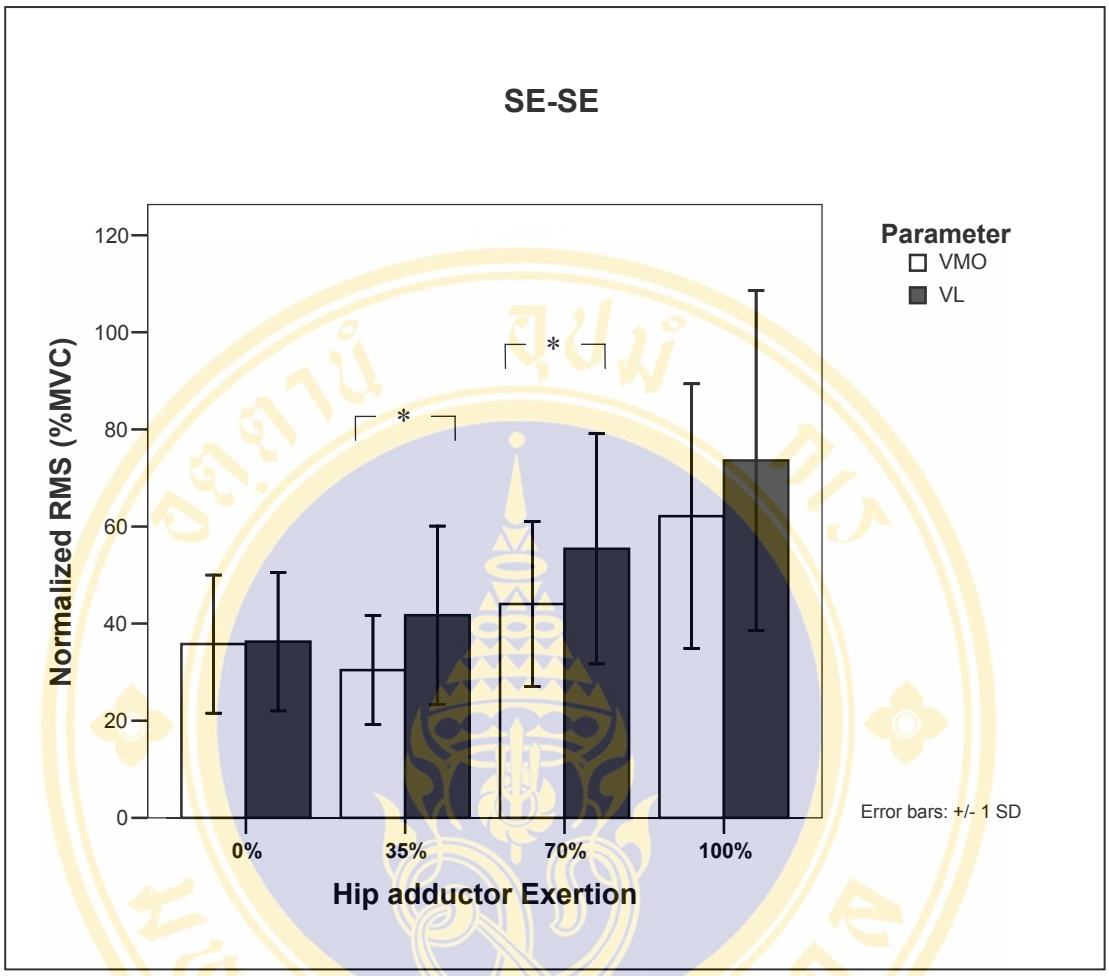


Figure 4.4 Comparison of normalized root mean square (RMS) between the vastus medialis oblique (VMO) and vastus lateralis (VL) muscles during squeeze (hip adduction) entire time-squat exercise (SE-SE) with 0%, 35%, 70% and 100% hip adductor exertion (*denoted statistically significant difference at $p < 0.05$).

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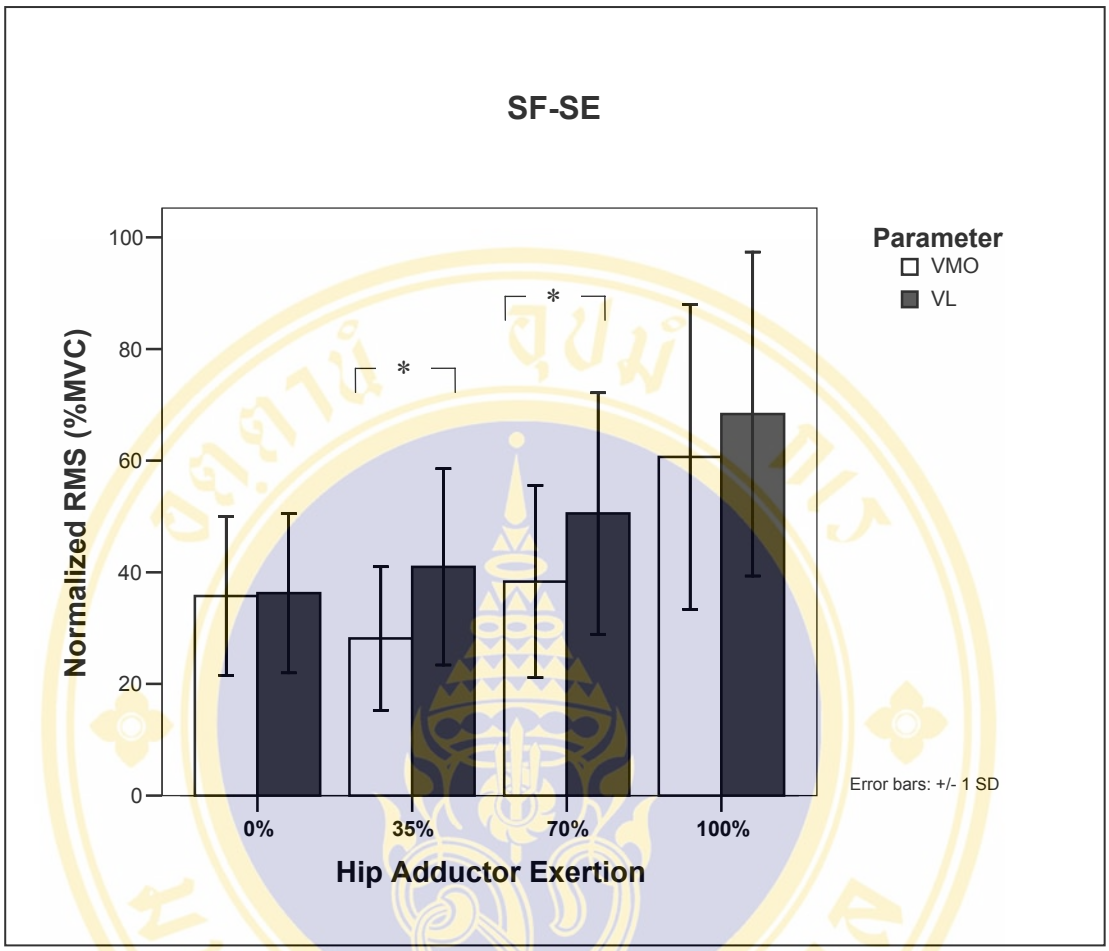


Figure 4.5 Comparison of normalized root mean square (RMS) between the vastus medialis oblique (VMO) and vastus lateralis (VL) muscles during squeeze (hip adduction) at 60 degrees knee flexion-squat exercise (SF-SE) with 0%, 35%, 70% and 100% hip adductor exertion (*denoted statistically significant difference at $p < 0.05$).

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CHAPTER V
DISCUSSION

5.1 Effects of Two Types of Exercise Sequence (SE-SE and SF-SE) on EMG Activity of the VMO and VL Muscles during Squat with Hip Adduction Exercise

Based on hypothesis that in SE-SE (hip adduction whole range of squat) the length of the VMO muscle was stretched and its origin was stabilized by hip adduction at the beginning and continuously for whole range of squat. Those effects were provided while squatting at 60 degrees knee flexion in SF-SE (hip adduction when squat at 60 degrees knee flexion). Thus, this study hypothesized that activation of the VMO and VL muscles during two exercise sequences were difference. Unexpected, the result showed no significant effects of two types of exercise sequence on activity of the VMO, VL and VMO:VL ratio. Nearly all conditions of hip adductor exertion, the VMO, VL activities and VMO:VL ratio during SE-SE were slightly but not significantly greater than that of the SF-SE. All of subjects reported that SF-SE was more comfortable than that of the SE-SE.

This study believed that SE-SE could provide greater enhancing activation of the VMO and VL muscles than during SF-SE. However, SE-SE was more difficult exercise, thus subjects provided larger effort for completing this exercise. They exerted hip adduction at the beginning and continuously for entire range of squat. This might lead to fatigue of the overall quadriceps and hip adductor muscles when squat at 60 degrees knee flexion. In addition, subjects might lean their back on the wall. Therefore, increasing activity of the VMO and VL muscle during SE-SE were faded off. This led to no significant difference of the VMO and VL activities between SE-SE and SF-SE.

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There has been no study attempt to compare between the two exercise sequences (SE-SE and SF-SE) before. Thus, comparisons of this result with others were limited. The previous studies related to squat with hip adduction exercise mostly focused on determining effect of combination of hip adduction with squat on activity of the vasti muscles. Most researches considered the activity of the VMO and VL muscles or VMO:VL ratio between the normal squat and the squat combined with hip adduction (5, 6, 8, 9, 21). The squat with hip adduction in those studies could be divided into two types of exercise sequence that performing hip adduction whole range of the squat (5, 6) and performing hip adduction when squatting at specific knee flexion (8, 21) as two types of exercise sequence in this study; SE-SE and SF-SE, respectively. Some previous studies did not describe the exercise sequence (9, 23). Cerny (1995) and Earl et al (2001) reported that the squat with maximum isometric hip adduction presented higher the VMO and VL activities than the squat without hip adduction but there was no significant difference in the VMO:VL ratio (5, 6). The squat with hip adduction exercise in those studies was executed following SE-SE. Previous studies related to SF-SE believed that the squat with isometric hip adduction could promote balances recruitment between the VMO and VL muscles (8, 21). One study found significant greater VMO:VL ratio during squat with 15%, 50% and 100% of MVC of the hip adductor muscles than squat without hip adduction (8). The other study showed no significant difference between the VMO and vastus lateralis longus (VLL) activities during squat with maximum isometric hip adduction in both healthy and PFPS but significant difference during squat without hip adduction in PFPS subjects (21).

Varied results in those previous studies might be caused by the differences in exercise sequences when subjects executed the squat with hip adduction exercise. The results from this study could answer that there were no differences between two types of exercise sequence on activity of the VMO and VL muscles.

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5.2 Effects of Hip Adductor Exertion (0%, 35%, 70% and 100%) on EMG Activity of the VMO and VL Muscles during Squat Exercise

Due to hypothesis that variability in the VMO activity during combining isometric hip adduction and knee extension exercise may be due to the difference in amount of effort each subject put into the hip adduction (6, 11). This study expected that during squat with hip adduction exercise, the activity of the VMO and VL muscles tends to increase according to hip adductor exertion. Due to hypothesis that addition of hip adduction with knee extension exercise could promote balance activity between the VMO and VL muscles which refers to the VMO:VL ratio exceed 1 or no difference between activity of the VMO and VL muscles. This study expected that the VMO:VL ratio should be exceed 1 or no difference between activity of the VMO and VL muscles under squat with each level of isometric hip adduction (35%, 70% and 100%).

The results showed greatest the VMO:VL ratio (1.06) when squat without hip adduction exercise. The outcome of paired t-test presented no significant difference between the VMO and VL activities during this exercise. Although squat without hip adduction exercise provided the VMO activity less than squat with 70% and 100% hip adductor exertion but it made minimal the VL activity. Thus, it could be indicated that squat without hip adduction does not promote lateral force acting by the VL muscle. This exercise could encourage balanced activity between the VMO and VL muscles.

Unexpected outcome showed less VMO activity but higher VL activity during squat with 35% hip adductor exertion exercise than the squat without hip adduction exercise. This resulted in the lowest VMO:VL ratio (0.76 for SF-SE and 0.80 for SE-SE) during squat with 35% hip adductor exertion exercise. The paired t-test showed strong significant difference between the VMO and VL activities under this exercise. It can be implied that the squat with 35% hip adductor exertion exercise enhanced activation of the VL muscle but not VMO muscle. Balance activation of the VMO and VL muscles were not produced with this exercise. Therefore, this exercise

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should not be applied because it may promote progressively lateral tracking of the patella.

In case of squat with 70% hip adductor exertion, both the VMO and VL activities were higher than the squat without hip adduction. Activation of the VMO muscle was enhanced less than the VL muscle thus the VMO:VL ratio during this exercise showed less value (0.85 for SF-SE and 0.87 for SE-SE) than the squat without hip adduction (1.06). It can be interpreted that the squat with 70% hip adductor exertion could promote recruitment of the VMO and VL muscles but it could not enhance balance activities between the VMO and VL muscles. Therefore, this exercise should be trained when enhanced recruitment of the VMO muscle was an aim, but the imbalance activation of the VMO and VL muscles should be carefully observed.

The squat with maximum (100%) hip adductor exertion showed the greatest VMO and VL activities; however, the VMO:VL ratio showed no significant difference from squat without hip adduction. Those findings are in accordance with the result of paired t-test that demonstrated no significant difference between the VMO and VL activities under condition of squat with maximum hip adductor exertion and squat without hip adduction. Those outcomes could be interpreted that the squat with maximum hip adductor exertion and the squat without hip adduction could encourage balance recruitment of the VMO and VL muscles. The squat with maximum hip adduction could promote greater recruitment of the VMO and VL muscles than the squat without hip adduction. Thus, this exercise should be applied to enhance recruitment of the VMO and VL muscle incorporate with balance activation between two muscles.

During squat plus hip adduction exercise, subjects required more effort for finishing the exercise when percentage of hip adductor exertion was increased. Thus, recruitment of overall quadriceps components; rectus femoris, vastus intermedius, vastus medialis, VL, were enhanced (6, 8). For these reasons, the results of this study showed increasing in VL activity as increasing percentage of hip

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adductor exertion. However, enhanced recruitment of the oblique part of the vastus medialis (VMO) was affected by mechanism of incorporated contraction of hip adductor muscles during squat (7, 8, 11, 16).

Recruitment of the VMO muscle could be enhanced by adding hip adduction during knee extension exercise. This could be explained by the anatomical relation between the adductor muscles and the VMO muscle. Due to the VMO muscle originates from the adductor muscles, contraction of the adductor muscles could provide a stable origin for the VMO muscle (7). In addition, according to the length tension properties, the VMO muscle could be stretched by adding of hip adduction which would alter length tension properties. This would enhance stronger contraction of the VMO muscle than that of with no adductor muscle contraction (16).

Hanten and Schulthies (1990) reported origin of the VMO muscle could be stabilized by strong contraction of hip adductor muscles (7). Hodges and Richardson (1993) described that high load of hip adduction could promote a functional role of the adductor muscles, which is essential action for the VMO muscle contraction (8). These information supported results of this study that activation of the VMO muscle was enhanced by squat with strong sub-maximum (70%) and maximum (100%) hip adduction but not weak sub-maximum (35%) hip adduction. Moreover, the results showed greater activity of the VMO muscle during squat with 100% hip adductor exertion than that of during squat with 70% hip adductor exertion. This indicated that under condition of squat combined with strong hip adduction, contraction of the hip adductor muscle could provide a greater magnitude of physiological stretch to the VMO muscle when hip adduction effort was increased.

This study found unexpected results that the VMO activity was lowest under squat with 35% hip adductor exertion exercise. On the other hand, squat with 35% hip adductor exertion exercise provided less the VMO activity than that of during squat without hip adduction exercise. Moreover, the VMO activity during squat without hip adduction showed less value than that of during squat with 70% and 100% hip adductor exertion. This study hypothesized that these results might be

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affected by a different type of contraction of the hip adductor muscle in each exercise. During squat with 35% hip adductor exertion, subjects usually exerted hip adduction harder than 35% hip adductor exertion at the beginning. Then they attended to release hip adduction effort down to 35% hip adductor exertion at the end. This indicated during squat at 60 degrees knee flexion, the hip adductor muscle acted as eccentric contraction, muscle was lengthened during contraction (25, 35). This study hypothesized that eccentric contraction of the hip adductor muscle during squat with 35% hip adduction might provide unstable origin for the VMO (7). In addition, this could not transfer physiological stretch to the VMO muscle (16). Thus, activation of the VMO mscle was decreased during squat with 35% hip adductor exertion.

Though during squat without hip adduction exercise, hip adduction against the elastic ball was not exerted; however, hip adductor muscle worked with isometric contraction which proposed to maintain neutral position of the hip in transverse plane (31, 100). This study hypothesized that small magnitude of isometric contraction of the hip adductor muscle could provide stable origin for the VMO during squat without hip adduction but this could not transfer physiological stretch to the VMO muscle (7, 16). Thus, activation of the VMO muscle during squat without hip adduction was greater than that of during squat with 35% hip adduction.

During squat with strong hip adduction (70% and 100% hip adductor exertion), subjects required greater effort for completing these exercise. Whole exercise period, they attempted to increase or maintain the magnitude of hip adductor exertion, which aimed for sustain 70% and 100% hip adductor exertion during exercise. This revealed that hip adductor muscle worked with concentric or isometric contraction at entire exercise time. This study hypothesized that a strong concentric or isometric contraction of the hip adductor muscle during squat with 70% and 100% hip adductor exertion could provide more stable origin for the VMO muscle (6-9). In addition, this could transfer a larger physiological stretch to the VMO muscle (6, 8, 9, 16). Therefore, during squat with 70% and 100% hip adductor exertion the VMO muscle was activated greater than during squat without hip adduction and squat with 35% hip adductor exertion.

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In summary, this study hypothesized the VMO muscle was stabilized its origin by isometric or concentric contraction but not eccentric contraction of the hip adductor muscle when perform squat combined with hip adduction exercise. Moreover, strong concentric or strong isometric contraction of the hip adductor muscle could transfer a larger stretching to the VMO muscle. This would alter the length tension properties of the VMO, thus contributing to enhanced activation of the VMO muscle.

This experiment presented greater activation of both VMO and VL muscles during squat with maximum (100%) hip adductor exertion than the squat without hip adduction exercise. Those findings are in accordance with previous studies that reported an increase in activity of the VMO and VL muscles when combined squat with maximum isometric hip adduction (5, 6, 8, 21). This investigation showed less VMO activity but higher VL activity during squat with weak sub-maximum (35%) hip adductor exertion when compared with the squat without hip adduction. The squat with strong sub-maximum (70%) hip adductor exertion showed higher in both the VMO and VL activities than the squat without hip adduction. Only one previous study reported the VMO and VL activities during squat with sub-maximum hip adduction (15% and 50% MVC), which significant higher in the VMO activity but that difference had not been found in the VL activity when compared with squat without hip adduction (8). However, their EMG data were not normalized.

Hodges and Richardson (1993) reported the VMO:VL ratio for squat with isometric hip adduction conditions compared with squat without hip adduction were statistical significant difference in all conditions (15%, 50% and 100% MVC of hip adduction). Each condition of squat with hip adduction showed greater the VMO:VL ratio than the squat without hip adduction (8). However, their study had weak point that the EMG data were not normalized. Conversely, present result and some previous researches demonstrated no significant changes in the VMO:VL ratio during squat with and without maximum isometric hip adduction (5, 6, 9). This present study found significant difference in the VMO:VL ratio of squat with each sub-maximum hip

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adduction (35% and 70%) when compared with squat without hip adduction, whereas squat with sub-maximum hip adduction showed lower VMO:VL ratio than the squat without hip adduction.

Normalization of the raw EMG is an important requirement if study compared data collected from different muscles, different individuals or different situation. Due to anthropomorphic differences between recording sites and between individuals such as thickness of subcutaneous adipose tissue, impedance of the skin, muscle resting length, muscle cross-sectional area, interelectrode distance and attachment between the electrodes and skin, might affect accuracy of those comparisons. Normalization is the technique used to control those variables (24). The most common method of normalization is the maximum voluntary isometric contraction (MVC) (24). Here, subjects were instructed to make the MVC for the muscle group of interest. Then, muscle activity during tested situation was calculated as a percentage of the MVC (%MVC). Normalization by MVC was more sensitive to contractions that require more effort (24) such as squat and squat plus hip adduction exercise as this study and previous studies (6, 9, 21, 22).

This experiment and some previous studies investigated the difference between activity of the VMO and VL muscles. This experiment and Coqueiro et al (2005) found no significant difference between activity of the VMO and VL muscles under both squat without hip adduction and squat with maximum hip adduction in healthy subjects (21). In contrary, previous studies demonstrated a significant difference between the two muscles during squat without hip adduction exercise and that difference had not been found during squat with maximum isometric hip adduction exercise in healthy (9) and PFPS subjects (21). Therefore, the squat with maximum isometric hip adduction exercise could promote a greater muscle balance between activity of the VMO and VL muscles and could be beneficial for the treatment of PFPS patients.

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5.3 Elastic Ball with Pressure Biofeedback and Quadriceps Muscles Exercise

This study used the elastic ball with pressure biofeedback in training of the VMO muscle with isometric hip adduction exercise. This equipment consists of an elastic ball, a hallow rubber tube, a valve pump, and a sphygmomanometer. They are in-house instrument that the total cost are about 400 baht. It was easily built up by connecting one end of a hallow rubber tube with a valve pump and an elastic ball and another end connect with a sphygmomanometer. The elastic ball can be used as the resistant of hip adduction. The sphygmomanometer provides feedback of hip adductor exertion (in mmHg) during exercise. Comfortable level and size of a ball could be adjusted by modify pressure into the elastic ball.

However it was observed that subject's height affected both legs position when squatting with the elastic ball between the knees. Taller subjects should place their feet closer than shorter stature subjects when they squeezed the elastic ball. All subjects in this study used the same elastic ball with the pressure biofeedback unit in order to avoid variation of the measurement and instrument. It was found that all subjects whose height between 150-180 cm could use this elastic ball (diameter 16.5 cm) without complaining of pain or uncomfortable feeling.

In the clinical setting, a towel roll, a bolster, an elastic band or a mechanical resistant are usually applied for isometric hip adduction resistant when one performs the combination of knee extension and hip adduction exercise or isolated hip adduction exercise. These tools are inconvenient and uncomfortable to use (6, 21). Excessive valgus posture of the knee occur when exercise with the small towel roll, soft bolster or weak elastic band. The elastic ball with the pressure biofeedback unit is convenient and comfortable to use. None of all subjects complained of any uncomforted symptoms by this unit. The therapist can generate objective command about hip adductor effort during exercise training.

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5.4 Clinical Implications

The squat with hip adduction exercise in this study should be applied for recovering function of the VMO muscle in individuals with PFPS who presented VMO insufficiency. Due to the PFPS has multifactor causing; (1) over use/ over load, (2) malalignment of lower extremities and/ or malalignment of the patella and (3) muscular dysfunction such as decreasing activation of the VMO muscle, increasing activation of the VL muscle and decreasing flexibility of the iliotibial band, etc (1, 13, 42). Before applying squat combined with hip adduction exercise, insufficiency of the VMO muscle should be indicated. Moreover, applying this exercise should be proposed to prevent patellofemoral dysfunction associated with VMO insufficiency in individuals with predisposing factors such as excessive foot pronation, high Q angle, genu valgum, genu varum, genu recurvatum, femoral neck anti version, patellar alta, patellar infera, general quadriceps muscle weakness and iliotibial band tightness, etc.

In clinical setting, exercise for retraining the VMO should focus on increasing the VMO recruitment. Moreover, recruitment of the VL should be less enhanced during training. This is due to the VMO and VL muscles function as dynamics balance of the patellofemoral joint. The VMO muscle is only dynamic medial stabilizer which counteracts the lateral force produced by the VL muscle and the iliotibial band in order to maintain normal tracking of the patella (1, 2, 4, 31). In general, insufficiency of the VMO muscle is frequent association cause of lateral tracking of the patella (1, 38).

Based on the results of this study, the squat with hip adduction could be trained with any exercise sequences, SE-SE or SF-SE. Due to the SF-SE is easier and more comfortable exercise, thus it should be applied in initial training period which for familiarization with squat plus hip adduction exercise. The symptomatic PFPS should be trained with SF-SE first which proposed to avoid pain aggravation. Due to the VMO, VL activities and VMO:VL ratio during SE-SE seemed slightly greater than SF-SE. However, this sequence is harder and less comfortable exercise. Thus, applying the SE-SE should aim for increasing performance of the VMO muscle in

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healthy individuals, athlete, asymptomatic PFPS or last training period in symptomatic PFPS.

Applying of squat with hip adduction exercise for training the VMO muscle should start with squat without hip adduction. The aim is to train balance activities of the VMO and VL muscles and to prepare the muscles to perform the squat with isometric hip adduction. The training should progress to squat with 70% hip adductor exertion for increasing recruitment of the VMO muscle. This exercise is easy to perform and is proposed to avoid fatigue and pain aggravation (when compare with squat with maximum hip adduction). Though this exercise could increase the VMO activation, but the imbalance activation of the VMO and VL muscles should be carefully observed. Finally, the training should progress to squat with maximum hip adductor exertion for increasing recruitment of the VMO muscle and promote balance activity of the VMO and VL muscles. It is noted that the squat with 35% hip adductor exertion should not be trained. This exercise induces low VMO recruitment and imbalance activity between the VMO and VL muscles. This may promote lateral tracking of the patella which is one of abnormal movement of patella.

This study and some previous studies found an increasing in both the VMO and VL activities during squat with sub-maximum (8) and maximum isometric hip adduction (5, 6, 21). This was implied that these exercises could enhance strengthening of the overall quadriceps muscle. According to results of this study, the squat combined with 70% and 100% hip adductor exertion should be set into exercise program for strengthening the overall quadriceps muscle.

Using the elastic ball with pressure biofeedback for training the VMO muscle or overall quadriceps muscle by combining knee extension with isometric hip adduction exercise, training hip adductor muscles by isolated hip adduction exercise is recommended. This unit could be applied in other exercises which require feedback during training. It could be used for creating progressive exercise programs and evaluate progression of clients. This instrument is recommended in physical therapy clinics, sport clubs or at homes.

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5.5 Limitation of the Study

This experiment examined the magnitude of the EMG but did not investigate timing parameters. Two exercise sequences and hip adductor exertions may influence the onset of VMO and VL. However, the testing protocol in this study did not designed to obtain the onset data.

This study tested in healthy subjects thus the results was limited for generalization to the group with no knee pain. Individuals with anterior knee pain were excluded from this study. This is due to many exercises testing in this study may aggravate pain. In addition, the maximum effort of the subjects may be limited by pain. However, results from this study should be used to select the exercise conditions and create exercise program for training the VMO muscle. This aims to prevent patellofemoral dysfunction associated with VMO insufficiency in individuals with predisposing factors.

The subjects in this study were not trained thus the results were restricted to answer that the exercises could retrain insufficiency of the VMO muscle. However, results of this study could be used to create the exercise program for further study, which should purpose to examine effects of exercise training for recovering the VMO insufficiency.

5.6 Further Study

Further investigation should purpose to test the effects of training program base on the squat without hip adduction progressively to the squat with 70% and 100% hip adductor exertion in individuals with patellofemoral dysfunction associated with VMO insufficiency. The training program could modify any exercise sequences, performing hip adduction whole range of the squat or performing hip adduction when squatting at specific knee flexion. The elastic ball with pressure biofeedback should be used for training.

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In electromyographic study the electrodes placement of the VMO and VL muscles should be applied while the tested knee is in a slightly flexed position. Electrodes placement position of all electrodes should be rechecked whether they remained over the muscle bellies during exercise. In order to avoid proximal migration of the VMO electrodes when knee flexion is increased during stand to squat (6).

Every study proposed to investigate EMG activity of the VMO and VL muscles during squatting with hip adduction exercise should concern on pressure effect on EMG activities. While subjects adduct their hip against an elastic ball, the elastic ball may press over the VMO electrodes which will affect the muscle activities. The EMG should be recorded while the investigator presses the elastic ball over the electrodes during relaxation. If there will be any change of muscle activity, the electrodes placement should be redone until no muscle activity occurs.

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CHAPTER VI
CONCLUSION

This study was purposed to examine the effects of hip adductor exertion and two exercise sequences on EMG activity of the VMO and VL muscles during squatting exercise in healthy individuals. The result showed as follows.

1. The two exercise sequences that performing hip adduction whole range of the squat and performing hip adduction when squatting at 60 degrees knee flexion showed no differences in activity of the VMO and VL muscles.
2. The hip adductor exertion that 0%, 35%, 70% and 100% showed significant effects on activity of the VMO and VL muscles. The squat with 35% hip adductor exertion presented small VMO activity but high in VL activity than the squat without hip adduction, which was the most imbalanced activity between two muscles. The squat with 70% hip adductor exertion provided higher VMO and VL activities than the squat without hip adduction but activity of two muscles were still in the imbalanced condtion. The squat with maximum hip adduction presented highest in the VMO and VL activities, which showed balance activity between two muscles. The squat without hip adduction created less in the VMO and VL activities but activity of two muscles was balanced.
3. An elastic ball with pressure biofeedback is easy and comfortable to use, which provide feedback during exercise. It consists of inexpensive in-house instruments. It works with the squat with isometric hip adduction training, which should be recommended for physical therapy clinics, sport clubs or at homes.

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The squat with isometric hip adduction could be applied by any exercise sequences (performing hip adduction whole range of the squat or performing hip adduction when squatting at 60 degrees knee flexion). Activation of the VMO and VL muscles should be considered before selecting the hip adduction effort when ones apply the squat plus hip adduction exercise.



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APPENDIX

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APPENDIX A
THE ETHICAL COMMITTEE ON RESEARCH INVOLVING HUMAN SUBJECT



COA. No. MU-IRB 2008/038.1706

Documentary Proof of Mahidol University Institutional Review Board

Title of Project: Effects of Hip Adductor Exertions and Two Exercise Sequences on Vastus Medialis Oblique (VMO) and Vastus Lateralis (VL) EMG Activities during Squat Exercises
(Thesis for Master Degree)

Principle Investigator: Miss Sirirat Kaewwichai

Name of Institution: Faculty of Physical Therapy and Applied Movement Sciences

This is to certify That Mahidol University Institutional Review Board is in full compliance with International Guidelines for Human Research Protection such as Declaration of Helsinki, The Belmont Report, CIOMS Guidelines and the International Conference on Harmonization in Good Clinical Practice (ICH-GCP)

Date of Approval: 17 June 2008

Date of Expiration: 16 June 2009

Signature of Chairman: .....
(Professor Shusee Visalyaputra)

Signature of Head of the Institute: .....
(Associate Professor Sansanee Chaiyaroj)
Vice President for Research and Academic Affairs

Office of the President, Mahidol University, 999 Phuttamonthon 4 Rd., Salaya, Phuttamonthon District, Nakhon Pathom 73170. Tel. (662) 8496241-6 Fax. (662) 8496247

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APPENDIX B.1
PARTICIPANT INFORMATION SHEET
(เอกสารชี้แจงอาสาสมัครผู้เข้าร่วมการวิจัย)

ในเอกสารนี้อาจมีข้อความที่ท่านอ่าน แล้วยังไม่เข้าใจ โปรดสอบถามหัวหน้าโครงการวิจัย หรือผู้แทนให้ช่วยอธิบายจนกว่าจะเข้าใจดี ท่านอาจจะขอเอกสารนี้กลับไปอ่านที่บ้านเพื่อปรึกษาหารือกับญาติพี่น้อง เพื่อนสนิท แพทย์ประจำตัวของท่าน หรือแพทย์ท่านอื่น เพื่อช่วยในการตัดสินใจเข้าร่วมการวิจัย

ชื่อโครงการวิจัย ผลของแรงหุบสะโพกและลำดับการหุบสะโพกสองแบบต่อคลื่นไฟฟ้ากล้ามเนื้อ vastus medialis oblique (VMO) และ vastus lateralis (VL) ขณะออกกำลังกายท่าย่อเข่า
Effects of hip adductor exertions and two exercise sequences on vastus medialis oblique (VMO) and vastus lateralis EMG activities during squat exercises

ชื่อหัวหน้าโครงการ นางสาวศิริรัตน์ แก้ววิชัย

สถานที่วิจัย ห้องวิจัย 602 ชั้น 6 คณะกายภาพบำบัดและวิทยาศาสตร์การเคลื่อนไหว ประยุกต์ มหาวิทยาลัยมหิดล ศาลายา เลขที่ 999 ถ.พุทธมณฑลสาย 4 ต.ศาลายา อ.พุทธมณฑล จ.นครปฐม 73170

ผู้ให้ทุน ไม่มี

โครงการวิจัยนี้จัดทำขึ้นเพื่อศึกษาผลของแรงหุบสะโพกและลำดับการหุบสะโพกสองแบบต่อคลื่นไฟฟ้ากล้ามเนื้อหน้าขาด้านในมัดเฉียง (vastus medialis oblique; VMO) และกล้ามเนื้อหน้าขาด้านนอก (vastus lateralis; VL) เมื่อหุบสะโพกด้วยแรงต่างกันและลำดับต่างกัน ขณะออกกำลังกายท่าย่อเข่า ในผู้ที่มีสุขภาพดี เพื่อให้ทราบแรงหุบสะโพกและลำดับการหุบสะโพกที่

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สามารถเพิ่มการทำงานของกล้ามเนื้อหน้าขาด้านในมัดเฉียง (VMO) และส่งเสริมความสมดุลระหว่างกล้ามเนื้อหน้าขาด้านในมัดเฉียง (VMO) และกล้ามหน้าขาด้านนอก (VL) ขณะออกกำลังกายท่าย่อเข่าร่วมกับหุบสะโพก อีกทั้งเพื่อทดลองใช้อุปกรณ์ชนิดใหม่ คือ ลูกบอลพลาสติกประยุกต์เข้ากับเครื่องวัดความโลหิต ในการออกกำลังกายดังกล่าว

โครงการวิจัยนี้จะมีผู้เข้าร่วมการวิจัยทั้งสิ้น 30 คน ระยะเวลาที่ใช้ในการวิจัยประมาณ 1 ปี

ท่านได้รับเชิญเข้าเป็นอาสาสมัครในการศึกษานี้ เนื่องจากท่านเป็นผู้มีสุขภาพดีและไม่มีปัญหาทางระบบกระดูกและกล้ามเนื้อในบริเวณเข่าหรือบริเวณตั้งแต่สะโพกถึงปลายเท้า (ระยางค์ส่วนล่าง)

เมื่อท่านเข้าร่วมการวิจัยแล้วสิ่งที่ท่านหรือเอกสารฉบับนี้เรียกว่า "ผู้เข้าร่วมวิจัย" จะต้องปฏิบัติตามขั้นตอนการดำเนินการวิจัย ดังต่อไปนี้

การเก็บข้อมูลผู้เข้าร่วมวิจัยแต่ละท่านใช้เวลาทั้งสิ้นประมาณ 60 นาที (ขั้นตอนการเตรียมการ 15 นาที ขั้นตอนการเก็บข้อมูล 45 นาที)

1) ขั้นตอนการเตรียมการ

อาสาสมัครรับการสัมภาษณ์เกี่ยวกับข้อมูลทั่วไปและข้อมูลด้านสุขภาพ ตามแบบสัมภาษณ์ข้อมูลผู้เข้าร่วมวิจัย โดยใช้เวลาประมาณ 5 นาที จากนั้นรับการตรวจความตึงตัวของกล้ามเนื้อหุบสะโพกและกล้ามเนื้อกางสะโพกทั้ง 2 ข้าง ข้างละ 1 ครั้ง ด้วยวิธีทางกายภาพบำบัด (Hip adduction contracture test และ Ober's test) และประเมินการผิดรูปของข้อเข่าทั้ง 2 ข้าง โดยใช้สายวัดและไม้วัดองศาการเคลื่อนไหววัดมุมข้อเข่า ขณะผู้ร่วมวิจัยยืนตรง เข่าและเท้าทั้งสองข้างชิดกัน โดยส่วนนี้ใช้เวลาประมาณ 10 นาที ทั้งนี้เพื่อคัดอาสาสมัครเข้าร่วมการศึกษา ซึ่งต่อไปนี้จะเรียกว่า "ผู้เข้าร่วมวิจัย"

2) ขั้นตอนการเก็บข้อมูล

ผู้เข้าร่วมวิจัยจะได้รับการตรวจคลื่นไฟฟ้ากล้ามเนื้อหน้าขาด้านในมัดเฉียง (VMO) และกล้ามเนื้อหน้าขาด้านนอก (VL) ของขาข้างถนัด ขณะย่อเข่าพร้อมกับหุบสะโพกด้วยแรงและลำดับต่าง ๆ กัน ตามรูปแบบการหุบสะโพก 2 แบบ ด้วยวิธีติดขั้วไฟฟ้าที่ผิวหนัง

ก่อนการตรวจวัดผู้เข้าร่วมวิจัยอบอุ่นร่างกายด้วยการยืดกล้ามเนื้อ ดังนี้ กล้ามเนื้อเหยียดเข่า (quadriceps) กล้ามเนื้อต้นขาด้านหลัง (hamstring) กล้ามเนื้อกางขา (iliotibial band) กล้ามเนื้อหุบสะโพก (hip adductor) และกล้ามเนื้อน่อง (calf muscles) จากนั้นจะได้รับการอธิบาย สาธิตขั้นตอน และฝึกออกกำลังกายจนถูกต้องและคุ้นเคย

ผู้เข้าร่วมวิจัยรับการทดสอบแรงหุบสะโพกมากที่สุด เพื่อนำไปคำนวณ ร้อยละ 35 และร้อยละ 70 ของแรงหุบสะโพกมากที่สุด ซึ่งแรงหุบสะโพกมากที่สุดถูกวัดเป็นแรงกดโดยเครื่องวัด

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ความดันโลหิตที่นำมาประยุกต์เข้ากับลูกบอลพลาสติก ท่าเริ่มต้นการทดสอบคือ ยืนตรงหลังแตะผนัง มือวางผสานกันบนหน้าอก เท้าวางห่างกันเท่ากับความกว้างช่วงสะโพก ปลายเท้าชี้ตรงไปข้างหน้า ส้นเท้าอยู่ห่างจากกำแพงประมาณ 10 เซนติเมตร จากท่าเริ่มต้นผู้เข้าร่วมวิจัยย่อเข่าลงตรง ๆ จนเข่างอ 60 องศา (วัดโดยอุปกรณ์วัดองศาการเคลื่อนไหว) แล้วหุบสะโพกเข้าหากันด้วยแรงมากที่สุดค้างไว้ 2 วินาที ผู้วิจัยกระตุ้นด้วยเสียงให้ออกแรงเต็มที่ (เตรียมพร้อม! ระวัง! เริ่ม! หนีบ! หนีบ! หนีบ! ผ่อนคลาย) และตลอดการทดสอบผู้เข้าร่วมวิจัยควบคุมส่วนต่าง ๆ ดังกล่าวให้แตะกับผนังเท่านั้น ผู้เข้าร่วมวิจัยต้องทดลองทำการทดสอบจนคุ้นเคย และพักอย่างน้อย 2 นาที ก่อนการเก็บข้อมูลจริง การทดสอบแรงหุบสะโพกมากที่สุดเก็บข้อมูล 3 ครั้ง พักระหว่างครั้งอย่างน้อย 2 นาที

ผู้เข้าร่วมวิจัยรับการโกนขนด้วยมีดโกนแบบใช้แล้วทิ้ง ทำความสะอาดผิวหนัง และเช็ดด้วยแอลกอฮอล์บริเวณที่จะติดขั้วไฟฟ้า ตำแหน่งขั้วไฟฟ้ากล้ามเนื้อหน้าขาด้านในมัดเฉียง (VMO) คือ เหนือขอบบนของกระดูกสะบ้า 3 เซนติเมตร และในต่อขอบบนด้านใน 4 เซนติเมตร วางทำมุมกับแนวต้นขา 55 องศา (ขนานกับใยกล้ามเนื้อ) ตำแหน่งขั้วไฟฟ้ากล้ามเนื้อหน้าขาด้านนอก (VL) คือ เหนือขอบบนของกระดูกสะบ้า 10 เซนติเมตร และห่างจากเส้นที่ลากผ่านจุดกึ่งกลางกระดูกสะบ้าและปุ่มกระดูกเชิงกรานด้านหน้า (ASIS) ไปด้านนอก 6-8 เซนติเมตร วางทำมุม 15 องศากับเส้นดังกล่าว (พิจารณาร่วมกับการคลำกล้ามเนื้อ เมื่อผู้เข้าร่วมวิจัยเกร็งกล้ามเนื้อเหยียดเข่า) (24, 99) ระยะระหว่างขั้วไฟฟ้า 2.2 เซนติเมตร ขั้วอ้างอิงติดที่ต้นขาด้านหน้า ขั้วทั้งหมดติดทับด้วยกระดาษกาวและพันด้วยผ้ายืดเพื่อป้องกันการเคลื่อนและให้ขั้วไฟฟ้าแนบสนิทกับผิวหนัง ทั้งนี้การติดขั้วไฟฟ้ากล้ามเนื้อจะกระทำขณะผู้เข้าร่วมวิจัยยืนงอเข่าเล็กน้อย

จากนั้นผู้เข้าร่วมวิจัยรับการวัดคลื่นไฟฟ้ากล้ามเนื้อหน้าขาด้านในมัดเฉียง (VMO) และกล้ามเนื้อหน้าขาด้านนอก (VL) ขณะออกแรงเหยียดเข่าเต็มที่ (maximum voluntary isometric contraction; MVIC) เพื่อนำไปคำนวณค่ามาตรฐานคลื่นไฟฟ้ากล้ามเนื้อ (normalization) ท่าเริ่มต้นการทดสอบคือ ผู้เข้าร่วมวิจัยนั่งบนเก้าอี้ทดสอบในท่างอสะโพก 90 องศา งอเข่า 60 องศา ผ้าพันลำตัวบริเวณเชิงกรานและรอบต้นขาติดกับเก้าอี้ทดสอบ เพื่อความปลอดภัยและป้องกันการเคลื่อนไหวที่ไม่ต้องการ ผ้าอีกผืนพันที่ขาท่อนล่างเหนือปุ่มกระดูกข้อเท้าด้านใน 2.5 เซนติเมตร เพื่อต้านการเหยียดเข่า ผู้เข้าร่วมวิจัยทดลองออกแรงเหยียดเข่า 4 ครั้ง พักอย่างน้อย 2 นาทีก่อนเก็บข้อมูลจริง การเก็บข้อมูลผู้เข้าร่วมวิจัยออกแรง เหยียดเข่าด้วยแรงมากที่สุดค้างไว้ 2 วินาที ผู้วิจัยกระตุ้นด้วยเสียงให้ออกแรงเต็มที่ (เตรียมพร้อม! ระวัง! เริ่ม! เต็มที่! เต็มที่! เต็มที่! ผ่อนคลาย) พร้อมทั้งเก็บข้อมูลคลื่นไฟฟ้ากล้ามเนื้อตั้งแต่เริ่มต้น การเก็บข้อมูลส่วนนี้ทำ 3 ครั้ง พักระหว่างครั้งอย่างน้อย 2 นาที ค่าเฉลี่ยของทั้ง 2 ครั้งที่คลื่นไฟฟ้ากล้ามเนื้อคงที่ที่สุดจะถูกนำไปใช้

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ผู้เข้าร่วมวิจัยรับการวัดคลื่นไฟฟ้ากล้ามเนื้อขณะออกกำลังกาย 7 แบบ ได้แก่

1) ย่อเข่าโดยไม่หุบสะโพก
2) ย่อเข่าร่วมกับหุบสะโพกด้วยแรงร้อยละ 35 โดยหุบสะโพกตลอดเวลา
3) ย่อเข่าร่วมกับหุบสะโพกด้วยแรงร้อยละ 70 โดยหุบสะโพกตลอดเวลา
4) ย่อเข่าร่วมกับหุบสะโพกด้วยแรงมากที่สุด โดยหุบสะโพกตลอดเวลา
5) ย่อเข่าร่วมกับหุบสะโพกด้วยแรงร้อยละ 35 โดยย่อเข่าที่มุม 60 องศาแล้วหุบสะโพก
6) ย่อเข่าร่วมกับหุบสะโพกด้วยแรงร้อยละ 70 โดยย่อเข่าที่มุม 60 องศาแล้วหุบสะโพก
7) ย่อเข่าร่วมกับหุบสะโพกด้วยแรงมากที่สุด โดยย่อเข่าที่มุม 60 องศาแล้วหุบสะโพก

การออกกำลังกายย่อเข่าร่วมกับหุบสะโพกโดยหุบสะโพกตลอดเวลา ผู้เข้าร่วมวิจัยยืนในท่าเริ่มต้น (ท่าเริ่มต้นเดียวกับการทดสอบแรงหุบสะโพกมากที่สุด) หุบสะโพกเข้าหากัน ต้านกับบอลพลาสติกที่อยู่ระหว่างต้นขาด้วยแรงที่กำหนด (ผู้เข้าร่วมวิจัยทราบแรงหุบสะโพกโดยตรงจากเครื่องวัดความดันโลหิต) และย่อเข่าลงจนงอ 60 องศา ค้างไว้ 2 วินาที จากนั้นลุกขึ้นสู่ท่าเริ่มต้น

การออกกำลังกายย่อเข่าร่วมกับหุบสะโพกโดยย่อเข่าที่มุม 60 องศา แล้วหุบสะโพก ผู้เข้าร่วมวิจัยยืนในท่าเริ่มต้น ย่อเข่าจนงอ 60 องศา แล้วหุบสะโพกเข้าหากันด้วยแรงที่กำหนด (ผู้เข้าร่วมวิจัยทราบแรงหุบสะโพกโดยตรงจากเครื่องวัดความดันโลหิต) ค้างไว้ 2 วินาที จากนั้นลุกขึ้นสู่ท่าเริ่มต้น

การออกกำลังย่อเข่าโดยไม่หุบสะโพก ผู้เข้าร่วมวิจัยยืนในท่าเริ่มต้นจากนั้นย่อเข่าจนงอ 60 องศา ค้างไว้ 2 วินาที จากนั้นลุกขึ้นสู่ท่าเริ่มต้น

การออกกำลังกายแต่ละแบบเก็บข้อมูล 3 รอบ ระหว่างรอบพักอย่างน้อย 2 นาที ค่าเฉลี่ยของ 2 ครั้งที่คลื่นไฟฟ้ากล้ามเนื้อคงที่ที่สุดจะถูกนำไปคำนวณ และวิเคราะห์ทางสถิติ

เนื่องจากการวัดคลื่นไฟฟ้ากล้ามเนื้อด้วยวิธีติดขั้วไฟฟ้าที่ผิวหนังมีการนำมาใช้อย่างแพร่หลายในการวิจัย และไม่มีการรายงานอันตรายที่เกิดขึ้นในการวิจัยที่วัดคลื่นไฟฟ้ากล้ามเนื้อ แต่อย่างไรก็ตามผู้เข้าร่วมวิจัยอาจมีอาการแพ้ของผิวหนังบริเวณติดขั้วไฟฟ้า ดังนั้นหลังจากติดขั้วไฟฟ้าผู้วิจัยจะรอสังเกตอาการแพ้ของผิวหนังบริเวณติดขั้วไฟฟ้าก่อน ซึ่งหากผู้เข้าร่วมวิจัยเริ่มมีอาการแพ้ เช่น แสบ คัน หรือรู้สึกไม่สบายบริเวณผิวหนังใต้ขั้วไฟฟ้า ผู้วิจัยจะนำขั้วไฟฟ้าออก ทำความสะอาดผิวหนัง เช็ดแอลกอฮอล์ และหยุดการวิจัยทันที นอกจากนี้ผู้เข้าร่วมวิจัยอาจมีความเมื่อยล้าในการออกกำลังเพื่อวัดคลื่นไฟฟ้ากล้ามเนื้อ ผู้วิจัยจะป้องกันโดยให้มีการอบอุ่นร่างกาย ยืดกล้ามเนื้อที่เกี่ยวข้องก่อนการตรวจวัด และจัดให้พักระหว่างการตรวจวัดอย่างเพียงพอ หากผู้เข้าร่วมวิจัยมีความรู้สึกไม่สบายเกิดขึ้นที่ส่วนใดส่วนหนึ่งของร่างกาย ผู้วิจัยจะหยุดการวิจัยทันที

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และให้นอนพัก โดยผู้วิจัยจะคอยสังเกตอาการจนกว่าอาการผิดปกตินั้นจะหายไปและไม่อนุญาตให้ผู้เข้าร่วมวิจัยท่านนั้นเข้าร่วมโครงการวิจัยต่อไป

หากผู้เข้าร่วมวิจัยมีปัญหาหรือข้อสงสัยใด ๆ ที่เกี่ยวข้องกับการวิจัยนี้ สามารถติดต่อผู้วิจัยตามที่อยู่และหมายเลขติดต่อ ดังนี้ นางสาวศิริรัตน์ แก้ววิชัย คณะกายภาพบำบัดและวิทยาศาสตร์-การเคลื่อนไหวประยุกต์ มหาวิทยาลัยมหิดล ศาลายา ชั้น 2 เลขที่ 999 ถ.พุทธมณฑลสาย 4 ต. ศาลายา อ.พุทธมณฑล จ.นครปฐม 73170 โทรศัพท์ 0-2441-5450 ในวันและเวลาราชการ หรือ 08-6095-6616 ได้ทุกเวลา

การวิจัยนี้ไม่มีค่าตอบแทนที่จะได้รับ รวมถึงค่าเดินทาง และไม่มีค่าใช้จ่ายใด ๆ ที่ผู้เข้าร่วมการวิจัยจะต้องรับผิดชอบ

หากมีข้อมูลเพิ่มเติมทั้งด้านประโยชน์และโทษที่เกี่ยวข้องกับการวิจัยนี้ ผู้วิจัยจะแจ้งให้ทราบโดยรวดเร็วไม่ปิดบัง

ข้อมูลส่วนตัวของผู้เข้าร่วมวิจัยจะถูกเก็บรักษาไว้ ไม่เปิดเผยต่อสาธารณะเป็นรายบุคคล แต่จะรายงานผลการวิจัยเป็นข้อมูลส่วนรวม ข้อมูลของผู้เข้าร่วมวิจัยเป็นรายบุคคลอาจมีคณะบุคคลบางกลุ่มเข้ามาตรวจสอบได้ เช่น ผู้ให้ทุนวิจัย สถาบันหรือองค์กรของรัฐที่มีหน้าที่ตรวจสอบ คณะกรรมการจริยธรรมฯ เป็นต้น

ประโยชน์ที่คาดว่าจะเกิดขึ้นต่อผู้ยินยอมตนให้ทำการวิจัยและต่อผู้อื่น

1) เพื่อที่จะได้ค่าแรงหุบสะโพกและหาลำดับการหุบสะโพกที่เหมาะสม ขณะออกกำลังกายท่าย่อเข่าร่วมกับหุบสะโพก ซึ่งเป็นการออกกำลังกายเพื่อเพิ่มการทำงานของกล้ามเนื้อหน้าขาด้านในมัดเฉียง (VMO) และส่งเสริมความสมดุลระหว่างกล้ามเนื้อหน้าขาด้านในมัดเฉียง (VMO) และกล้ามเนื้อหน้าขาด้านนอก (VL) ในผู้ที่กล้ามเนื้อหน้าขาด้านในมัดเฉียง (VMO) อ่อนแรง เช่น กลุ่มอาการปวดเข่าทางด้านหน้าที่มีพยาธิสภาพของข้อต่อกระดูกต้นขาและกระดูกสะบ้า ผู้ที่มีการบาดเจ็บเรื้อรังของกล้ามเนื้อหรือเอ็นข้อเข่า กล้ามเนื้อเหยียดเข่าอ่อนแรง เข่าเสื่อม เด็กพิการทางสมอง ผู้ป่วยอัมพาตครึ่งซีก หรือ พากินสัน เป็นต้น

2) เพื่อนำเสนอการใช้อุปกรณ์ชนิดใหม่ คือ ลูกบอลพลาสติกประยุกต์เข้ากับเครื่องวัดความ-ดันโลหิต ซึ่งเป็นอุปกรณ์ที่ใช้ง่าย สามารถประดิษฐ์ได้เองโดยใช้อุปกรณ์ที่หาซื้อได้ง่ายและราคาไม่แพง ในการออกกำลังกายดังกล่าว

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ผู้เข้าร่วมการวิจัยมีสิทธิ์ถอนตัวออกจากโครงการวิจัยเมื่อใดก็ได้ โดยไม่ต้องแจ้งให้ทราบล่วงหน้าและการไม่เข้าร่วมการวิจัยหรือถอนตัวออกจากโครงการวิจัยนี้จะไม่มีผลกระทบต่อการบริการและการรักษาที่สมควรจะได้รับแต่ประการใด

ข้าพเจ้าได้อ่านรายละเอียดในเอกสารนี้ครบถ้วนแล้ว

ลงชื่อ...../วันที่.....

(……)

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APPENDIX B.2
INFORMED CONSENT FORM
(หนังสือแสดงเจตนายินยอมเข้าร่วมโครงการการวิจัย)

วันที่.....เดือน.....พ.ศ. .....

ข้าพเจ้า..... อายุ.....ปี

อาศัยอยู่บ้านเลขที่.....ถนน.....แขวง/ตำบล.....

เขต/อำเภอ.....จังหวัด.....รหัสไปรษณีย์.....

โทรศัพท์.....

ขอแสดงเจตนายินยอมเข้าร่วมโครงการวิจัย เรื่อง “ผลของแรงหุบสะโพกและลำดับการหุบสะโพกสองแบบต่อคลื่นไฟฟ้ากล้ามเนื้อ vastus medialis oblique (VMO) และ vastus lateralis (VL) ขณะออกกำลังกายท่าย่อเข่า” (Effects of hip adductor exertions and two exercise sequences on vastus medialis oblique (VMO) and vastus lateralis (VL) EMG activities during squat exercises)

ก่อนที่จะลงนามในใบยินยอมให้ทำการวิจัยนี้ ข้าพเจ้าได้รับการอธิบายจากผู้วิจัยถึงวัตถุประสงค์ของการวิจัย วิธีการวิจัย อันตราย หรืออาการที่อาจเกิดขึ้นจากการวิจัย รวมทั้งประโยชน์ที่จะเกิดขึ้นจากการวิจัยอย่างละเอียด และมีความเข้าใจดีแล้ว

หากข้าพเจ้ามีข้อข้องใจเกี่ยวกับขั้นตอนของการวิจัย หรือหากเกิดผลข้างเคียงที่ไม่พึงประสงค์จากการวิจัยขึ้นกับข้าพเจ้า ข้าพเจ้าจะสามารถติดต่อกับ นางสาวศิริรัตน์ แก้ววิชัย เบอร์โทรศัพท์ 08-6095-6616 ได้ทุกเวลา

ผู้วิจัยรับรองว่าจะตอบคำถามต่าง ๆ ที่ข้าพเจ้าสงสัยด้วยความเต็มใจ ไม่ปิดบังซ่อนเร้น จนข้าพเจ้าพอใจ

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ข้าพเจ้ามีสิทธิที่จะบอกเลิกการเข้าร่วมโครงการวิจัยนี้เมื่อใดก็ได้ และเข้าร่วมโครงการวิจัยนี้ โดยสมัครใจ และการบอกเลิกการเข้าร่วมการวิจัยนี้ จะไม่มีผลต่อการรักษาโรคที่ข้าพเจ้าจะพึงได้รับต่อไป

ผู้วิจัยรับรองว่าจะเก็บข้อมูลเฉพาะที่เกี่ยวกับตัวข้าพเจ้าเป็นความลับ และจะเปิดเผยได้เฉพาะในรูปที่สรุปผลการวิจัย การเปิดเผยข้อมูลเกี่ยวกับตัวข้าพเจ้าต่อหน่วยงานต่าง ๆ ที่เกี่ยวข้องกระทำได้เฉพาะกรณีจำเป็นด้วยเหตุผลทางวิชาการเท่านั้น

ผู้วิจัยรับรองว่าหากเกิดอันตรายใด ๆ จากการวิจัยดังกล่าวข้าพเจ้าจะได้รับการรักษาพยาบาล โดยไม่คิดมูลค่าตามมาตรฐานวิชาชีพ และจะได้รับการชดเชยรายได้ที่สูญเสียไประหว่างการรักษาพยาบาลดังกล่าว ตลอดจนเงินทดแทนความพิการที่อาจเกิดขึ้น

ผู้วิจัยรับรองว่าหากมีข้อมูลเพิ่มเติมที่ส่งผลกระทบต่อการวิจัย ข้าพเจ้าจะได้รับการแจ้งให้ทราบ โดยไม่ปิดบังซ่อนเร้น

ข้าพเจ้าได้อ่านข้อความข้างต้นแล้ว มีความเข้าใจดีทุกประการ และได้ลงนามในใบยินยอมนี้ด้วยความเต็มใจ

ในกรณีที่ผู้ยินยอมตนให้ทำการวิจัยไม่สามารถอ่านหนังสือได้ จะต้องได้รับการยินยอมในขณะที่ยังมีสติสัมปชัญญะ และระบุข้อความไว้ตามนี้ ข้าพเจ้าไม่สามารถอ่านหนังสือได้ แต่ผู้วิจัยได้อ่าน ข้อความในใบยินยอมนี้ให้แก่ข้าพเจ้าฟังจนเข้าใจดีแล้ว ข้าพเจ้าจึงลงนาม หรือประทับลายนิ้วแม่มือขวาของข้าพเจ้าในใบยินยอมนี้ด้วยความเต็มใจ

ลงนาม ..... ผู้ยินยอม

ลงนาม ..... พยาน

ลงนาม ..... พยาน

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APPENDIX C.1
SCREENING QUESTIONNAIRE (IN THAI)
(แบบสัมภาษณ์ข้อมูลผู้เข้าร่วมการวิจัย)

เลขที่.....

ส่วนที่ 1 ข้อมูลทั่วไปของผู้เข้าร่วมวิจัย

เพศ ( ) ชาย ( ) หญิง วัน-เดือน-ปี เกิด..... อายุ..... ปี.....เดือน
น้ำหนัก..... กิโลกรัม ส่วนสูง..... เซนติเมตร ขาข้างที่ถนัด ( ) ซ้าย ( ) ขวา
ที่อยู่.....
โทรศัพท์..... E-mail address.....

ส่วนที่ 2 ข้อมูลด้านสุขภาพ

1. ขณะนี้ท่านมีปัญหาทางด้านสุขภาพ และ/หรือมีภาวะเครียด หรือไม่ ( ) ไม่มี ( ) มี ระบุ.....
2. ท่านมีโรคประจำตัวหรือเคยมีประวัติการเจ็บป่วยที่สำคัญ หรือไม่ ( ) ไม่มี ( ) มี ระบุ.....
3. ท่านเคยปวดข้อเข่าขณะทำกิจกรรมดังต่อไปนี้หรือไม่
   - ขึ้น-ลงบันได ( ) ไม่เคย ( ) เคย
   - ยืนย่อเข่า ( ) ไม่เคย ( ) เคย
   - วิ่ง ( ) ไม่เคย ( ) เคย
   - กระโดด ( ) ไม่เคย ( ) เคย
   - นั่งเป็นเวลานาน ( ) ไม่เคย ( ) เคย
4. ท่านเคยมีประวัติกระดูกสะบ้าหลวมหรือเลื่อนหลุด หรือไม่ ( ) ไม่เคย ( ) เคย
5. ท่านเคยได้รับบาดเจ็บบริเวณเข่า หรือไม่ ( ) ไม่เคย ( ) เคย ระบุ.....
6. ท่านเคยได้รับการผ่าตัดบริเวณเข่าหรือระยางค์ส่วนล่าง หรือไม่ ( ) ไม่เคย ( ) เคย ระบุ.....
7. (สำหรับอาสาสมัครหญิง) ขณะนี้ท่านกำลังมีรอบเดือน หรือไม่ ( ) ไม่มี ( ) มี

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APPENDIX C.2

DATA COLLECTING FORM

ID……. Date……..

1. Personal information

Sex ( ) F ( ) M Age…….years….. months Dominant leg ( ) Rt. ( ) Lt.

Weight……...kg Height…….cm BMI…….kg/m^2

2. Physical examination

2.1 Adduction contracture test ( ) Positive ( ) Negative

2.2 Ober's test ( ) Positive ( ) Negative

2.3 Knee alignment

2.3.1 Genu valgus ( ) Present ( ) No present

2.3.2 Genu varus ( ) Present ( ) No present

2.3.3 Genu recurvatum ( ) Present ( ) No present

3. Maximum hip adductor exertion test

	1st trial	**2nd trial**	**3rd trial**	**Max.**
Maximum hip adductor exertion				

4. EMG data collecting

3.1 Maximum Voluntary Isometric Contraction (MVC) Test

	1st trial		**2nd trial**		**3rd trial**		**Average (2 selected trial)**	
MVC	vmo	vl	vmo	vl	vmo	vl	vmo	vl

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3.2 Exercise testing

Exercise	Normalized RMS											
	1st trial			2nd trial			3rd trial			Average (2 selected trial)		
	vmo	vl	vmo:vl	vmo	vl	vmo:vl	vmo	vl	vmo:vl	vmo	vl	vmo:vl
SE (0%)												
35%SE-SE												
70%SE-SE												
100%SE-SE												
35%SF-SE												
70%SF-SE												
100%SF-SE												

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APPENDIX D
SAMPLE SIZE CALCULATION

The sample size for this study was calculated by the following equation

$$n = \left[\frac{(Z_{\alpha/2} + Z_{\beta})\sigma}{(\mu_1 - \mu_2)} \right]^2$$

n = sample size of the study

$Z_{\alpha/2}$ = Z-value when setting the confident level equal to 95% or significant level equal to 0.05 ($\alpha = 0.05$) = 1.96

Z_{β} = Z-value when setting the power of testing equal to 80% ($\beta = 0.2$) = 0.84

$\mu_1 - \mu_2$ = the difference of mean of parameter between the exercise conditions in the pilot study

σ = estimated standard deviation

From the sample size calculation, the normalized root mean square (RMS) of the vastus medialis oblique (VMO) muscle was concerned for this study. Below is sample size calculation using the normalized RMS of the VMO between the squeeze at 60 degrees knee flexion-squat exercise (SF-SE) with 0% and 70% hip adductor exertion resulted from a pilot study (n= 10).

$$n = \left[\frac{(Z_{\alpha/2} + Z_{\beta})\sigma}{(\mu_1 - \mu_2)} \right]^2$$

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$$n = \left[\frac{(1.96 + 0.84)\ 21.21}{(34.54 - 46.62)} \right]^2$$

$$= 24.22$$

The result of sample size calculation was 24.22. Hence sample size of this study was set as 30 subjects.

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APPENDIX E
DATA DISTRIBUTION

Kolmogorov-Sminov Goodness of Fit test was performed to test normal distribution of all parameters. Normal distribution of all parameters ($p > 0.05$) are presented in Table E.1.

Table E.1 Determination of data distribution tested by Kolmogorov-Sminov Goodness of Fit test (the values in this table are shown in form of *p*-value)

Parameter	*p*-value						
	SE	SF-SE			SE-SE		
	0%	35%	70%	100%	35%	70%	100%
Normalized RMS							
VMO	0.591	0.633	0.468	0.609	0.966	0.714	0.749
VL	0.536	0.941	0.667	0.865	0.515	0.306	0.079
VMO:VL ratio	0.970	0.293	0.645	0.709	0.691	0.831	0.699

significant difference at $p < 0.05$

SE = squat without hip adduction exercise

SF-SE = squeeze (hip adduction) at 60 degrees knee flexion–squat exercise

SE-SE = squeeze (hip adduction) entire time–squat exercise

0%, 35%, 70% and 100% = percentage hip adductor exertion

VMO = vastus medialis oblique muscle

VL= vastus lateralis muscle

VMO:VL ratio = ratio between normalized RMS of the VMO and VL muscles

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APPENDIX F
RAW DATA OF THE STUDY

Table F.1 Characteristics and maximum hip adductor exertion of the subjects

ID	Gender (M, F)	Age (yrs)	Weight (kg)	Height (cm)	BMI (kg/m^2)	Hip adductor exertion (mmHg)
1	M	18	65	173	21.72	285
2	M	21	56	174	18.50	250
3	M	20	68	173	22.72	250
4	M	26	72	180	22.22	285
5	M	23	66	173	22.05	280
6	M	26	61	170	21.11	300
7	M	20	65	175	21.22	285
8	F	20	52	165	19.10	246
9	F	26	50	163	18.82	220
10	F	23	62	165	22.77	250
11	F	25	58	163	21.83	215
12	M	24	64	174	21.14	270
13	M	24	64	168	22.68	276
14	M	19	60	178	18.94	285
15	M	19	70	180	21.60	285
16	F	22	42	157	17.04	246
17	M	19	51	169	17.86	264
18	M	19	55	172	18.59	260
19	F	24	48	156	19.72	255
20	F	20	55	165	20.20	246
21	F	19	55	160	21.48	264
22	F	20	43	162	16.38	228
23	F	20	53	153	22.64	255
24	F	19	46	155	19.15	250
25	F	19	45	150	20.00	250
26	F	19	52	167	18.65	250
27	F	18	51	153	21.79	250
28	F	20	50	159	19.78	240
29	M	22	54	158	21.63	260
30	M	19	70	175	22.86	240

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Table F.2 Root mean square (RMS) of the vastus medialis oblique (VMO) muscle

ID	RMS of the VMO muscle (mV)															
	100% MVC		0%SE		SF-SE						SE-SE					
					35%		70%		100%		35%		70%		100%	
	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2
1	329.76	349.35	112.29	97.65	37.63	39.93	81.79	90.98	118.62	127.41	75.44	63.49	110.20	129.01	103.58	111.22
2	477.61	362.05	91.93	101.58	69.50	79.36	75.24	93.59	74.79	72.54	59.43	64.24	57.08	60.49	71.82	61.28
3	195.26	218.80	51.15	58.96	27.71	25.81	74.62	51.69	134.88	129.84	61.80	72.92	76.84	67.07	73.13	104.55
4	153.86	174.55	48.57	56.20	79.16	75.46	130.95	139.85	164.38	189.22	76.27	68.89	56.74	121.12	164.01	175.41
5	595.19	596.61	123.91	105.45	88.69	89.83	130.51	139.34	153.44	166.82	102.09	100.30	152.84	159.70	266.66	248.24
6	266.12	255.34	118.87	112.05	78.18	87.76	104.92	143.72	212.53	242.20	105.19	99.24	133.83	131.68	134.39	168.06
7	348.05	388.76	91.79	100.47	139.43	121.81	134.79	127.50	183.36	170.28	145.42	133.35	126.11	136.39	151.33	164.03
8	223.92	205.13	51.78	50.06	33.20	29.99	58.43	70.23	70.38	81.13	63.84	57.71	67.55	83.53	87.84	88.59
9	177.07	141.73	52.45	45.08	52.08	62.82	67.63	60.86	59.19	56.74	29.04	26.04	62.84	59.70	101.91	83.98
10	142.51	150.27	60.27	54.95	81.68	82.75	134.99	106.20	123.43	122.76	71.26	65.17	86.53	88.92	114.06	97.21
11	66.20	61.53	42.31	42.55	30.04	27.21	32.86	30.15	43.70	45.85	27.73	31.26	29.53	32.64	30.41	31.67
12	175.80	195.38	64.16	59.18	21.99	27.12	39.97	34.34	72.61	61.79	63.26	66.63	125.77	148.96	174.78	210.40
13	166.11	154.07	43.13	46.89	35.77	32.43	65.51	55.60	114.00	124.35	41.91	44.34	59.66	53.44	121.07	111.34
14	315.45	335.03	33.90	34.32	64.29	73.40	89.42	84.57	257.54	228.97	41.11	36.85	126.11	130.30	206.00	222.18
15	241.56	229.68	107.12	115.84	68.80	62.96	77.37	74.54	189.80	189.65	94.23	88.88	120.92	136.98	116.29	159.46

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Table F.2 Root mean square (RMS) of the vastus medialis oblique (VMO) muscle (Continued)

ID	RMS of the VMO muscle (mV)															
	100% MVC		0%SE		SF-SE						SE-SE					
					35%		70%		100%		35%		70%		100%	
	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2
16	234.03	254.29	98.42	96.68	59.75	66.97	65.12	53.57	66.04	75.52	58.69	59.15	66.23	62.14	67.60	60.18
17	436.59	398.59	111.04	114.30	76.70	82.48	64.67	65.59	107.47	97.18	95.95	103.65	119.33	107.43	191.66	202.02
18	274.00	407.71	124.24	109.63	67.92	55.99	89.61	89.01	118.06	118.97	57.14	68.40	78.63	81.34	139.14	161.17
19	488.44	541.38	186.59	152.18	221.52	232.60	271.63	247.21	315.69	271.90	226.04	219.99	209.40	216.47	326.03	313.96
20	162.54	168.26	76.74	77.81	46.25	50.67	58.35	59.73	112.81	115.35	43.90	40.80	52.39	53.16	74.62	77.46
21	66.97	90.61	56.12	50.86	23.82	22.40	40.68	39.46	55.35	53.19	32.15	33.52	60.55	65.68	99.26	107.47
22	303.45	285.98	84.88	98.15	74.80	101.25	115.43	114.69	160.28	160.45	88.28	80.21	116.34	127.75	164.40	143.75
23	97.93	101.16	52.68	48.65	35.82	32.81	35.33	32.29	81.47	70.20	31.59	37.20	68.51	71.00	103.18	123.50
24	386.46	392.68	112.14	109.16	109.79	104.59	195.87	196.37	264.39	288.91	142.64	128.43	195.04	191.12	345.61	330.89
25	346.86	387.13	146.80	141.55	33.72	41.09	54.89	64.94	150.12	145.13	42.01	47.87	60.40	62.22	115.65	137.90
26	210.32	212.92	41.35	43.73	40.17	44.31	71.12	70.86	125.31	158.51	66.53	61.56	85.91	64.56	94.00	91.10
27	112.03	106.88	54.06	60.74	46.94	36.19	43.22	43.86	134.09	150.76	31.92	37.50	52.35	62.91	72.57	101.48
28	97.28	103.54	64.67	65.51	52.00	46.25	56.82	63.80	103.62	117.05	38.48	44.51	78.01	69.32	99.80	94.86
29	214.41	209.58	39.71	45.21	30.04	34.67	57.79	57.26	90.65	80.16	29.89	32.26	135.70	112.99	156.48	158.57
30	418.78	407.31	145.24	152.24	208.49	185.33	255.65	272.95	313.20	254.44	212.29	205.55	248.17	211.10	283.55	268.34

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Table F.3 Root mean square (RMS) of the vastus lateralis (VL) muscle

ID	RMS of the VL muscle (mV)															
	100% MVC		0%SE		SF-SE						SE-SE					
					35%		70%		100%		35%		70%		100%	
	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2
1	148.13	154.17	60.67	48.93	36.89	41.13	47.99	55.09	70.45	71.71	53.99	55.72	61.63	63.08	68.29	62.53
2	277.55	153.93	32.98	36.79	23.02	24.42	22.66	30.42	24.31	26.40	61.25	58.54	59.28	61.66	51.72	52.83
3	105.40	123.92	24.32	25.94	35.30	32.73	29.91	28.51	43.18	55.22	33.91	34.76	30.68	38.58	37.39	51.81
4	153.54	152.74	65.87	65.29	90.28	80.97	139.97	149.36	159.75	167.60	94.19	84.06	65.30	136.29	192.55	201.94
5	287.74	223.52	102.94	86.09	84.97	78.25	98.71	123.41	120.53	132.30	96.37	83.04	112.40	112.73	185.07	162.92
6	193.86	204.02	82.87	95.46	68.56	85.25	124.75	142.12	228.26	217.65	120.63	130.21	144.55	144.41	159.32	158.60
7	285.80	290.50	66.78	79.30	89.30	89.41	90.83	96.24	140.90	133.99	104.63	95.18	93.87	102.55	122.06	126.84
8	73.91	82.95	16.61	16.55	16.80	16.21	23.50	27.83	27.89	31.21	22.12	17.77	21.90	30.53	29.61	34.12
9	138.05	125.09	49.17	44.82	58.39	67.99	70.91	73.61	59.51	58.13	34.08	29.04	75.45	70.95	112.91	94.98
10	235.03	236.90	77.33	67.60	102.94	92.56	189.76	159.67	204.05	154.43	93.76	90.51	125.78	140.22	155.63	177.26
11	43.83	41.79	23.40	25.29	29.76	26.00	31.70	36.96	54.58	44.47	28.66	30.46	24.72	27.18	30.90	31.08
12	181.80	194.36	49.93	50.65	77.34	83.55	107.02	118.56	197.94	204.93	107.76	99.52	102.87	110.75	124.49	150.88
13	228.20	275.42	45.79	42.56	37.34	41.04	60.31	52.44	77.48	81.03	38.01	36.06	59.29	55.45	87.03	91.80
14	155.28	145.62	25.17	24.57	30.76	38.12	38.49	34.93	85.21	100.09	24.40	24.35	53.96	54.89	73.00	75.27
15	133.45	120.32	40.22	44.13	38.37	41.60	55.03	54.88	73.06	78.70	37.00	51.03	62.44	67.83	83.22	88.89

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Table F.3 Root mean square (RMS) of the vastus lateralis (VL) muscle (Continued)

ID	RMS of the VL muscle (mV)															
	100% MVC		0%SE		SF-SE						SE-SE					
					35%		70%		100%		35%		70%		100%	
	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2
16	46.08	54.63	28.1	27.68	40.69	44.27	55.34	50.37	46.08	54.63	28.1	27.68	40.69	44.27	55.34	50.37
17	100.15	95.99	57.75	66.44	80.27	68.09	105.72	95.11	100.15	95.99	57.75	66.44	80.27	68.09	105.72	95.11
18	86.03	82.57	67.2	70.55	101.9	98.77	111.82	109.53	86.03	82.57	67.2	70.55	101.9	98.77	111.82	109.53
19	180.16	157.14	102.94	115.22	113.11	128.96	182.7	172.29	180.16	157.14	102.94	115.22	113.11	128.96	182.7	172.29
20	105.29	134.8	61	61.46	114.74	100	171.28	163.64	105.29	134.8	61	61.46	114.74	100	171.28	163.64
21	63.3	62.86	40.37	41.94	62.3	66.19	91.19	100.19	63.3	62.86	40.37	41.94	62.3	66.19	91.19	100.19
22	63.94	64.37	33.99	34.18	48.61	49.5	71.75	68.51	63.94	64.37	33.99	34.18	48.61	49.5	71.75	68.51
23	84	87.94	39.45	39.68	52.82	52.24	79.6	92.88	84	87.94	39.45	39.68	52.82	52.24	79.6	92.88
24	70.85	71.16	53.92	55.01	73.55	82.95	103.62	78.83	70.85	71.16	53.92	55.01	73.55	82.95	103.62	78.83
25	62.89	49.46	41.65	42.28	61.91	59.71	70.2	100.53	62.89	49.46	41.65	42.28	61.91	59.71	70.2	100.53
26	102.64	78.23	87.29	74.37	94.28	87.89	134.87	127.28	102.64	78.23	87.29	74.37	94.28	87.89	134.87	127.28
27	63.41	68.3	26.36	25.15	43.2	44.98	52.31	67.1	63.41	68.3	26.36	25.15	43.2	44.98	52.31	67.1
28	65.27	60.57	35.88	49.88	49.76	36.57	62.5	46.53	65.27	60.57	35.88	49.88	49.76	36.57	62.5	46.53
29	157.27	168.48	80.91	79.97	197.2	179.02	188	185.54	157.27	168.48	80.91	79.97	197.2	179.02	188	185.54
30	278.07	245.55	204.91	211.5	262.02	225.41	249.83	245.98	278.07	245.55	204.91	211.5	262.02	225.41	249.83	245.98

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Table F.4 Normalized root mean square (RMS) of the vastus medialis oblique (VMO) muscle

ID	Normalized RMS (%MVC) of the VMO muscle													
	0%SE		SF-SE						SE-SE					
			35%		70%		100%		35%		70%		100%	
	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2
1	33.07	28.76	11.08	11.76	24.09	26.79	34.93	37.52	22.22	18.70	32.45	37.99	30.50	32.76
2	21.90	24.20	16.55	18.90	17.92	22.29	17.82	17.28	14.16	15.30	13.60	14.41	17.11	14.60
3	24.71	28.48	13.38	12.47	36.04	24.97	65.15	62.71	29.85	35.22	37.11	32.40	35.32	50.50
4	29.58	34.22	48.21	45.95	79.75	85.17	100.11	115.23	46.45	41.96	34.55	73.76	99.88	106.82
5	20.79	17.70	14.88	15.07	21.90	23.38	25.75	28.00	17.13	16.83	25.65	26.80	44.75	41.66
6	45.59	42.98	29.98	33.66	40.24	55.12	81.51	92.89	40.35	38.06	51.33	50.51	51.55	64.46
7	24.91	27.27	37.85	33.06	36.59	34.61	49.77	46.22	39.47	36.20	34.23	37.02	41.08	44.52
8	24.14	23.33	15.48	13.98	27.24	32.74	32.81	37.82	29.76	26.90	31.49	38.94	40.95	41.30
9	32.90	28.28	32.67	39.41	42.43	38.18	37.13	35.60	18.22	16.34	39.42	37.45	63.94	52.68
10	41.17	37.54	55.79	56.53	92.21	72.54	84.32	83.86	48.68	44.51	59.11	60.74	77.91	66.40
11	66.25	66.62	47.03	42.60	51.45	47.21	68.42	71.78	43.41	48.94	46.24	51.11	47.62	49.58
12	34.57	31.89	11.85	14.61	21.54	18.50	39.12	33.29	34.09	35.90	67.77	80.26	94.18	113.36
13	26.94	29.29	22.35	20.26	40.92	34.73	71.21	77.67	26.18	27.70	37.26	33.38	75.63	69.54
14	10.42	10.55	19.77	22.57	27.49	26.00	79.18	70.40	12.64	11.33	38.78	40.06	63.34	68.31
15	45.46	49.16	29.20	26.72	32.84	31.64	80.55	80.49	39.99	37.72	51.32	58.14	49.35	67.68

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Table F.4 Normalized root mean square (RMS) of the vastus medialis oblique (VMO) muscle (Continued)

ID	Normalized RMS (%MVC) of the VMO muscle													
	0%SE		SF-SE						SE-SE					
			35%		70%		100%		35%		70%		100%	
	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2
16	40.31	39.60	24.47	27.43	26.67	21.94	27.05	30.93	24.04	24.22	27.13	25.45	27.69	24.65
17	26.59	27.37	18.37	19.75	15.49	15.71	25.74	23.27	22.98	24.82	28.58	25.73	45.90	48.38
18	36.45	32.16	19.93	16.43	26.29	26.12	34.64	34.90	16.76	20.07	23.07	23.86	40.82	47.28
19	36.24	29.55	43.02	45.17	52.75	48.01	61.31	52.81	43.90	42.72	40.67	42.04	63.32	60.97
20	46.40	47.04	27.96	30.63	35.28	36.11	68.20	69.74	26.54	24.67	31.68	32.14	45.12	46.83
21	71.22	64.55	30.23	28.43	51.63	50.08	70.24	67.51	40.80	42.55	76.85	83.36	125.98	136.41
22	28.80	33.30	25.38	34.36	39.17	38.92	54.38	54.44	29.95	27.22	39.47	43.35	55.78	48.78
23	52.92	48.88	35.98	32.96	35.49	32.44	81.84	70.52	31.74	37.37	68.82	71.33	103.65	124.06
24	28.78	28.02	28.18	26.85	50.28	50.41	67.87	74.16	36.61	32.97	50.07	49.06	88.71	84.94
25	40.00	38.57	9.19	11.20	14.96	17.70	40.91	39.54	11.45	13.04	16.46	16.95	31.51	37.58
26	19.54	20.66	18.98	20.94	33.61	33.48	59.21	74.90	31.44	29.09	40.59	30.51	44.42	43.05
27	49.39	55.49	42.89	33.07	39.48	40.07	122.51	137.74	29.17	34.26	47.83	57.47	66.30	92.71
28	64.41	65.25	51.79	46.06	56.59	63.54	103.20	116.58	38.32	44.33	77.70	69.04	99.40	94.47
29	18.73	21.33	14.17	16.36	27.26	27.01	42.76	37.81	14.10	15.22	64.01	53.30	73.81	74.80
30	35.16	36.86	50.48	44.87	61.89	66.08	75.83	61.60	51.40	49.76	60.08	51.11	68.65	64.97

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Table F.5 Normalized root mean square (RMS) of the vastus lateralis (VL) muscle

ID	Normalized RMS (%MVC) of the VL muscle													
	0%SE		SF-SE						SE-SE					
			35%		70%		100%		35%		70%		100%	
	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2
1	40.14	32.37	24.41	27.21	31.75	36.45	46.61	47.45	35.72	36.87	40.78	41.74	45.18	41.37
2	15.28	17.05	10.67	11.32	10.50	14.10	11.27	12.24	28.39	27.13	27.48	28.58	23.97	24.49
3	21.21	22.62	30.79	28.55	26.08	24.86	37.66	48.16	29.58	30.31	26.76	33.64	32.61	45.19
4	43.01	42.64	58.95	52.87	91.40	97.53	104.31	109.45	61.51	54.89	42.64	88.99	125.74	131.86
5	40.27	33.68	33.24	30.61	38.62	48.28	47.15	51.75	37.70	32.49	43.97	44.10	72.40	63.73
6	41.65	47.99	34.46	42.85	62.71	71.44	114.74	109.41	60.63	65.45	72.66	72.59	80.09	79.72
7	23.17	27.52	30.99	31.03	31.52	33.40	48.90	46.50	36.31	33.03	32.58	35.59	42.36	44.02
8	21.18	21.10	21.42	20.67	29.96	35.48	35.57	39.80	28.21	22.66	27.92	38.92	37.75	43.50
9	37.37	34.06	44.38	51.67	53.89	55.95	45.23	44.18	25.90	22.07	57.35	53.93	85.81	72.19
10	32.77	28.65	43.62	39.23	80.42	67.67	86.47	65.44	39.73	38.36	53.31	59.42	65.96	75.12
11	54.66	59.08	69.51	60.74	74.06	86.35	127.51	103.87	66.94	71.16	57.75	63.50	72.18	72.60
12	26.55	26.93	41.12	44.42	56.90	63.04	105.24	108.96	57.29	52.91	54.69	58.89	66.19	80.22
13	18.18	16.90	14.83	16.30	23.95	20.82	30.77	32.18	15.09	14.32	23.54	22.02	34.56	36.46
14	16.73	16.33	20.45	25.34	25.58	23.22	56.64	66.53	16.22	16.19	35.86	36.49	48.52	50.03
15	31.70	34.78	30.24	32.79	43.37	43.26	57.58	62.03	29.16	40.22	49.21	53.46	65.58	70.06

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Table F.5 Normalized root mean square (RMS) of the vastus lateralis (VL) muscle (Continued)

ID	Normalized RMS (%MVC) of the VL muscle													
	0%SE		SF-SE						SE-SE					
			35%		70%		100%		35%		70%		100%	
	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2
16	44.61	40.73	51.50	54.60	41.20	46.50	51.43	60.97	31.37	30.89	45.41	49.40	61.77	56.21
17	45.21	47.01	49.14	48.88	52.54	59.13	72.60	69.58	41.87	48.16	58.19	49.36	76.64	68.95
18	33.35	31.43	26.73	21.78	25.90	27.24	35.12	33.71	27.44	28.80	41.60	40.33	45.65	44.72
19	40.69	30.21	55.53	54.06	52.73	58.65	71.40	62.27	40.80	45.66	44.83	51.11	72.41	68.28
20	55.27	50.58	61.47	59.00	64.66	72.06	95.02	121.64	55.05	55.46	103.54	90.24	154.56	147.66
21	77.13	85.29	62.17	60.18	77.64	65.98	89.76	89.14	57.24	59.47	88.34	93.85	129.31	142.07
22	44.49	50.38	64.30	69.19	54.75	57.83	90.71	91.32	48.21	48.48	68.96	70.22	101.79	97.18
23	25.29	27.35	35.00	32.54	33.34	36.36	71.82	75.19	33.73	33.93	45.16	44.67	68.06	79.41
24	34.97	36.60	30.11	25.57	33.79	35.71	42.90	43.08	32.65	33.31	44.53	50.22	62.74	47.73
25	31.19	33.14	21.25	21.02	29.94	33.87	38.82	30.53	25.71	26.10	38.22	36.86	43.34	62.07
26	55.69	67.92	94.51	86.00	94.08	100.20	133.68	101.88	113.68	96.86	122.78	114.47	175.65	165.77
27	24.13	27.36	48.49	44.89	49.72	56.59	70.82	76.27	29.43	28.08	48.25	50.23	58.42	74.94
28	43.91	38.34	37.26	44.06	40.45	51.27	74.40	69.04	40.89	56.85	56.71	41.68	71.24	53.04
29	34.56	33.31	45.19	43.31	68.30	69.78	96.47	103.35	49.63	49.05	120.96	109.81	115.32	113.81
30	28.42	31.86	46.28	40.61	70.78	69.17	71.82	63.42	52.92	54.63	67.68	58.22	64.53	63.53

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Table F.6 VMO:VL ratio (normalized root mean square (RMS))

ID	VMO:VL ratio (normalized RMS; %MVC)													
	0%SE		SF-SE						SE-SE					
			35%		70%		100%		35%		70%		100%	
	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2
1	0.824	0.888	0.454	0.432	0.759	0.735	0.750	0.791	0.622	0.507	0.796	0.910	0.675	0.792
2	1.433	1.419	1.551	1.670	1.706	1.581	1.581	1.412	0.499	0.564	0.495	0.504	0.714	0.596
3	1.165	1.259	0.435	0.437	1.382	1.004	1.730	1.302	1.009	1.162	1.387	0.963	1.083	1.118
4	0.688	0.803	0.818	0.869	0.873	0.873	0.960	1.053	0.755	0.764	0.810	0.829	0.794	0.810
5	0.516	0.525	0.448	0.492	0.567	0.484	0.546	0.541	0.454	0.518	0.583	0.608	0.618	0.654
6	1.095	0.896	0.870	0.786	0.642	0.772	0.710	0.849	0.665	0.582	0.706	0.696	0.644	0.809
7	1.075	0.991	1.221	1.066	1.161	1.036	1.018	0.994	1.087	1.096	1.051	1.040	0.970	1.012
8	1.140	1.106	0.723	0.676	0.909	0.923	0.922	0.950	1.055	1.187	1.128	1.000	1.085	0.949
9	0.880	0.830	0.736	0.763	0.787	0.682	0.821	0.806	0.703	0.740	0.687	0.694	0.745	0.730
10	1.256	1.310	1.279	1.441	1.147	1.072	0.975	1.281	1.225	1.161	1.109	1.022	1.181	0.884
11	1.212	1.128	0.677	0.701	0.695	0.547	0.537	0.691	0.648	0.688	0.801	0.805	0.660	0.683
12	1.302	1.184	0.288	0.329	0.378	0.294	0.372	0.306	0.595	0.679	1.239	1.363	1.423	1.413
13	1.482	1.733	1.507	1.243	1.708	1.668	2.314	2.414	1.735	1.934	1.583	1.516	2.188	1.908
14	0.623	0.646	0.967	0.891	1.075	1.120	1.398	1.058	0.779	0.700	1.081	1.098	1.305	1.365
15	1.434	1.414	0.966	0.815	0.757	0.731	1.399	1.298	1.371	0.938	1.043	1.088	0.753	0.966

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Table F.6 VMO:VL ratio (normalized root mean square (RMS)) (Continued)

ID	VMO:VL ratio (normalized RMS; %MVC)													
	0%SE		SF-SE						SE-SE					
			35%		70%		100%		35%		70%		100%	
	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2	Trial1	Trial2
16	0.904	0.972	0.475	0.502	0.647	0.472	0.526	0.507	0.766	0.784	0.597	0.515	0.448	0.438
17	0.588	0.582	0.374	0.404	0.295	0.266	0.355	0.334	0.549	0.515	0.491	0.521	0.599	0.702
18	1.093	1.023	0.745	0.754	1.015	0.959	0.986	1.035	0.611	0.697	0.554	0.592	0.894	1.057
19	0.891	0.978	0.775	0.836	1.000	0.819	0.859	0.848	1.076	0.936	0.907	0.823	0.874	0.893
20	0.840	0.930	0.455	0.519	0.546	0.501	0.718	0.573	0.482	0.445	0.306	0.356	0.292	0.317
21	0.923	0.757	0.486	0.472	0.665	0.759	0.783	0.757	0.713	0.715	0.870	0.888	0.974	0.960
22	0.647	0.661	0.395	0.497	0.715	0.673	0.600	0.596	0.621	0.561	0.572	0.617	0.548	0.502
23	2.093	1.787	1.028	1.013	1.064	0.892	1.140	0.938	0.941	1.101	1.524	1.597	1.523	1.562
24	0.823	0.765	0.936	1.050	1.488	1.412	1.582	1.721	1.121	0.990	1.124	0.977	1.414	1.780
25	1.283	1.164	0.432	0.533	0.499	0.522	1.054	1.295	0.445	0.500	0.431	0.460	0.727	0.605
26	0.351	0.304	0.201	0.243	0.357	0.334	0.443	0.735	0.277	0.300	0.331	0.267	0.253	0.260
27	2.047	2.028	0.884	0.737	0.794	0.708	1.730	1.806	0.991	1.220	0.991	1.144	1.135	1.237
28	1.467	1.702	1.390	1.045	1.399	1.239	1.387	1.689	0.937	0.780	1.370	1.657	1.395	1.781
29	0.542	0.640	0.314	0.378	0.399	0.387	0.443	0.366	0.284	0.310	0.529	0.485	0.640	0.657
30	1.237	1.157	1.091	1.105	0.874	0.955	1.056	0.971	0.971	0.911	0.888	0.878	1.064	1.023

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BIOGRAPHY

NAME	Miss Sirirat Kaewwichai
DATE OF BIRTH	31 March 1984
PLACE OF BIRTH	Sisaket, Thailand
INSTITUTIONS ATTENED	Mahidol University, 2002-2006 Bachelor of Science (Physical Therapy) with the first class honors Mahidol University, 2006-2009 Master of Science (Physical Therapy)
SCHOLARSHIP	Grant to Support Graduate Students in Academic Presentations in Thailand Academic Year 2008
HOME ADDRESS	187 Moo 3, Nong-Ngu-Lueam, Benchalak, Sisaket, 33110 Tel. 08-6095-6616 E-mail: sirirat_kwch@hotmail.com
PLUBLICATION	Oral Presentation at the 11th National Graduate Research Conference 17-18 December 2008, Graduate School, Valaya Alongkorn Rajabhat University, Pathumthani
AWARD	Outstanding Oral Presentation Award at the 11th National Graduate Research Conference 17-18 December 2008, Graduate School, Valaya Alongkorn Rajabhat University, Pathumthani

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